

RESPIRATORY DISEASES



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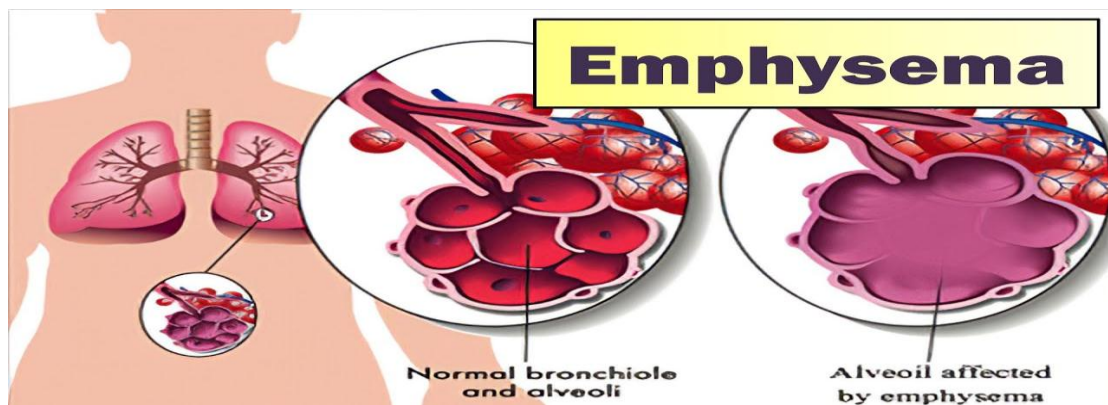
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Chronic Obstructive Pulmonary Disease (COPD)

Definitions:

- **Chronic Obstructive Pulmonary Disease (COPD):** is a common, preventable, and treatable heterogeneous lung condition characterized by chronic respiratory symptoms (dyspnea, cough, sputum production, exacerbations) due to abnormalities of the airways (bronchitis, bronchiolitis) and/or alveoli (emphysema) that cause persistent, often progressive, airflow obstruction usually caused by significant exposure to noxious particles or gases.
- **Chronic Bronchitis:** the presence of cough and sputum production for at least 3 months in each of two consecutive years, after exclusion of other causes of chronic cough.
- **Emphysema:** pathological term in which abnormal permanent enlargement of the airspaces distal to the terminal bronchioles with destruction of their wall without obvious fibrosis.



Causes and risk factors:

COPD is the end-result of complex, cumulative and dynamic gene-environment interactions over the lifetime that can damage the lungs and/or alter their normal developmental or aging processes.

I-Environmental risk factors

1-Cigarette smoking: The main risk factor for COPD is tobacco smoking it accounts for 80 to 90% of the risk of developing COPD than non-

smokers. Other types of tobacco (e.g., pipe, cigar, water pipe) and marijuana are also risk factors for COPD. Passive exposure to cigarette smoke, also known as environmental tobacco smoke (ETS), may also contribute to respiratory symptoms and COPD.

2-Occupational exposure: including organic and inorganic dusts, chemical agents, and fumes

3- Air pollution: Air pollution typically consists of particulate matter (PM), ozone, oxides of nitrogen or sulfur, heavy metals, and other greenhouse gases, is a major worldwide cause of COPD also increases the risk of COPD exacerbations, hospitalizations, and mortality

II- Genetic risk factor

Documented genetic risk factor for COPD are genetic mutations that leads to the hereditary deficiency of α -1 antitrypsin (AATD) a major circulating inhibitor of serine proteases.

III-Age and sex

Aging (old age) is often listed as a risk factor for COPD and may due to cumulative exposures throughout life.as regard sex the prevalence of COPD is now equal in men and women, probably reflecting the changing patterns of tobacco smoking.

Prevalence of COPD

- Prevalence of COPD was higher in smokers and ex-smokers compared to non-smokers.
- In each country, the prevalence of COPD increased steeply with age, with the highest prevalence among those > 60 years.
- COPD is a major cause of chronic morbidity and mortality throughout the world.
- 3rd leading cause of death in the world.
- 3 million deaths annually.

Pathobiology

In patients with COPD pathological changes can be found in the airways, lung parenchyma, and pulmonary vasculature These include inflammatory and structural changes.

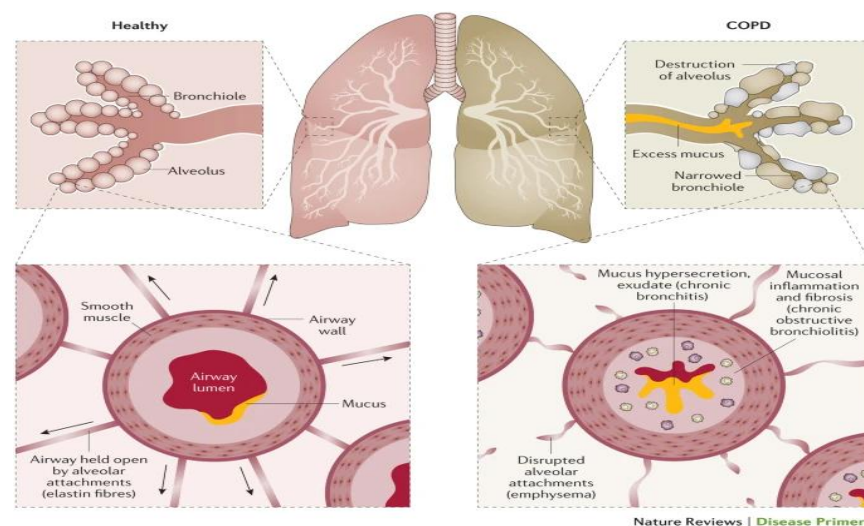
Inflammatory changes:

Inflammatory response to chronic irritants such as cigarette smoke includes:

- increased numbers of macrophages, increased activated neutrophils and increased lymphocytes
- release multiple inflammatory mediators attract inflammatory cells from the circulation
- amplify the inflammatory process (via proinflammatory cytokines), and induce structural changes (via growth factors)

Structural changes

- An imbalance in the lungs between proteases derived from inflammatory and epithelial cells that break down connective tissue components and antiproteases that counterbalance this action. Protease-mediated destruction of elastin, a major connective tissue component of the lung parenchyma, lead to emphysema
- Peribronchiolar fibrosis causing permanent development of small airways obstruction.



Pathophysiology

1-Airflow limitation and gas trapping: lead to exertional dyspnea in COPD

2-Mucus Hyper secretion: causes chronic cough and sputum production but does not occur in all COPD patients.

3-Gas exchange abnormalities: uneven distribution of ventilation-perfusion ratios leads to hypoxemia. In severe cases, hypercapnia due to alveolar hypoventilation is also observed.

4-pulmonary hypertension: in severe cases, the major cause is hypoxic pulmonary vasoconstriction may lead to right ventricular hypertrophy and eventually to right-side cardiac failure (cor-pulmonale).

5-Systemic features: Systemic inflammation characterized by increased inflammatory cytokine and C-reactive protein levels causing the following disorders:

1. Nutritional disorders leading to decreased fat mass and lean body mass.
2. Musculoskeletal disorders associated with decreased muscle mass and muscle strength.
3. Cardiovascular diseases, including myocardial infarction, angina pectoris, and cerebrovascular accidents.
4. Osteoporosis leading to vertebral compression fractures.
5. Diabetes mellitus
6. Sleep disorders
7. Anemia

Diagnosis of COPD

A-Clinical presentation:

Symptoms:

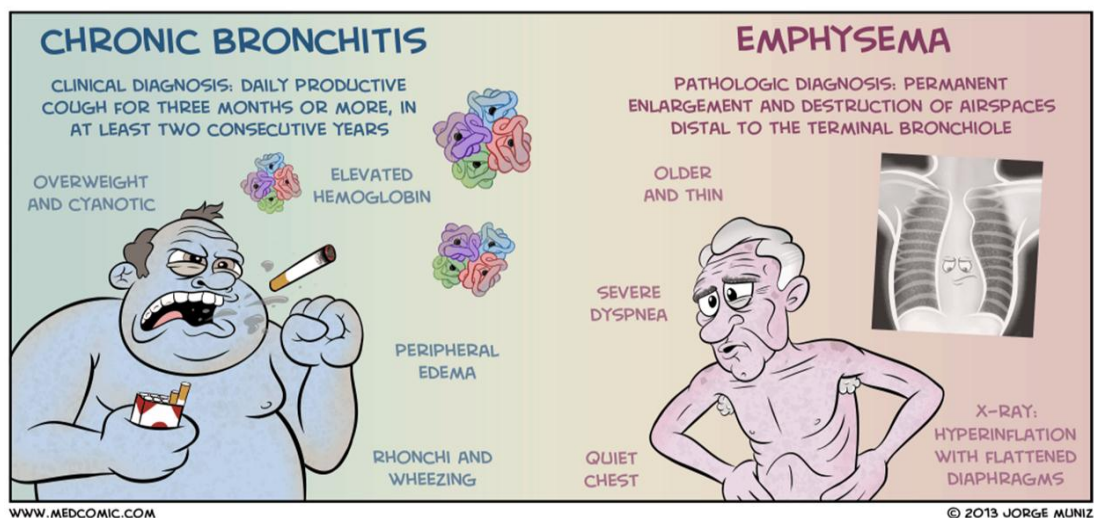
- 1- Chronic Cough: cough is often the first symptom of COPD, Initially, the cough may be intermittent, but subsequently may be present every day, often throughout the day.
- 2- Sputum production: expectorate small quantities of tenacious sputum with coughing.
- 3- Dyspnea is a cardinal symptom of COPD progressive and get worse with exertion.
- 4- Sputum production
- 5- Wheezing and chest tightness
- 6- Additional clinical features in severe disease: Fatigue, Weight loss, muscle mass loss, and anorexia

B-Physical examination

- **On inspection:** *Barrel shape* chest as a sign of hyperinflation.
- **Hoover's sign:** the horizontal position of diaphragm pull in the lower ribs during inspiration.
- **On percussion:** there is encroachment on hepatic and cardiac dullness due to overinflation, Hyperresonance note.
- **On auscultation:** vesicular breathing with prolonged expiratory phase
- **Adventitial sounds:** Ronchi, **generalized**, sonorous or sibilant mainly expiratory, and /or wet, inspiratory crepitations.
- **Clinical patterns of COPD:**

1- Pink puffer (A-type): emphysema more than chronic bronchitis patient is mostly thin and tall, tachypneic with signs of hyperinflation

2- Blue Bloater (B-type): chronic bronchitis more than emphysema, patient mostly obese and short, productive cough ,early complicated by respiratory failure and DCP.



Differential Diagnosis of COPD

Table 2.3

Diagnosis	Suggestive Features
COPD	Symptoms slowly progressive History of tobacco smoking or other risk factors
Asthma	Variable airflow obstruction Symptoms vary widely from day to day Symptoms worse at night/early morning Allergy, rhinitis, and/or eczema also present Often occurs in children Family history of asthma
Congestive heart failure	Chest X-ray shows dilated heart, pulmonary edema Pulmonary function tests indicate volume restriction, not airflow obstruction
Bronchiectasis	Large volumes of purulent sputum Commonly associated with bacterial infection Chest X-ray/HRCT shows bronchial dilation
Tuberculosis	Onset all ages Chest X-ray shows lung infiltrate Microbiological confirmation High local prevalence of tuberculosis
Obliterative bronchiolitis	Can occur in children Seen after lung or bone marrow transplantation HRCT on expiration shows hypodense areas
Diffuse panbronchiolitis	Predominantly seen in patients of Asian descent Most patients are male and nonsmokers Almost all have chronic sinusitis Chest X-ray & HRCT show diffuse small centrilobular nodular opacities & hyperinflation

These features tend to be characteristic of the respective diseases, but are not mandatory. For example, a person who has never smoked may develop COPD (especially in LMICs where other risk factors may be more important than cigarette smoking).

Other Causes of Chronic Cough

Table 2.2

INTRATHORACIC	EXTRATHORACIC
• Asthma • Lung Cancer • Tuberculosis • Bronchiectasis • Left Heart Failure • Interstitial Lung Disease • Cystic Fibrosis • Idiopathic Cough	• Chronic Allergic Rhinitis • Post Nasal Drip Syndrome (PNDS) • Upper Airway Cough Syndrome (UACS) • Gastroesophageal Reflux • Medication (e.g., ACE Inhibitors)

Investigations:

I. Diagnostic imagin:

I. A plain chest X-ray:

A: P-A view

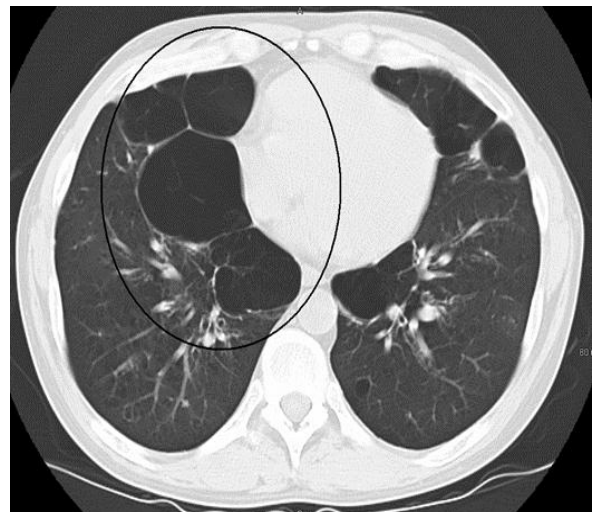
Findings include increased radiolucency of the lung fields, diminished peripheral blood vessel shadows in the lung fields, low & flattening of the diaphragm, decrease in the cardiothoracic ratio and widening of the intercostal spaces.

B: Lateral view

Findings include flattening of the diaphragm, widening of the intercostal spaces, and increase in the retrocardiac space.



II. CT chest: early detection of emphysematous lesions.



2. Pulmonary function tests:

A-Spirometry:

The diagnosis of COPD requires detection of an obstructive ventilation disorder by spirometry in which a post-bronchodilator FEV1/FVC less than 70%

- Assessing the **degree of reversibility** of airflow limitation by measuring FEV1 before and after bronchodilator to differentiate the diagnosis from asthma.
- An improvement of 12% over baseline FEV1 or FVC and an increase at least 200 ml has been suggested as significant bronchodilator response and diagnosis of asthma and exclusion of COPD.

B-Lung volumes: all lung volumes increased includes:

- Increase residual volume (RV): volume of air remains in the lung at the end of forced expiration. .(air trapping)
- Increase functional residual capacity (FRC): volume of air remain in the lung at end of normal expiration. .(air trapping)
- Increase total lung capacity (TLC): volume of air in the lung at maximum inspiration. .(hyperinflation)

C-pulmonary carbon monoxide diffusing capacity (DLCO):

Decreased surface area in COPD patients cause decrease diffusion.

3. Arterial blood gas analysis: is a useful means of assessing ventilation status, oxygenation capacity, and acid-base balance in a patient.

4-Alpha-1 antitrypsin deficiency (AATD) screening: A low concentration (< 20% normal) is highly suggestive of homozygous deficiency.

- It should be measured in all patients with early onset emphysema (age under 45years), or nonsmoker with emphysema.
- With positive family history

5-Other investigations:

- CBC: detect leukocytosis in acute exacerbation, polycythemia in chronic hypoxemia.
- ECG: detect arrhythmia

- ECHO: detect cor pulmonal & pulmonary hypertension.

Prevention of COPD:

1-Smoking Cessation: is the single most important step in prevention the progression of the disease, Nicotine replacement therapy can be used.

2-Vaccination: Influenza vaccine, Pneumococcal vaccine.

3-Avoidance of atmospheric pollution: stay indoors, wearing respiratory protective equipment (**masks**)

Treatment of COPD

Pharmacological therapy for COPD is used to: ►

- A. reduce symptoms,
- B. reduce the frequency and severity of exacerbations,
- C. improves exercise tolerance and health status.

I-PHARMACOLOGICAL THERAPY FOR STABLE COPD:

- Bronchodilators
- Anti-inflammatory agents
- Others

I-Bronchodilators

1-Beta2-agonists:

- **Action:** relax airway smooth muscle by stimulating beta2-adrenergic receptors, which increases cyclic AMP and produces functional antagonism to bronchoconstriction.
- There are short-acting (SABA) and long-acting (LABA) beta2-agonists.
- **SABAs** show duration of action within 4 to 6 hours as salbutamol, terbutaline, albuterol for regular and as-needed use.

- **LABAs** show duration of action of 12 or more hours as Formoterol and salmeterol, that improve lung function and symptoms.
- **Adverse effects:** sinus tachycardia, somatic tremor, hypokalemia

2-Antimuscarinic drugs (anticholinergic bronchodilators):

- **Action:** block the bronchoconstrictor effects of acetylcholine on M3 muscarinic receptors expressed in airway smooth muscle
- **Short-acting antimuscarinics (SAMAs)**, as ipratropium and oxitropium ‘
- **Long-acting muscarinic antagonists (LAMAs)**, such as tiotropium, have prolonged binding to M3 muscarinic receptors cause prolonging the duration of bronchodilator effect.
- **Adverse effects:** dryness of mouth, a bitter metallic taste.

3-Methylxanthines: act as non-selective phosphodiesterase inhibitors: as theophylline

Adverse effects. Toxicity is dose-related, which is a particular problem with xanthine derivatives because their therapeutic ratio is small and most of the benefit occurs only when near-toxic doses are given.

- atrial and ventricular arrhythmias, convulsions
- nausea, and heartburn
- These medications have significant interactions with commonly used medications such as erythromycin (but not azithromycin), certain quinolone antibiotics (ciprofloxacin)

4-Combination bronchodilator therapy: LABA and LAMA in a single inhaler, is more effective than long-acting bronchodilator monotherapy for preventing exacerbations.

II- Anti-inflammatory agents

a-Phosphodiesterase-4 (PDE4) inhibitors: Roflumilast is a once daily oral medication reduces moderate and severe exacerbations.

- **Adverse effects_**PDE4 inhibitors have more adverse effects than inhaled medications for COPD. The most frequent are diarrhea, nausea, reduced appetite, weight loss, abdominal pain, sleep disturbance, and headache.

B-Inhaled corticosteroids (ICS): In patients with moderate to very severe COPD and frequent exacerbations, an ICS combined with a LABA is more effective than either component alone in improving lung function, health status and reducing exacerbations, e.g. Beclomethasone, budesonide, fluticasone.

- **Adverse effects.** Oral candidiasis, hoarse voice and pneumonia.

III-Others

- **Alpha-1 antitrypsin augmentation therapy:** Intravenous augmentation therapy has been recommended for individuals with alpha-1 antitrypsin deficiency (AATD)
- **Mucolytics** (N-acetyl cysteine)

II. Non-pharmacologic therapy of COPD

I-Pulmonary rehabilitation: improve dyspnea, health status and exercise tolerance.

AIM: improve the physical and psychological condition and improve quality of life.

That include,

1. **Exercise training:** is the core of pulmonary rehabilitation upper and lower limbs ideally should be included as walking exercise; flexibility, inspiratory muscle training
2. **Education:** help patients deepen their understanding of the disease and gain the skills they will need for self-management during the stable period and exacerbations.
3. **Nutrition support:** nutritional supplementation promotes significant weight gain and leads to significant improvements in respiratory muscle strength.

II-Oxygen therapy: receive long-term oxygen therapy (LTOT) or home oxygen therapy (HOT) for more than 15 h/day improves the survival prognosis of COPD patients with severe chronic respiratory failure.

Indications of LTOT: 1- Resting room air Pao₂ <55 mmHg or oxygen saturation < 88%

2- Resting room air Pao₂ 56 to 60 mmHg or oxygen saturation > 88% to 89% with supporting evidence of chronic hypoxemia such as

- Polycythemia
- Pulmonary hypertension
- Cor-pulmonale or
- Psychological impairment

Oxygen administered by nasal cannula with the flow rate adjusted to maintain a resting saturation greater than 90%

III Interventional and Surgical treatment of COPD

- Lung volume reduction surgery (LVRS).). LVRS is a procedure leads to reduce hyperinflation of the lung can be done surgically or by bronchoscopy.
- Bullectomy. Removal of a large bulla that does not contribute to gas exchange.
- Lung transplantation: In appropriately selected patients with very severe COPD.

Complications of COPD

- 1-Acute exacerbation of COPD (the most common)
- 2- Hypercapnic Respiratory failure
- 3- Pneumonia
- 4-Cor-pulmonal
- 5- Pneumothorax
- 6-Lung cancer
- 7- Arrhythmias
- 8-Pulmonary embolism

Acute Exacerbation of COPD

Definition:

An exacerbation of chronic obstructive pulmonary disease (ECOPD) is defined as an event characterized by increased dyspnea and/or cough and sputum that worsens in < 14 days which is often associated with increased local and systemic inflammation caused by infection, pollution, or other insult to the airways.

Exacerbations are classified after the event has occurred as:

- ▶ Mild (treated with short acting bronchodilators only, SABDs)
- ▶ Moderate (treated with SABDs and oral corticosteroids ± antibiotics)
or
- ▶ Severe (patient requires hospitalization or visits the emergency room).
Severe exacerbations may also be associated with acute respiratory failure.

Investigations:

- Chest x-ray
- Arterial blood gases (ABG)
- CBC, ECG, ECHO

Indications for hospitalization

- Presence of high-risk co-morbid conditions, including congestive heart failure, diabetes mellitus, renal or liver failure
- Inadequate response of symptoms to outpatient management
- Marked increase in dyspnea
- Worsening hypoxemia
- Worsening hypercapnia
- Inability of the patient to care for her/himself
- Inadequate home care

Indications for ICU admission :

- Respiratory rate: > 30 breaths per minute; using accessory respiratory muscles;
- acute changes in mental status;
- hypoxemia not improved with supplemental oxygen via Venturi mask or requiring $\text{FiO}_2 > 40\%$;
- hypercarbia i.e., PaCO_2 increased compared with baseline or elevated > 60 mmHg or the presence of respiratory acidosis ($\text{pH} \leq 7.25$).

Management of Exacerbations

I-Pharmacological treatment

- **Bronchodilators:** that short-acting inhaled beta2-agonists, with or without short-acting anticholinergic, are the initial bronchodilators for acute treatment of a COPD exacerbation

Using metered dose inhalers (MDI) or nebulizers to deliver the agent.

- **Glucocorticoids:** A dose of 30-60 mg prednisone per day for 5 days is recommended.
- **Antibiotics:** the infectious agents in COPD exacerbations can be viral or bacterial (*Hemophilus influenza*, *Moraxella catarrhalis*, *Streptococcus pneumoniae*)
 - antibiotics should be given to patients with exacerbations of COPD who have three cardinal symptoms: increase in dyspnea, sputum volume, and sputum purulence; have two of the cardinal symptoms, if increased purulence of sputum is one of the two symptoms; or require mechanical ventilation (invasive or noninvasive)
 - Choice of antibiotic: Usually **initial empirical treatment** is an aminopenicillin with clavulanic acid, macrolide, or quinolone.
 - In patients with frequent exacerbations, severe airflow limitation and/or exacerbations requiring mechanical ventilation **cultures** from sputum or other materials from the lung should be performed, as gram-negative bacteria (e.g., *Pseudomonas* species) or resistant

pathogens that are not sensitive to the above-mentioned antibiotics may be present.

- The route of administration (oral or intravenous) depends on the patient's ability to eat and the pharmacokinetics of the antibiotic,
- Improvements in dyspnea and sputum purulence suggest clinical success.

- **Respiratory support**

- A-Oxygen therapy**

- B-Ventilatory support** in an exacerbation can be provided by either noninvasive (nasal or facial mask) or invasive ventilation.

- **Adjuvant therapies**

Depending on the clinical condition of the patient, , use of diuretics when clinically indicated, anticoagulants, treatment of comorbidities.

Bronchial Asthma

Definition:

Asthma is a chronic inflammatory disease of the bronchial tree characterized by hyper responsiveness to broncho-constricting stimuli with episodic air flow limitation that is reversible either spontaneously or with therapy.

Etiology:

** The etiology of Asthma is not fully understood. Asthma appears to be a multi factorial disorder with both **Genetic** and **Environmental** causes.

** It is likely to be a Clinical Syndrome in which various precipitating factors result in similar clinical and pathologic manifestation. e.g.

- Childhood-onset atopic asthma
- Adult-onset asthma in a middle-aged ex-smoker without apparent allergies,
- Asthma developing in the setting of occupational exposure to toluene di-isocyanate.

** Genetic factors: Asthma is familial. Asthma is frequently associated with clinical manifestation of Atopy (Familial allergy to common allergens) and elevated serum total IgE concentration, but some evidence suggests that asthma and allergy may be inherited independently.

Pathogenesis:

1. The Eosinophil is a major effector cell in asthmatic airway inflammation, but the Mast cell and T lymphocyte also play crucial roles.

2. The inflammatory response in asthmatic airways is extremely complex. It is characterized by: -

- Varying degrees of mononuclear cell and eosinophil infiltration
- Epithelial desquamation
- Mucus hyper secretion (airway plugging)
- Smooth muscle hyperplasia
- Airway remodeling with sub epithelial fibrosis

3. Mediators in the biology of Asthma, include:

A- Cytokines:

-Principal cytokines implicated in asthma pathogenesis are Interleukin-1B (IL-1B), IL5, IL3, IL4, and tumor necrosis factor- α (TNF α). They are produced by mast cells, T lymphocyte (TH2), macrophages and epithelial cells.

Cytokines have the following effects:

- 1) Cytokines regulate inflammation and the immune response.
- 2) Recruit eosinophils to the airways and perpetuating eosinophilic inflammation in asthma.
- 2) Increase expression of adhesion molecule on endothelium which enhances inflammatory cell migration into the airway.
- 3) IL4 plays a key role in initiating IgE synthesis and TH2 lymphocyte development.

B- Histamine:

- It is released on activation of mast cells primarily through IgE-receptor interaction
- Histamine has direct Vaso-active and smooth muscle spasmogenic effect

C- Arachidonic acid metabolites i.e., Lipid mediators (PGs and Leukotrienes):

-They are produced by various inflammatory cell types, including mast cells, eosinophils, basophiles, and macrophages.

-PGD₂ and PGF₂ α and leukotrienes have the following effects:

1) Inflammation,

2) Increased capillary permeability.

4.Airway smooth muscles:

-Bronchospasm remains an important component of asthma, particularly acute asthma.

-Smooth muscle is innervated by branches of both the parasympathetic (cholinergic) and sympathetic (adrenergic) nervous systems.

-The parasympathetic system is excitatory and vagal stimulation narrows the airways.

-The sympathetic system is inhibitory, and stimulation of this system's β -receptors results in smooth muscle relaxation.

Clinical features:

The spectrum of disease severity:

Some individuals with asthma may be entirely asymptomatic between attacks provoked by specific exposures. Other asthmatics may have nearly continuous symptoms and frequent severe attacks. Asthma may present at any age including the extremes of life.

The Cardinal symptoms of Asthma include:

Episodic wheezing, Dyspnea, Chest tightness, Cough and Sputum production (sputum is small amount, sticky, white or clear usually, colored if infection, rarely profuse and frothy). *Other presentation of*

asthma such as isolated persistent unexplained cough (Cough Variant Asthma), Or Exertional chest tightness Or Dyspnea.

Provoking Factors:

- 1) Allergy: Over 90% of childhood asthmatic and 50% of adult are atopic.

Episodes of wheezing and Dyspnea provoked by inhalation of airborne allergens.

- 2) Infection: Viral infection
- 3) Air pollution
- 4) Smoking
- 5) Psychological factors
- 6) Gastro-esophageal reflux
- 7) Exercise (exercise induced asthma)
- 8) Aspirin induced asthma: characterized by
Asthma + Nasal polyposis + Aspirin Sensitivity.

NB: Atopy: a syndrome caused by a genetic predisposition to develop allergic hypersensitivity reactions to allergens. Patients with atopy develop excessive IgE reactions when exposed to allergens.

Signs:

- The characteristic clinical finding in asthmatic patient is **wheeze**.

Wheezing is caused by high velocity, turbulent airflow through narrowed airways; it is usually present during expiration, although it may be present during inspiration as well. Wheezing is typically Polyphonic, indicating their origin from many separate airways (non-uniformity of airflow). Wheezing may be absent in very mild and very severe airway obstruction.

- Clinical signs of Rhinitis, Sinusitis and Nasal polyps are seen more commonly in patients with asthma.

- Patients with asthma may have entirely Normal physical examination In-between Exacerbations.

Investigations:

A- Pulmonary function tests:

Pulmonary function tests are important for:

- *Confirming the diagnosis.
- *Establishing the severity of the disease.
- *Monitoring the response to therapy.

1) Spirometry: Pattern of airflow limitation are:

Features of obstructive disorders

Diagnostic features: ↓ FEV 1 /FVC ratio < 70%

Other features:

- ↓ FEV 1
- Normal FVC (can be ↓ in severe cases)

The definition of a significant response to bronchodilators is increase in FEV1 and/or FVC by >12% and >200 ml in the post-bronchodilator study, in this case we can diagnose **bronchial asthma**.

If partially reversible or irreversible, in this case we can diagnose **COPD**

2) Lung volumes: Residual volume (RV), Functional residual capacity (FRC), and Total lung capacity may increase in acute asthmatic episodes.

3) Diffusing Lung Capacity: Usually Normal in asthma

B- Arterial Blood Gases (ABG):

1-Typically, Normal ABG in patients **with chronic stable asthma**.

2 -**During an acute episode**

Hypoxemia is present + **Hypocapnia**; (arterial PCO₂ is typically reduced owing to hyperventilation).

- Severe obstruction leads to Arterial PCO₂ rise

C- Radiography:

-Because the chest radiograph is generally unremarkable in patients with uncomplicated asthma, it is used primarily to exclude other causes of respiratory symptoms.

- Over inflation has been reported in asthmatics with severe disease, chronic, persistent, rather than intermittent, and during acute exacerbation.

D- Electrocardiography:

Asthma in remission is usually not associated with ECG abnormalities.

E- Blood tests:

1. Allergic disorders (e.g. allergic asthma) are classically characterized by the presence of eosinophils in blood.

2. Elevated total IgE serum level is frequently seen in allergic patients, but the findings are not specific for asthma

F- Sputum examination:

Sputum eosinophilia is more characteristically found in asthmatics than in patients with COPD.

G- Allergy tests:

-Tests to determine whether the patient is allergic and to investigate the role of specific allergens as a cause of asthma.

-Skin Prick test is performed.

H- Bronchial Challenge testing:

Bronchial hyper responsiveness may be detected by broncho provocation tests with an inhaled pharmacologic bronchoconstrictor such as histamine or methacholine.

Differential Diagnosis:

- 1- COPD: Chronic bronchitis, Emphysema
- 2- Laryngeal dysfunction.
- 3- Congestive feature failure.
- 4- Pulmonary eosinophilia.
- 5- Pulmonary embolism.
- 6- Chronic sinusitis and Gastro esophageal reflux.
- 7- Acute bronchiolitis.
- 8- Localized anatomic lesions obstructing airway, e.g., tumors, FB.

Management of Asthma:

A-Non pharmacologic therapy:

Non pharmacologic management strategies including patient education & avoidance of asthma triggers.

1. Education: the goal of asthma education is to improve patient understanding of the disease and its management and consequently to improve adherence to treatment recommendations
2. Environmental control: Avoidance of aeroallergens, viral respiratory pathogens, air pollution and certain drugs (beta blockers, aspirin) can prevent exacerbation, reduce the need for drug treatment and decrease utilization of emergency facilities
3. Influenza vaccination

4- Allergen immunotherapy also appears to be of benefit in highly selected patients with defined allergic triggers.

B- Pharmacological Management: (Asthma Medications) Drugs available for treatment of asthma are:

a. Drugs currently available to treat asthma are classified, on the base of their principal pharmacodynamic and clinical effect, into: -

1- Quick-relief medications or “relievers”

Short-acting bronchodilators such as: inhaled short acting beta agonists or short- acting anticholinergics (Antimuscarinic drugs) are considered quick-relief medications.

2- Long-term control medications or “controllers”

Corticosteroids, long-acting B2 agonists, leukotriene pathway inhibitors (antileukotriene), mast cell stabilizers e.g. (cromolyn sodium, nedocromil sodium), sustained-release theophylline, and Anti-IgE (omalizumab) are considered long-term control medications, since they are used to achieve and maintain control of symptoms and are usually used daily on a long-term basis.

I- β -adrenergic agents (Agonists):

1. Non selective β -adrenergic agonists stimulate both α & β receptors. i.e., Adrenaline, Nor-Adrenaline, Ephedrine, Isoproterenol (onset <5 min., duration 2-3 hrs.)

2. Selective β_2 -adrenergic agonists:

A-Short acting β_2 -agonists i.e., Albuterol, Terbutaline, Salbutamol

-Method of administration: Oral (tablets, suspension),

Inhalation: MDI, Solution (Nebulizer), Dry powder

- **Inhaled short acting β 2-agonists are the drugs of Choice for relief of symptoms due to acute airway spasm**, onset within 10-15 min. of administration. - The duration of action agents is 4-6 hrs.

B- Long acting and sustained release β 2-agonists:

- Long acting inhaled β 2-agonists (Salmeterol, Formoterol) which have Onset 1-2 hrs. And at least a Duration of 12 hrs. of action
- Oral sustained release β 2-agonists (Albuterol) have one advantage of BID oral administration.

II- Methylxanthines (Theophylline):

*It has the following properties (*Advantages*):

- Weak bronchodilator by inhibiting the Phospho-diesterase enzyme (PDE)
- Anti-inflammatory effect

**Route of administration:*

Oral sustained release tab., syrup.

Parenteral: I.V, I.M.

Rectal: suppositories

III-Anti-cholinergic drugs (Antimuscarinic drugs):

Anti-cholinergic agents induce airway smooth muscle relaxation by blocking muscarinic receptors on airway smooth muscle thus inhibiting vagally mediated cholinergic tone.

***Short-acting anticholinergic agent**, e.g. ipratropium bromide

Method of administration: Inhalation by MDI or solution (Nebulizer)

IV-Corticosteroids:

***Corticosteroids are the most effective agents available for treating persistent asthma.** Steroid efficacy is generally attributed to anti-inflammatory effects.

***Corticosteroids are available for systemic or inhalation use:**

- Inhaled glucocorticoids include: Beclomethasone dipropionate, Budesonide, Fluticasone.

-Systemic glucocorticoids include:

a- Oral preparations e.g. Prednisolone, prednisone, cortisone, betamethasone.

b- Parenteral preparations e.g. Hydrocortisone, methyl prednisolone, betamethasone, dexamethasone

Side effects Corticosteroids include:

a- Systemic e.g. Adrenal suppression, Osteoporosis, Hypertension, DM, Myopathy, Growth suppression, and Psychiatric reaction

b- Local e.g. Oral Candidiasis and Dysphonia

V- Mast cell stabilizers: -

Cromolyn sodium and nedocromil sodium are classified as controller agents

VI- Leukotriene modifiers (antileukotriene/ leukotriene receptor antagonist):

Leukotriene synthesis inhibitor (e.g. Zileuton) and **leukotriene receptor antagonist** (e.g. Zafirlukast, montelukast,) are oral medications.

Treatment regimens based on classification of asthma severity are:

1- Mild intermittent asthma:

- ***Clinical features:** Cough, wheezing, dyspnea,
Daytime symptoms ≤ 2 times/week
Nocturnal symptoms ≤ 2 times/ month
- ***Peak flow or FEV1** ≥ 80 % predicted (when asymptomatic)
- ***Therapy:** Short acting β -agonist, as-needed basis (1-2 puffs by MDI) (**reliever**)

2- Mild persistent asthma:

- ***Clinical features:**
Daytime symptoms > 2 times/ wk but not daily
Nocturnal symptoms > 2 times/ mo
- ***Peak flow or FEV1** ≥ 80 % predicted.
- ***Therapy:**
Reliever: Short acting β_2 -agonist, as needed, low dose.
Controllers: Long acting β -agonist. OR Cromolyn sodium. OR Inhaled glucocorticoids. OR Leukotriene modifiers
Preferable Controllers: inhaled corticosteroids (low dose) plus Long acting β -agonist

3. Moderate persistent asthma:

- ***Clinical features:**
Daytime symptoms $\rightarrow$ Daily.
Nocturnal symptoms > 1 times/wk.
- ***Peak flow or FEV1** > 60 % to < 80 % predicted.
- ***Therapy:**
Reliever: Short acting β -agonist $\rightarrow$ as needed, medium to high dose.
Controllers: Long acting β -agonist. **PLUS**
Inhaled glucocorticoids. OR Sustained release Theophylline.

Preferable Controllers: inhaled corticosteroids (medium dose) plus long acting β 2-agonist

4. Severe persistent asthma

***Clinical features:**

Daytime symptoms, Continuous

Nocturnal symptoms, Frequent

***Peak flow or FEV1** $\leq$ 60 % predicted.

***Therapy**

Reliever: Short acting β -agonist, as needed, high dose.

Controllers: Long acting β -agonist.

Inhaled glucocorticoids.

Sustained release Theophylline or leukotriene modifier

Preferable Controllers: inhaled corticosteroids (high dose) plus long acting β -2 agonist

Patients not controlled with inhaled glucocorticoids and one or more long-acting bronchodilator are often considered candidates for systemic glucocorticoid therapy 0.5 mg/kg/day for 8-21 days, followed by a tapering course to the lowest dose that maintain control.

Acute Severe Asthma

Definition

Acute severe attacks represent progression of an attack of bronchospasm to the point where the patient is breathless at rest and has signs of cardiac stress.

Symptoms:

- a. Increasing breathlessness.
- b. Difficulty in talking or inability to do anything other than breathing.
- c. Anxiety
- d. The patient sits up in bed and chair.
- e. Wheezes are audible without stethoscope and the chest is visibly over inflated.

Treatment: - (The patient should be hospitalized)

1. Oxygen therapy

Oxygen is given and titrated to achieve values above 92%

2. Short acting β 2-agonists

Inhaled (Salbutamol, Albuterol) by Nebulizer.

3. Corticosteroids

Prednisone, Methyl prednisolone.

High dose I.V. (Methyl prednisolone 60-120 mg every 6 h.

4. Anti-Cholinergic e.g. Ipratropium bromide

5. Theophylline

Loading dose (I.V. 5 mg /kg over 30 min).

Maintenance dose (I.V. 0.5 mg /kg /hr).

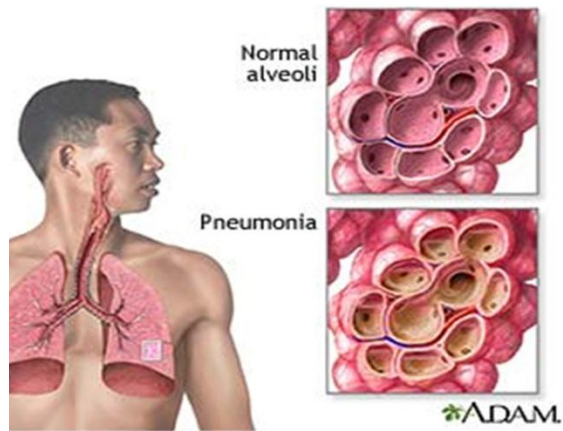
6. Antibiotics, when needed.

7. Mechanical ventilation; in severe un-responding or deteriorating cases.

Pneumonia

Lecture Outline

- Definition
- Classifications
- CAP / HAP
- Risk Factors & pathophysiology
- Pneumococcal pneumonia
- Atypical pneumonia
- Staphylococcus & Gram negative pneumonia
- Viral pneumonia



Definition:

- Pneumonia is an acute respiratory illness associated with recently developed radiological pulmonary shadowing.
- It is an inflammatory process of the lung parenchyma that is commonly caused by infectious agents.

Classification of pneumonia

I)According to etiology:

Infective (Microbiological):

- Bacterial (the most common cause of pneumonia) e.g. streptococcus
- Viral pneumonia
- Fungal pneumonia
- Parasitic / T.B

Non infective

- Physical: radiation pneumonitis
- Chemical pneumonia (ingestion of kerosene or inhalation of irritating substance), lipid Inhalation pneumonia (aspiration pneumonia)
- Allergic: Loffler's syndrome.

II) Anatomical classification:

- **Lobar pneumonia**; if one or more lobe is involved. It is radiological and pathological term.
- **Bronco-pneumonia**; the pneumonic process has originated in one or more bronchi and extends to the surrounding lung tissue.
- **Segmental or sub-segmental**

III) Clinical classification:

- Community Acquired Pneumonia (CAP)
- Nosocomial/Hospital Acquired Pneumonia.
- Pneumonia in immuno-compromised host.

I- Community Acquired Pneumonia (CAP)

Definition: CAP is pneumonia in person not hospitalized or residing in a long-term care facility for ≥ 14 days.

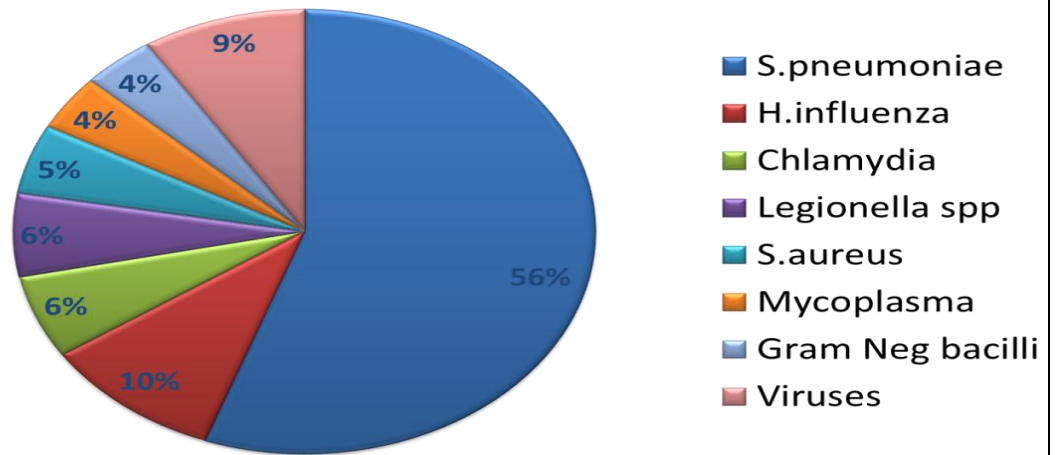
Predisposing factors:

- Immuno-suppressed patients
- Cigarette smoking
- Difficult swallowing (due to stroke, dementia, parkinsons disease, or other neurological conditions)
- Impaired consciousness (loss of brain function due to dementia, stroke, or other neurological conditions).
- Chronic lung disease (COPD, bronchiactasis)
- Old age
- Other serious illness such as heart disease, liver cirrhosis, renal disease and DM
- Recent cold, laryngitis or flu
- Splenectomy, functional asplenia or hyposplenia.

CAP; The Two Types of Presentations:

- Classical vs Atypical

The pathogen involved: commonest is **Streptococcus pneumoniae** followed by **Haemophilus influenza** then viruses and atypical bacteria.



Lobar (pneumococcal) pneumonia

- **Etiology** : **Streptococcus pneumoniae** (pneumococci): G+ve diplococci
- The most common cause of CAP.
- **Typical presenting symptoms** are. malaise, shaking chills, fever, rusty sputum, pleuritic chest pain, cough)
- Lobar infiltrate on **chest X- ray** (CXR).
- May be Immuno- suppressed host
- 25% of patients will have bacteremia – serious **complications**

Pathogenesis:

- Routes of infection

- Inhalation
- Aspiration
- Hematogenous

- Pathological stages

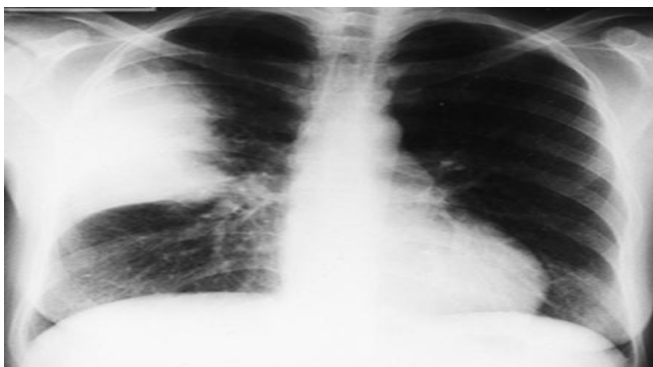
- Stage of congestion
- Stage of red hepatization
- Stage of grey hepatization
- Stage of resolution

These stages take about **7-10 days**.

Clinical manifestations:

- Acute onset usually preceded by mild coryza or URT symptom
- Shaking chills
- Rapidly rising fever (39.5 to 40.5 degree)
- Tachypnea, nasal flaring
- Flushed cheeks
- Loss of appetite, low energy, and fatigue
- Stabbing chest pain aggravated by respiration and coughing due to pleurisy.
- Patient is very ill and lies on the affected side to decrease pain
- Cough with purulent, blood tinged or rusty sputum
- Shortness of breath
- Recovery by crisis or lysis.
- **General signs:**
 - Patient looks ill, sweaty and flushed.
 - Fever, Tachycardia and Tachypnia.
 - Herpes Labialis
 - Central cyanosis if developed respiratory failure.
- **Local signs:**
 - Signs of consolidation taking the topography of a lobe
 - Diminished respiratory movements
 - Increased TVF (Bronchophony)
 - Impaired note on percussion
 - Tubular breathing, crackles, pleural rub.
 - Use of accessory muscles of respiration e.g. abdominal and intercostal muscles

Diagnostic tests:



Chest x ray:

- Usually needed to establish diagnosis.
- It is a prognostic indicator.
- To rule out other disorders.
- May help in etiological diagnosis.
- Typically shows opacity with lobar distribution.

Laboratory Tests:

- **Oxygen saturation & ABG** to detect oxygenation status and acid base disturbance
- **CBC** with Differential count; typically shows leukocytosis with neutrophilia.
- **BUN and serum creatinine**

Microbiological:

- **Gram stain of sputum**
- **Culture and sensitivity of sputum sample.**
- **Pre -treatment blood cultures** in selected cases.
- **Culture of pleural fluid** if any.
- **Serology and PCR**
- **The pneumococcal urinary antigen test** may be used to supplement blood and sputum cultures in adults. This assay is an immunochromatographic membrane test to detect the pneumococcal cell-wall polysaccharide. The test has sensitivity of 50%-80% (70%-90% if bacteremia) and a specificity of about 90% in adults (lower in children).

N.B. According to latest ATS guideline:

- We recommend not obtaining sputum Gram stain and culture nor blood cultures routinely in adults with CAP managed in the outpatient setting but recommended in severe disease as well as in all in-patients empirically treated for MRSA or P. aeruginosa.
- If the CAP is not severe and no other risk factors testing urine for pneumococcal antigen or legionella antigen is not recommended.

Validated definition of severe CAP includes either one major criterion or three or more minor criteria

I) Minor criteria:

- ❖ Respiratory rate >30 breaths/min
- ❖ PaO₂/FIO₂ ratio <250
- ❖ Multilobar infiltrates

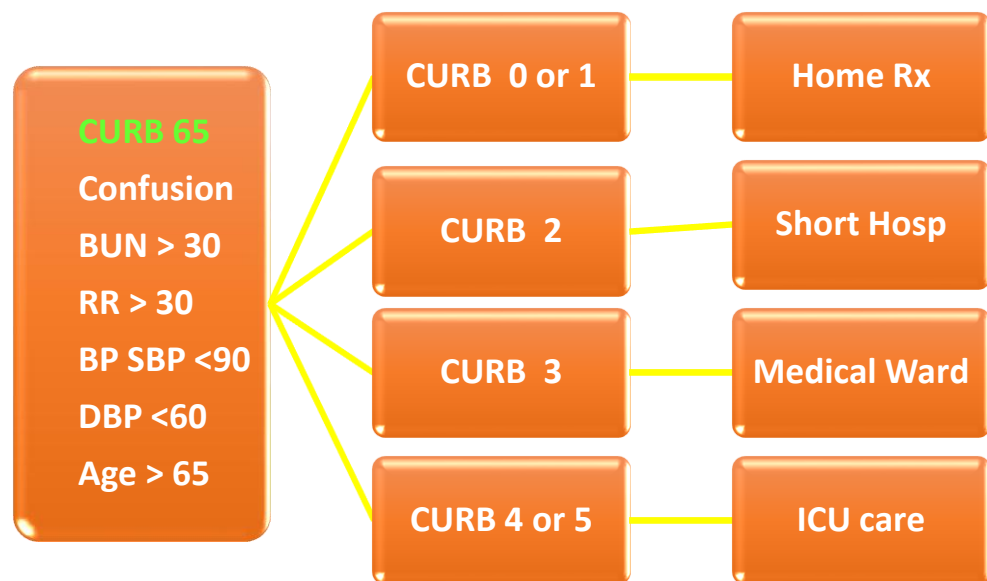
- ❖ Confusion/disorientation
- ❖ Uremia (blood urea nitrogen level >20 mg/dl)
- ❖ Leukopenia* (white blood cell count, 4,000 cells/ml)
- ❖ Thrombocytopenia (platelet count, 100,000/ml)
- ❖ Hypothermia (core temperature less than 36 C)
- ❖ Hypotension requiring aggressive fluid resuscitation

II) Major criteria:

- ❖ Septic shock with need for vasopressors
- ❖ Respiratory failure requiring mechanical ventilation.

CURB-65 severity of illness score:

- **Confusion:** new mental confusion
- **Urea:** new raised > 7 mmol/L
- **Respiratory rate:** >30/min
- **Blood pressure:** low blood pressure (*systolic blood pressure* < 90 mm Hg and/or *diastolic blood pressure* <60 mm Hg)
- **Age:** ≥ 65 year



Complications:

Pulmonary :

- Delayed or failure of resolution
- Lung abscess
- Pleural; pleurisy, serous effusion or empyema.

Systemic:

- Haematogenous spread; pericarditis, endocarditis, toxic myocarditis & heart failure, peritonitis and arthritis.
- Bactremia in COPD, DM and women
- Disseminated intravascular coagulation (DIC), Syndrome of inappropriate antidiuretic hormone secretion (SIADH) and septic shock.

Differential Diagnosis:

- Fever with chest symptom.
- Bacteriological diagnosis.
- Consolidation Vs collapse, fibrosis and pleural effusion.

Treatment:

1-General: Bed rest and good nutrition and hydration.

2-Antibiotic:

- About 34% of pneumococcal isolates are penicillin-resistant.
- The mechanism of resistance: altered penicillin-binding protein.
- Resistant to amoxicillin-clavulanate.
- **Antibiotic of choice:-**
 - Advanced Macrolide e.g azithromycin or clarithromycin.
 - Amoxicillin-clavulanate.
 - Respiratory quinolone (moxifloxacin, levofloxacin).

Table 3. Initial Treatment Strategies for Outpatients with Community-acquired Pneumonia

	Standard Regimen
No comorbidities or risk factors for MRSA or <i>Pseudomonas aeruginosa</i> *	Amoxicillin or doxycycline or macrolide (if local pneumococcal resistance is <25%) [†]
With comorbidities [‡]	Combination therapy with amoxicillin/clavulanate or cephalosporin AND macrolide or doxycycline [§] OR monotherapy with respiratory fluoroquinolone

3-Symptomatic treatment.

4-Treatment of complications.

Prevention:

- **Smoking cessation**
- **Vaccination recommendations**
 - Influenza: annual viral influenza vaccine.
 - Pneumococcal vaccine

Indications of vaccination:

- 1- All patients with chronic illness
- 2- Immunocompetent ≥ 65 ys
- 3- Immunocompromised ≤ 64 y
- 4- All patients in long term care facilities
- 5- Revaccinate after 5-7 years

Causes of delayed or unresolved pneumonia

- 1- Inadequate treatment
- 2- Post obstructive pneumonia
- 3- Underlying lung pathology
- 4- Immuno-compromised patient

Atypical bacterial pneumonia

Etiology:

- Mycoplasma pneumoniae, Legionella, Chlamydia.
- Small free living lacking cell wall

Clinical picture:

- Younger populations (5-15 ys):
- Mild, upper respiratory symptoms; coryza and tracheobronchitis.
- Older populations:
- More serious. Common cause of CAP.
- 2-3 weeks incubation period.
- Gradual onset of symptoms over 2-4 days
- **Extrapulmonary symptoms**
 - **Legionella:** CNS, heart, liver, Gastrointestinal and Genitourinary
 - **Mycoplasma pneumonia:** upper respiratory tract (RT), Gastrointestinal, skin

X ray: Patchy, widespread or confluent shadows.

Blood picture: normal leucocytic count or mild increase.

Serology: IgM cold heamagglutinins against I on RBCs with titre $\geq 1/32$ or rising

Treatment:

- Macrolide antibiotics; Erythromycin, clarithromycin.
- Doxycycline and tetracycline.

Complication:

- **Autoimmune hemolysis.**
- **Extrapulmonary complication;** meningitis , transverse myelitis, pericarditis and fulminant renal failure 2ry to intravascular hemolysis.



Staphylococcal pneumonia

Etiology: Staphylococcus aureus; coagulase and other toxins producing.

- Less common but more serious cause of CAP.
- Important cause of nosocomial pneumonia.

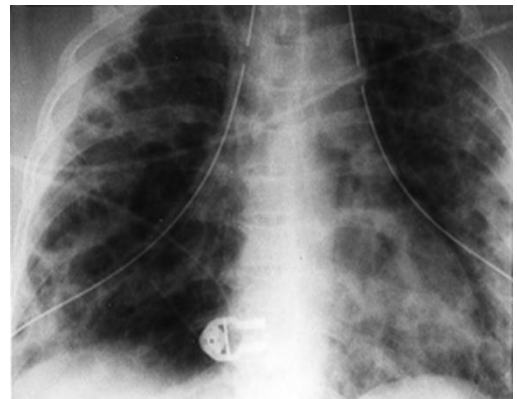
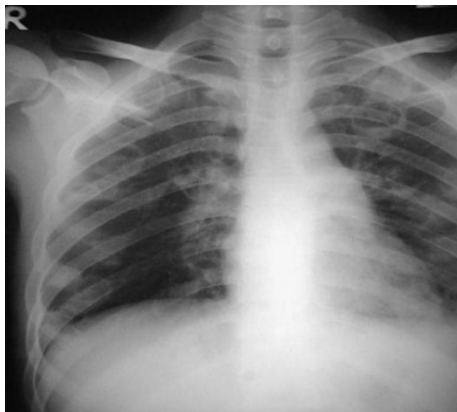
Pathogenic route:

- Aspiration is the usual pathogenic route.
- Haematogenous spread (usually in setting of endocarditis or infected IV line)

Clinical picture:

- Symptoms & signs of underlying endocarditis or infected IV line.
- Fever, dyspnea, cough and purulent sputum.

Investigations: **X-ray:** segmental or central consolidation, Pl effusion, empyema, cavities and abscesses.



Treatment Staph aureus pneumonia:

- **Methicillin sensitive staph aureus (MSSA);**
 - Methicillin.
 - Amoxycillin clavulinate, Ampicillin sulbactam.
 - 1st generation cephalosporins.
 - 2nd generation cephalosporins (Cefuroxime)
- **Methicillin resistant staph aureus (MRSA);**
 - Glycopeptides: (vancomycin, teicoplanin) is used for MRSA
 - Oxazolidinone : Linezolid is the first antibacterial drug in a new class of synthetic antibiotics called oxazolidinones

II- Hospital acquired pneumonia (HAP) (Nosocomial pneumonia)

Definition:

Hospital-acquired pneumonia (HAP) Occurs 48 hours or more after admission, this was not incubated at the time of admission. It includes ICU and ventilator associated pneumonia.

Gram negative Pneumonia

Etiology: Klebsiella Pneumonia., Pseudomonas aerogenosa, E coli.

- Important cause of nosocomial pneumonia

Route of infection: Aspiration or hematogenous spread from GIT or Genitourinary are the common route of infection.

Clinical Picture:

- Productive cough, pleuritic chest pain.
- Fever, rigors, Prostration and hypotension.
- Blood tinged sputum, Current jelly sputum in case of Klebsiella P.

X- ray:

- Consolidation, Pl effusion.
- Bowed fissure sign (important radiological sign for Klebsiella Pneumonia)

Treatment:

- Hospitalization.
- Antibiotic according to culture sensitivity
- Combination of antipseudomonal drugs in case of moderate and severe pseudomonas pneumonia as follow:-
- One of group A plus one of group B

Group A

1- Antipseudomonal cephalosporins:

- Cefepime 1–2 g every 12 h
- Ceftazidime 2 g every 8 h

2- Or Carbapenems: Meropenem 1g every 8 h

3- Or β -Lactam/ β -lactamase inhibitor: Piperacillin–tazobactam 4.5 g every 6h.

Group B

- 1- Aminoglycosides: Amikacin 20 mg/kg per d
- 2- Or Antipseudomonal quinolones :
 - Levofloxacin 750 mg every day
 - Ciprofloxacin 400 mg every 8 h

Viral pneumonia

- Viruses recently recognized as important pathogens in CAP due to improved diagnostic tests (PCR)
- Cause of 2 - 35% of CAP in adults (more in kids)
- Recent emergence of new viral respiratory pathogens
- Risk factors for viral pneumonia in adults:
 - Elderly: Higher rates of hospitalization and death from viral PNA in persons >60 years old.
 - COPD and asthma: frequently complicated by respiratory viral infections
 - Immuno-compromised patients at increased risk

Clinical syndromes:

- 1- Upper respiratory tract (cold, pharyngitis, bronchitis)
- 2- Bronchiolitis: acute inflammatory disorder of small airways
 - Obstruction with air trapping, hyperinflation, wheezing.
 - Most common < 2 years old
 - RSV most common, also influenza A and B viruses, parainfluenza viruses, rhinovirus, adenoviruses, measles virus, and human metapneumovirus
- 3- Pneumonia
 - Similar presentation to bacterial Pneumonia

Clinical picture: Flu like symptoms, skin rash, dry cough, dyspnea

X ray: diffuse infiltrates

Diagnosis:

- Nasal swab specimens, nasal aspirates, or combined nose and throat swab specimens
- Sputum, endotracheal aspirate samples, or BAL
- Rapid antigen detection, and PCR methods

Panel: Viruses linked to community-acquired pneumonia in children and adults

- Respiratory syncytial virus
- Rhinovirus
- Influenza A, B, and C viruses
- Human metapneumovirus
- Parainfluenza viruses types 1, 2, 3, and 4
- Human bocavirus*
- Coronavirus types 229E, OC43, NL63, HKU1, SARS
- Adenovirus
- Enteroviruses
- Varicella-zoster virus
- Hantavirus
- Parechoviruses
- Epstein-Barr virus
- Human herpesvirus 6 and 7
- Herpes simplex virus
- Mimivirus
- Cytomegalovirus†
- Measles†

*Mostly in children. †Mostly in developing countries.



Influenza

- Annual winter epidemics 6-8 weeks (year round in tropics)
- **Transmitted by** small particle aerosols
- 2-3 day **incubation period**
- Maximum **virus shedding** is at onset of illness, continues for 5 to 7 days
- **High risk populations:** infants, young kids, healthy adults 20-40s, pregnant/postpartum women, immunocompromised, obesity, DM, COPD, asthma
- Elderly less susceptible to H1N1 due to prior exposure
- **Secondary bacterial pneumonia** can occur mostly caused by pneumococci and staphylococci.
 - Most common in elderly, or underlying pulmonary or cardiac disease
 - Period of improvement followed by increased cough, sputum production, and consolidation
 - Most common Strep pneumonia, then Staph. aureus and Group A Streptococcus.

Prevention:

Vaccines:

- ✓ **Inactivated virus vaccines:** inactivated purified virions or partially purified HA and NA preparations
 - Efficacy 70% to 90% in healthy adults/children if good antigenic match
- ✓ **Live, attenuated vaccine:**
 - More effective in children
 - In adults equal or less effective than inactivated vaccine
 - Contraindicated in pregnant or immunosuppressed

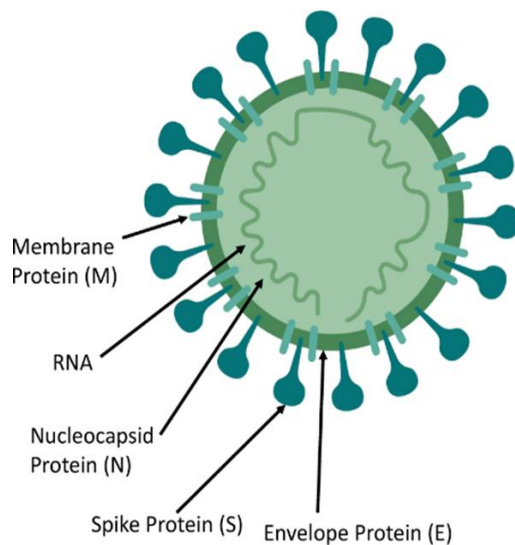
Treatment:

- Antivirals: Neuraminidase inhibitors (NIs) oseltamivir (Tamiflu) and zanamivir
- both influenza A and B

Respiratory syncytial virus (RSV)

- 2nd most common cause of viral PNA in older adults
- Common in winter (November – April, peak Jan-Feb)
- Major cause of serious lower respiratory tract infections in young children
- Primary RSV infection is nearly universal by age 2 and repeat infections are common due to incomplete immunity.
- Also important pathogen in adults, especially in elderly, chronic lung disease, or immune-compromised
- Approximately 10,000 deaths in persons > age 65 in the United States each year from RSV (2nd to influenza)

Coronavirus



Structural Protein	Function of Protein
Nucleocapsid Protein (N)	Bound to RNA genome to make up nucleocapsid
Spike Protein (S)	Critical for binding of host cell receptors to facilitate entry of host cell
Envelope Protein (E)	Interacts with M to form viral envelope
Membrane Protein (M)	- Central organiser of CoV assembly - Determines shape of viral envelope

- It has been noted that some CoVs do not need to have the full ensemble of structural proteins to make virions, highlighting that certain proteins may be dispensable or compensated by the function of non-structural proteins.

Coronaviruses are a large family of viruses that are known to cause illness ranging from the common cold to more severe diseases such as Middle East Respiratory Syndrome (MERS) and Severe Acute Respiratory Syndrome (SARS).

- 2019-novel coronavirus (COVID-19) was identified in Wuhan, China in late 2019. This is a new coronavirus that has not been previously identified in humans.
- The virus that causes COVID-19 is known as SARS-CoV-2.
- The outbreak has since spread across China to other countries around the world. By the end of January, the new coronavirus had been declared a public health emergency of international concern by the WHO.
- The most commonly reported symptoms include a fever, dry cough and tiredness, and in mild cases people may get just a runny nose, anosmia or a sore throat.
- In the most severe cases, people with the virus can develop difficulty breathing, and may ultimately experience organ failure. Some cases are fatal.
- The virus causes Acute Respiratory Distress Syndrome (ARDS).

Bronchiectasis

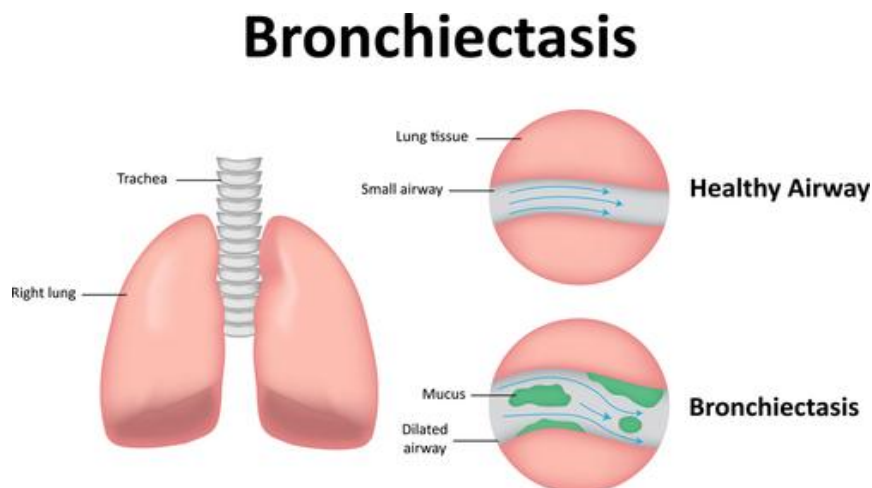
Definition:

Bronchiectasis is an abnormal and permanent dilatation of the medium sized bronchi.

Most cases of bronchiectasis are acquired but may result from an underlying genetic or congenital defect of airway defenses.

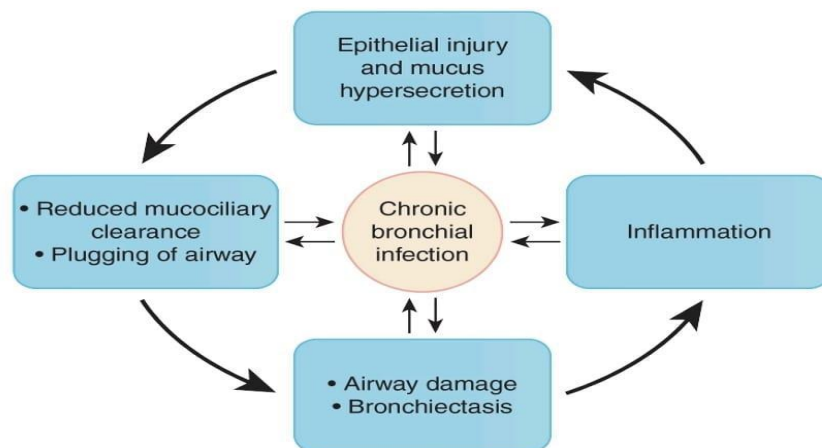
Morphological types:

- Cylindrical or tubular bronchiectasis
- Varicose
- saccular or cystic bronchiectasis *is considered the most severe form of bronchiectasis*



Etiology and pathogenesis of Bronchiectasis:

- Bronchiectasis is usually caused by chronic inflammation and infection in airways.
- Localized bronchiectasis may be due to bronchial distension resulting from the accumulation of pus beyond an obstructing bronchial lesion, such as enlarged tuberculous hilar lymph nodes, a bronchial tumor or an inhaled foreign body (e.g., an aspirated peanut).



Causes of Bronchiectasis

- Congenital
- Cystic fibrosis.
- Ciliary dysfunction syndromes:
 - Primary ciliary dyskinesia (immotile cilia syndrome)
 - Kartagener's syndrome (sinusitis and transposition of the viscera)
- Primary hypogammaglobulinemia
- Acquired children
 - Pneumonia (complicating whooping cough or measles)
 - Primary TB.
 - Inhaled foreign body.
- Acquired adults
 - Suppurative pneumonia.
 - Pulmonary TB.
 - Bronchial tumors
 - Allergic bronchopulmonary aspergillosis complicating asthma

Pathology

- The bronchiectasis cavities may be lined by granulation tissue, squamous epithelium, or normal ciliated epithelium.
- There may also be inflammatory changes in the deeper layers of the bronchial wall and hypertrophy of the bronchial arteries.
- Chronic inflammatory and fibrotic changes are usually found in the surrounding lung tissue.

Site of bronchiectasis:

Bronchiectasis may involve any part of the lungs, while bronchiectasis is predominant on the left side and in the lower lobes

the more efficient drainage by gravity of the upper lobes usually produces less serious symptoms & complications than when bronchiectasis involves the lower lobes.

Clinical features of bronchiectasis

Due to accumulation of pus in dilated bronchi

- Chronic productive cough usually worse in mornings and often brought on by changes of posture *is the most common presentation and is called Wet bronchiectasis."*
- Sputum often copious and persistently purulent *in advanced disease and during exacerbation.*
- Halitosis is a common accompanying feature
- Shortness of breathing/dyspnea

Due to inflammatory changes in lung and pleura surrounding dilated bronchi

- Fever, malaise and increased cough and sputum volume when spread of infection causes pneumonia, which may be associated with pleurisy.
- Recurrent pleurisy in the same site often occurs in bronchiectasis
- Hemoptysis: can be slight or massive and is often recurrent. Usually associated with purulent sputum or an increase in sputum purulence. Can, however, be *the only symptom in so-called 'dry bronchiectasis.'*
- Bad general health

When disease is extensive and sputum persistently purulent a decline in general health occurs with weight loss, anorexia, lassitude, low-grade fever, and failure to thrive in children. **In these patients digital clubbing is common.**

Physical signs in the chest may be unilateral or bilateral.

- If the bronchiectatic airways do not contain secretions and there is no associated lobar collapse, there are no abnormal physical signs.
- **Finger (digital) clubbing** has long been recognized as a sign of chronic suppurative lung disease
- **Coarse crackles** is the characteristic clinical finding in bronchiectasis and heard over the affected areas.
- **Signs of collapse and fibrosis** may be present in advanced cases (breath sounds are diminished if the airway is obstructed).
- Advanced disease may lead to scarring with associated overlying **bronchial breathing**.
- **Cyanosis** with occurrence of respiratory failure

When to suspect bronchiectasis?

- Chronic cough, sputum
- hemoptysis
- Coarse rales
- Persistent respiratory symptoms
- Recurrent pneumonia
- Progressive obstructive lung disease
- Clubbing

Investigations

a. Bacteriological and mycological examination of sputum

The most common organisms isolated from the respiratory tract are hemophiles influenza, and pseudomonas aeruginosa, In addition to common respiratory pathogens, such as Aspergillus and various Mycobacteria.

Frequent cultures are necessary to ensure appropriate treatment of resistant organisms.

b. Radiological examination

- 1- **Plain chest- Xray**: Bronchiectasis, unless very gross, is not usually apparent on a chest X-ray. In advanced disease, thickened airway walls, cystic bronchiectatic spaces, and associated areas of pneumonic consolidation or collapse may be visible.
- 2- **High resolution computed tomography (HRCT)** scan of chest is the most sensitive for diagnosis of bronchiectasis, and shows thickened dilated airways

c. Assessment of ciliary function

- A screening test can be performed in patients suspected of having a ciliary dysfunction syndrome by assessing the time taken for a small pellet of saccharin placed in the anterior chamber of the nose to reach the pharynx, when the patient can taste it. This time should not exceed 20 minutes and is greatly prolonged in patients with ciliary dysfunction.
- Ciliary beat frequency may also be assessed using biopsies taken from the nose.
- Structural abnormalities of cilia can be detected by electron microscopy.

Management:

a. Postural drainage

- Physiotherapy: Patients should be instructed how to perform regular daily physiotherapy to keep the dilated bronchi empty of secretions. Efficiently performed, **this is of great value both in reducing the amount of cough and sputum and in preventing recurrent episodes of bronchopulmonary infection.**
- Patients should adopt a position in which the lobe to be drained is uppermost. Deep breathing followed by forced expiratory man oeuvres (the 'active cycle of breathing' technique) is of help in allowing secretions in the dilated bronchi to gravitate towards the trachea, from which they

can be cleared by vigorous coughing. 'Percussion' of the chest wall with cupped hands may help to dislodge sputum, and several mechanical devices are available which cause the chest wall to oscillate, thus achieving the same effect.

- The optimum duration and frequency of physiotherapy depends on the amount of sputum, but 5-10 minutes once or twice daily is a minimum for most patients.
- In patients with airflow obstruction, inhaled bronchodilators and corticosteroids should be used to enhance airway patency

b. Antibiotic therapy

- Antibiotics which can be used in patients with bronchiectasis are: amoxycillin, clarithromycin, Co-amoxiclav).
- Some present difficult therapeutic problems because of secondary infection with bacteria such as staphylococci and Gram-negative bacilli, in particular *Pseudomonas* species. In these circumstances antibiotic therapy should be guided by the microbiological results but frequently requires the use of oral Quinolones as (ciprofloxacin 250-750 mg 12-hourly) or 3 rd. generation cephalosporin as (ceftazidime by intravenous injection or infusion 1-2 g 8-hourly).

c. Surgical treatment

Indication of surgery:

- the diagnosis has been confirmed radiographically
- the disease is sufficiently localized to one lung or to part of one lung to enable curative resection
- the residual lung is judged to be sufficiently healthy in terms of measured lung function

- the patient is significantly disabled by persistent sputum production, recurrent infective exacerbations or hemoptysis despite all reasonable medical measures having been taken over a period of at least 12 months
- there is no persisting systemic causal factor likely to result in the development of new disease in previously unaffected lung.

Prognosis:

- The disease is progressive when associated with ciliary dysfunction and cystic fibrosis, and eventually causes respiratory failure.
- In other patients the prognosis can be relatively good if postural drainage is performed regularly, and antibiotics are used judiciously.

Prevention:

- As bronchiectasis commonly starts in childhood following measles, whooping cough or a primary tuberculous infection, it is essential that these conditions receive adequate prophylaxis and treatment.
- The early recognition and treatment of bronchial obstruction are also particularly important.

Complications:

- 1- Infective exacerbation
- 2- Massive hemoptysis is reported frequently, especially during exacerbations.
- 3- Suppurative pneumonia, lung abscess are uncommon.
- 4-Pleurisy, empyema
- 5- Sinusitis.
- 6-Cor- pulmonale
- 7- Brain abscess.
- 8- Amyloidosis.

Lung Abscess

Definition:

Is a localized area of destruction of lung parenchyma in which infection by pyogenic organisms results in tissue necrosis and suppuration.

Lung abscesses may be single or multiple and they frequently contain air–fluid levels.

Etiology of Lung Abscess:

- **Aspiration:** seizure, coma, surgery, DM, sedatives, alcohol, neurologic diseases
- **Bronchial obstruction:** malignancy, F.B.
- **Septic emboli:** SBE, catheters, prostheses, pelvic thrombophlebitis
- **Direct Spread:** subphrenic, hepatic
- **Pneumonia complication:** *S. aureus*, *Klebsiella*, *pseudomonas*, etc.

Classifications:

Duration: Acute (< 4-6 week) and Chronic

Causes: Primary and Secondary

Symptoms of Lung Abscess:

- Cough and purulent sputum **are predominant symptoms**
- A putrid smell is indicative of anaerobic infection.
- Fever and chills
- Chest pain may be pleuritic
- Hemoptysis
- Dyspnea
- Anorexia
- Night sweat

Physical signs:

- Signs There are no signs specific for lung abscess.
- Digital clubbing may develop within a few weeks if treatment is inadequate.
- Dullness to percussion and diminished breath sounds may be present if the abscess is large and situated near the surface of the lung.
- There may be evidence of associated consolidation, and a pleural

friction rub is sometimes heard.

- The ‘amphoric’ or ‘cavernous’ breath sounds traditionally associated with large lung cavities.

Causes of lung abscess

Aspiration Abscess

- Most common cause
- Tonsillectomy, seizure, neurosurgery, alcoholism, etc.
- Organism identification in only 30-40%
- Mostly Anaerobic, mixed organisms
- “Putrid sputum”
- Dependent portions: Lower lobe, posterior & lateral basal segment. Upper lobe, posterior segment.
- Usually, single abscess cavity

Necrotizing Pneumonia

- Community: Staph. Aureus or Klebsiella
- Hospital: Pseudomonas or Proteus
- Aspiration pneumonias cause necrotizing infections
- Klebsiella predominant in alcoholics or DM

Secondary to Malignancy

- Bronchogenic cancer: Squamous Ca
- Lymphoma
- Leukemia
- Multiple Myeloma
- Metastatic Malignancies

Diagnosis:

- X-ray: Cavity with “air-fluid level”
- CBC: leukocytosis, Anemia, etc.
- Cultures: Sputum & Blood
- Anaerobic culture is important
- Chest CT
- Sputum cytology
- Sputum AFB
- Bronchoscopy to Rule out malignancy

Treatment:

- Medical treatment is the mainstay
- Penicillin, Cephalosporin

- Clindamycin, chloramphenicol, Metronidazole to cover for the Anaerobes
- Postural drainage
- Bronchoscope drainage

Complications:

1. Spread of infection: Direct, lymphatic & blood.
2. Rupture leads to hemoptysis and bronchopleural fistula.
3. Lung gangrene (due to putrefaction by saprophytic bacteria).
4. Lung fibrosis leads to pulmonary hypertension.
5. 2ry Amyloidosis: in chronic lung abscess.

Indications for Surgery

- Massive hemoptysis
- Refractory to Medical treatment
- Large cavity with thick walls
- Complicated by malignancy
- Empyema develops
- Chronicity, Recurrence
- Remaining residual cavity

Prognosis

- Relatively Favorable
- Underlying Disease is important
- Operation Rate: 15%
- Overall mortality rate: 10%



Tuberculosis

Definition:

Tuberculosis is an infectious disease caused by the pathogenic organism, mycobacterium tuberculosis complex: M.tuberculosis, M.Bovis, M.africanum.

Today, T.B remains the leading cause of death from infectious disease in the world.

Pathogenesis:

Tubercle bacillus:

Koch first described tubercle bacillus now known as mycobacterium tuberculosis in 1882,

Mycobacteria are now known to comprise a large group of **Acid fast, alcohol fast, aerobic or microaerophilic, nonspore-forming, non-motile bacilli. Of the many different mycobacteria, only M.tuberculosis, M.bovis, M.africanum are recognized as tubercle bacilli.**

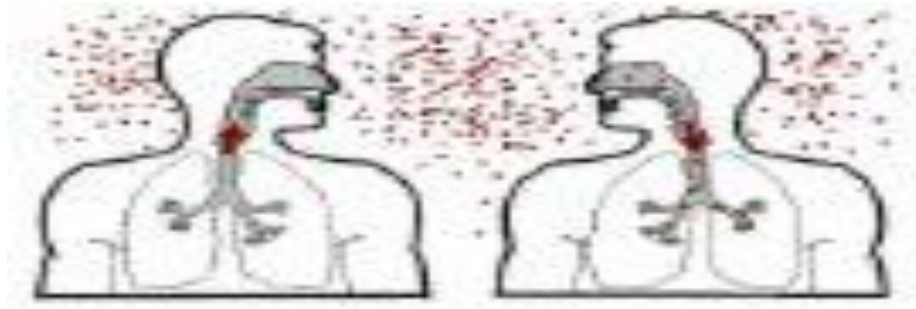
Tubercle bacilli are difficult to **stain** but once stained they strongly retain dye, which is not removed by acid-alcohol solution. this acid, alcohol fastness can be demonstrated by the **Ziehl Nielsen staining** procedure.

Tubercle bacilli can also be **stained by fluorescent dyes** such as auramine and rhodamine → bacilli fluoresce strongly.

-In **culture** tubercle bacilli grow slowly taking **2-6 weeks** to form colonies on **Lowenstein-Jensen medium**. Growth is optimal at temperature of 33-39 at PH of 6.5-6.8 in an atmosphere of 5-10% CO₂.

Transmission:

1-Inhalation is the commonest method. The **human bacilli** are inhaled in coughed droplets or dusts contaminated with sputum from a case of open pulmonary tuberculosis (infectious case). **The inhaled bacilli infect the tonsils or the lungs.**



2-Ingestion of raw milk contaminated with **bovine or human bacilli** infect the tonsils or the intestine.

3-Skin inoculation by handling infected materials as infected meat or dressing of tuberculous lesions.

Tuberculin skin testing (TST):

Tuberculin skin testing is **simple method used to screen for TB infection in high-risk population and in suspected TB cases.**

Technique of TST:

Mantoux test:

Is performed by **intra dermal injection of 0.1ml of PPD** (purified protein derivative) **containing 5TU** (tuberculin units) into the palmer surface of forearm using 27-gauge needle with a plastic or glass syringe. The injection should make just a wheal of 6-10mm in diameter.

The test is read at 48-72hour. The reaction consists of erythema and induration. The erythema should be disregarded, and **the diameter of Induration measured in millimeters.**

Interpretation of TST reaction:

Reactions in the following categories of patients should be considered **significant (positive TST) and indicative of tuberculous infection:**

Induration $\geq 5\text{mm}$: in

- HIV infected persons,
- persons who have had close recent contact with an infectious case of TB

Induration $\geq 10\text{mm}$: in persons with risk factors for TB other than the above including the following:

- IV drug users (IV drug addiction)

-Persons with medical conditions predisposed to TB e.g., silicosis, gastric surgery, renal failure, diabetes mellitus, immune suppressive therapy.

Induration $\geq 15\text{mm}$: Considered positive in **all persons** including those who have had previous BCG vaccination.

Causes of false negative test

1-Patient related factors:

- ◆Infections: overwhelming TB, miliary TB, TB meningitis, measles, typhoid, brucellosis, influenza.
- ◆Chronic renal failure, malnutrition, Hodgkin's disease, newborn, elderly.

2-Tuberculin-related factors:

- ◆Improper storage.
- ◆ Improper dilution.
- ◆Contamination.

3-Method of administration factors:

- ◆Injection of too little antigen.
- ◆Injection too deep.

4- Reading and recording factors:

- ◆Inexperience in reading.
- ◆Bias in reading.
- ◆Error in recording.

Types of tuberculosis according to the site of affection

1-Pulmonary tuberculosis:

- A- Primary pulmonary tuberculosis
- B- Secondary pulmonary tuberculosis.
- C- Miliary tuberculosis.

2-Extrapulmonary tuberculosis

A-Primary Pulmonary tuberculosis:

The first infection with the tubercle bacilli is known as primary tuberculosis and usually includes involvement of draining lymph node in addition to the initial lesion. This type is called **Childhood type**; it is frequently seen in young children (especially younger than 4 years). Infection is by inhalation of the human bacilli.

Pathological features:

Primary pulmonary complex; is a *triad of* Gohn's focus, tuberculous lymphangitis and tuberculous lymphadenitis of the regional lymph node (hilar lymph nodes).

Gohn's focus; an initial lesion occurs in the lung. The focus is situated at the periphery of lung underneath the pleura. It is may be found anywhere commonly in lower portions of upper lobe or upper portions of lower lobes.

Fate of primary pulmonary complex

I-Healing occurs with mild infection and good immunity.

II-Spread: occur with low resistance.

a- direct spread:

- Direct spread of Gohn's focus on the lung tissue → lead to tuberculous pneumonia.
- Direct spread to the pleura → lead to tuberculous pleurisy

b- Hematogenous spread:

- Small number of bacilli → destroyed
- Moderate number of bacilli → isolated organ tuberculosis
- large number of bacilli → miliary tuberculosis

c- Bronchial spread → tuberculous bronchopneumonia

III-Encapsulation and reactivation

Clinical picture:

- Great majority of primary tuberculous infections are probably symptomless or discovered only when they are investigated as contacts to an adult case.

- In few cases those with more severe infection the child may be unwell with loss of appetite, failure to gain weight, fretfulness

Cough may occur & Sputum production is rare in children.

◆-Auscultation of the chest is usually unrewarding but occasionally

- 1-crepitations may be heard over an extensive primary focus and
- 2-Wheezes may result from pressure of enlarged lymph node on bronchi lead to partial obstruction (*Localized rhonchi*)
- 3-More obvious physical signs may be present if there is segmental or lobar exudation or collapse.

Diagnosis:

A-Chest radiograph

- 1- lymphadenopathy and parenchymal lesion.
 - hilar lymph nodes most commonly involved also Para tracheal lymph node frequently involved (enlarged) both groups may enlarge together
- 2- There may be radiological evidence of segmental or lobar consolidation or *obstructive emphysema (due to pressure by enlarged lymph node on bronchus lead to partial obstruction)*

B-Tuberculin skin test strongly +ve

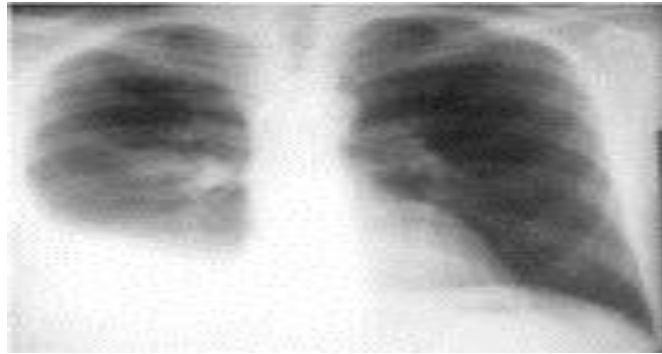
-ve test make diagnosis unlikely

C-Gastric washing; for direct smear for AFB and culture

D-PCR

Complications:

- Collapse or consolidation
- Bronchiectasis
- Obstructive emphysema
- Erythema nodosum
- Phlyctenular conjunctivitis
- Pleural effusion



B-Post primary pulmonary tuberculosis:

Post primary pulmonary tuberculosis is **the most important type of tuberculosis because it is **most frequent and** because smear positive sputum is the main source of infection responsible for the persistence of disease in community.**

Pathogenesis:

Post primary tuberculosis may arise in one of three ways:

- 1-direct progression of a primary lesion.
- 2-reactivation of quiescent primary or post primary lesion.
- 3-exogenous reinfection.
- Reactivation of the primary or post primary lesion.

This can occur at any time in a patient life, the original lesion need not necessarily be visible radiographically.

N.B. Worsening of individual defenses at any time of life may result in the development of post primary pulmonary tuberculosis.

Examples of predisposing factors:

- 1-Malnutrition.
- 2- Poor housing conditions with overcrowding.
- 3- Pulmonary silicosis, workers in occupation giving arise to pulmonary silicosis such as knife grinders, coal miners have greater risk to develop Tuberculosis.
- 4- Alcoholism

5- HIV infection

6- Cigarette smokers.

7- Steroid and other immunosuppressive drugs.

8- Disease associated with impaired cellular immunity such as Hodgkin's disease, leukemia, lymphoma, and AIDS,

Diabetes may predispose to tuberculosis and lead to a significant increase in the cavitory and smear positive disease.

Clinical features:

1-Symptoms:

1-Symptom-free discovered on routine radiology (active disease may be present with **no symptoms** at all

2-Many patients with TB present with **general symptoms** such as tiredness malaise, loss of appetite, weakness or loss of weight, night fever, night sweats

3-**Symptoms related to respiratory system:** Persistent cough with or without sputum, sputum may be mucoid, purulent or blood tinged

4-**Hemoptysis** is a classic symptom of pulmonary TB and may vary from blood staining of sputum to massive hemoptysis. Massive hemoptysis is usually due to erosion of one of the branches of bronchial artery, which bleeds at systemic pressure.

5-**Chest pain** may vary from dull ache or tightness to pleuritic pain.

6-**Breathlessness** is a feature of extensive pulmonary disease

7-**Localized wheeze**, as *endobronchial tuberculosis may result in localized wheeze* due to partial narrowing of the bronchus

8-Amenorrhea may be presenting symptoms, usually in sever tuberculosis

2- Physical sign:

There may be no physical sign in pulmonary tuberculosis even with relatively advanced disease, although there may be pallor, a hectic flush or cachexia in severe disease.

- Tachypnea with advanced disease
- There may be severe of varying degree with more extensive disease
- finger clubbing is unusual even in chronic disease, it suggests another diagnosis in particular carcinoma of the bronchus.
- With more advanced disease (severe or extensive disease) there may be pallor, hectic flush, cachexia, varying degree of fever, Tachypnea

In chest there may be no physical signs whatever in spite of extensive - radiological changes, the most common abnormality consists of ***POST TUSSIVE CREPITATIONS*** in the upper zone or apices.

- there may be signs of consolidation with pneumonic disease
- in chronic disease deviation of trachea may occur due to fibrosis, signs of pulmonary fibrosis may be present

classical physical signs of a cavity are seldom found even when large cavity - are evident in chest film

Investigations

1-Chest radiography:

A normal chest film does not completely exclude pulmonary tuberculosis.

Characteristic radiographic appearances.

- soft confluent shadows alone suggesting exudative process.
- linear shadows suggest fibrosis.

-bilateral upper zone fibro tic shadows, with shrinkage of the upper lobes and elevation of the pulmonary hila, distortion of fissures (trachea, mediastinum, diaphragm) this is evidence of fibrosis.

-calcification suggests healed disease.

-cavitations.

-tuberculoma is a cavity that may become blocked and fill with purulent or caseous material.

-Bronchiectasis in upper zones.

Enlargement of hilar or Para tracheal lymph nodes



Radiological classification of disease extent:

- 1- **Minimal**: there are infiltration, lesion is non-cavitary and not extending more than the anterior end of the second rib on one side.
- 2- **Moderately advanced**: here the lesion is cavitory, diameter of cavity is not more than 4 cm and lesion are not extending more than fourth rib anteriorly on one side.
- 3- **Far advanced**, if diameter of cavities or extension of lesions is more than that of moderately advanced disease.

2- Sputum examination:

Sputum examination is of great value in diagnosis of pulmonary TB and in following response to treatment.

-by **direct smear using the ziehl-nielsen stain** or fluorescence stain.

-**culture on Lowenstein Jensen medium** , growth is visible on culture after 2-6 weeks.

■ If sputum not available:

◆Gastric aspirate.

◆laryngeal swabs.

◆Fiberoptic bronchoscopic specimens (bronchial wash & bronchial brushing or TBB).

◆Fine needle aspiration biopsy. ◆Mediastinoscopy.

3-Newer diagnostic techniques:

1- **Serological** diagnosis of T.B by using enzyme linked-immunosorbent assay.

2- Polymerase chain reaction (PCR) can detect a single organism in a specimen of sputum.

4-Tuberculin skin testing: As explained previously.

Most patients with post primary T.B have a strongly positive (+ve) tuberculin test.

5-Liver function tests:

Before start anti-TB treatment and during course of the treatment because some anti-TB drugs have hepatotoxic effect

6-Blood picture:

◆Normocytic normochromic anemia is common in pulmonary tuberculosis.

◆Normal white blood cell count.

Complication of Post primary pulmonary tuberculosis:

- Pleurisy; pleural rub may be heard.
- Pleural effusion
- Tuberculous empyema.
- Tuberculous laryngitis occurs in pts. With +ve sputum and relatively severe disease.
- Cor pulmonal (with advanced disease).
- Chronic obstructive airway disease occurs in pt. with severe fibrotic pulmonary disease.
- Amyloidosis.
- Aspergillomas.
- Carcinoma of the bronchus.
- Pulmonary tuberculoma.

2- Miliary tuberculosis:

Miliary tuberculosis is **produced by acute dissemination of tubercle bacilli via the blood stream.**

The term miliary derives from radiographic picture of diffuse discrete nodular shadows about size of millet seed (2mm) that are characteristic of classical disease.

Clinical picture:

♥-**Symptoms** include the following:

- In **children** the onset may be associated with an acute or subacute febrile illness. but
- in **adult** the onset is insidious with gradual development of vague, ill health, Weakness, fatigue, malaise, anorexia, weight loss and fever (cough, breathless, hemoptysis and night sweats) are less common.

- Headache as a feature suggests associated tuberculous meningitis.

♥-Signs:

- Apart from fever there may be no physical signs.
- Chest frequently normal on auscultation although crepitations may develop in late stages
- Hepatomegaly, ●splenomegaly, ●generalized lymphadenopathy,
- nuchal rigidity.
- Choroidal tubercles→ over 90%of children with military T.B.

♠Diagnosis

1-Chest Radiography:

- Abnormal shadows evenly distributed and may vary from faint shadows 1-2mm in diameter to large dense shadows up to 5 or 10mm.
- Bil pleural effusion.



2-Tuberculin skin test→ is usually (+ve) except in advanced disease (–ve TST)

3-Direct smear of sputum or gastric lavage.

4-Fibreoptic bronchoscopy (FOB) (samples→ if necessary).

5-Liver function tests

6-Blood picture: -normocytic normochromic anemia is common in pulmonary tuberculosis

Treatment of tuberculosis

New TB cases were defined as TB patients who had denied having had any prior anti-TB treatment or who received anti-TB treatment for 30 days or less.

Retreatment TB cases were TB patients who had been receiving TB treatment for at least 30 days or who had had documented evidence of prior treatment from the case report or surveillance database.

Retreatment TB cases included relapses, treatment after failure, treatment after default and chronic cases.

MDR-TB was defined as TB with resistance to at least isoniazid and rifampin.

XDR-TB was defined as TB with resistance to at least isoniazid, rifampin, a fluoroquinolone (e.g. ofloxacin, levofloxacin) and one of three injectable second-line drugs (capreomycin, kanamycin, and amikacin)

Antituberculous drugs:

1-First line drugs are the drugs used in initial and maintained chemotherapy unless drug resistance is known.

Include 5 drugs **Isoniazid** (INH), **Rifampicin** (RIF), **Pyrazinamide** (PZA), **Ethambutol** (EMB), **Streptomycin** (SM)

1-Isoniazid (INH):

Is a first line agent for ttt of all forms of tuberculosis caused by organisms known or presumed to be susceptible to the drug, it has profound early **bactericidal activity** against rapidly developing cells.

-Dose:

- Adult: 5mg/kg (300mg/daily). 15mg/kg (900mg once, twice or three times weekly (intermittent)).
- children: 10-15 mg/kg (300mg) daily, 20-30mg/kg (900) twice weekly.

-Preparations: Tablets (50mg, 100mg, 300mg), syrup (50mg/5ml), aqueous solution for IV., IM injection.

Adverse effects:

1-Drug induced hepatotoxicity can vary from asymptomatic elevations in the liver enzymes to liver failure.

2-Peripheral neurotoxicity: The risk is increased in patients with other conditions such as **diabetes mellitus, malnutrition, renal failure**. **Vitamin B6 (Pyridoxine) supplementation** is needed in dose (25mg/day) to prevent neuropathy.

3-Seizures, irritability, dysarthria.

4-Hypersensitivity reactions such as fever, rash, Steven – Jensen syndrome, hemolytic anemia.

2-Rifampicin (RIF):

Is a first line agent. It has activity against organisms that are dividing rapidly (early **bactericidal activity**) and against semidormant bacterial populations thus account for its sterilizing activity rifampicin is an essential component of all short course regimen.

-Dose:

- Adult→ <50 kg→ 450mg daily.
→ >50 kg→ 600mg daily.
- Intermittent→ 10mg/kg twice weekly or three times weekly.
- Children →10-20 mg/kg (600mg) once daily or twice weekly.

Preparations: Capsules (150mg, 300mg), oral suspension, aqueous solution for parenteral administration

Adverse effects:

1-Orange discoloration of body fluids (sputum, urine, sweat, tears).

2- Drug induced hepatotoxicity can vary from asymptomatic elevations in the liver enzymes to liver failure.

3-Gastrointestinal side effects→ nausea, anorexia, abdominal pain , vomiting , diarrhea.

4-Cutaneous reactions→ purities

5-Severe immunological reactions→ thrombocytopenia, hemolytic anemia

3-Pyrazinamide (PZA):

Is a first line agent. The drug has **bactericidal activity**. The drug is believed to exert greatest activity against the population of dormant or semidormant organisms contained within macrophages or the acidic environment of caseous material.

-Dose:

- Adult→ <50 kg 1.5 gm daily.
→ 50-75 kg 2 gm daily.

→ >75 kg 2.5 gm daily.

Children → 15-30 (40) mg/kg (2gm) once daily

→ 50 mg/kg twice weekly (2 gm).

Preparations: Tablets (500 mg, scored)

Adverse effects:

1-Drug induced hepatotoxicity can vary from asymptomatic elevations in the liver enzymes to liver failure

2-Symptomatic hyperuricemia (drug induced).

3-Non gouty polyarthralgia.

4-Acute gouty arthritis, Gouty attacks have been associated with taking pyrazinamide due to pyrazinamide induced hyperuricemia. So, **the use of pyrazinamide is contraindicated in patients with acute gout.**

5-GIT side effects → nausea, anorexia, vomiting.

4- Ethambutol (EMB):

Is generally not recommended for routine use in children whose visual acuity cannot be monitored, it is a bacteriostatic drug.

-Dose:

- Adult → 15-20 mg/kg daily or → 25mg/kg daily for 2 months then 15mg/kg.
- Children → 15-20mg/kg daily in older children. (**not used in children less than 5 years of age**).

Preparations: Tablets (100 mg, 400mg) for oral administration.

Adverse effects:

1-Optic neuritis, (more specifically retro bulbar neuritis) causing blurred vision, decreased visual acuity, **not recommended for routine use in children whose visual acuity cannot be monitored** (during follow up)

2-Peripheral neuritis.

3-Cutaneous reactions.

5-Streptomycin (SM):

Is bactericidal in an alkaline environment.

-Dose:

Adult → <30 kg → 750mg.

→ >30 kg → 1gm.

→ >40 yrs → 750mg.

(15mg/ kg/day).

♦ It is given **parenterally by intramuscular injection**

Children → 20-40 mg/kg/day.

NB: Streptomycin (SM) is contraindicated in pregnancy because of the risk of fetal hearing loss.

Preparations: Aqueous solution in vials of 1 gm.

Adverse effects:

1-Ototoxicity: Most important one & including vestibular and hearing disturbances (Eight cranial nerve). **Contraindicated in pregnancy as it causes fetal hearing loss**

2-Nephrotoxicity: occur less commonly with SM than with amikacin, kanamycin, or capreomycin.

3-Neurotoxicity.

NB: Rifabutin or Rifapentine can be used as alternative drug to rifampicin in some situations

Rifabutin:

-Dose:

Adult: → <50 kg → 150mg.

→ >50 kg → 300mg/daily.

Preparations: Capsules 150 mg.

Side effects: Hepatotoxicity → hepatitis.

Rifapentine:

-Dose:

Adult: → 10mg/kg (600mg once weekly).

Preparations: Tablets → 150 mg.

Side effects: Hepatotoxicity → hepatitis.

2-

SECOND LINE DRUGS (reserve drugs):

Reserve drugs are used in the treatment or retreatment of patients with known or suspected mycobacterial drug resistance.

1-Cycloserine.

2-Ethionamide.

3-para-aminosalicylic acid.

4-amikacin, kanamycin.

5-Capreomycin.

6-fluoroquinolones: levofloxacin, gatifloxacin have the most activity against mycobacterium tuberculosis, ciprofloxacin, ofloxacin

♣-Principles of antituberculosis chemotherapy:

Treatment of tuberculosis is a Combination chemotherapy (NO single drug use for treatment of tuberculosis disease):

♥-The primary goals of antituberculosis chemotherapy are to:

1-kill tubercle bacilli rapidly.

2-Prevent the emergency of drug resistance.

3-Eliminate persistent bacilli from the host's tissues to prevent relapse.

To achieve these goals multiple antituberculosis drugs must be taken for a sufficiently long time.

♣-Chemotherapy regimens for pulmonary tuberculosis:

Each regimen has an initial phase of 2 months followed by the continuation phase of either 4-7 months.

♥-Standardized 6-months short course chemotherapy:

☉ -Initial phase:

two months include 4 drugs

isoniazid (INH) plus rifampicin (RIF), plus Pyrazinamide (PZA) plus [Streptomycin (SM) or Ethambutol (EMB)].

☉ - continuation phase:

4 months of 2 drugs rifampicin (RIF) plus isoniazid (INH).

♦ Isoniazide → 300mg orally (for 6 months)

♦ Rifampicin → 600mg > 50kg orally (for 6 months).
→ 450 mg < 50kg

♦ Pyrazinamide → 1.5gm < 50 kg. orally (for 2 months).

→2.0gm 50-70 kg.

→2.5gm> 75mg.

♦ **Ethambutol** → 15-25mg/kg **orally (for 2 months).**
or

♦ **Streptomycin** → 750 mg 6 days out of 7 **IM (for 2 months)**

All drugs administrated orally should be taken fasting **30 min before breakfast** since concomitant food significantly decreases the bioavailability of RIF and INH.

♥-STANDERED 9-MONTHS CHEMOTHERAPY:

☉ -Initial phase:

for 2 months include 3 drugs RIF plus INH, plus (ETB or SM).

☉ Continuation phase: 7 months of 2 drugs INH plus RIF.

♦ Isoniazid → **(for 9 months)**

♦ Rifampicin → **(for 9 months)**

♦ Ethambutol → **(for 2 months)**

or

♦ Streptomycin→ **(for 2 months)**

♠-DOT: Directly Observed Therapy for tuberculosis:

DOT refers to the process whereby **a health care worker or trained persons watches while a patient swallows antituberculosis drugs over the six (6) to nine (9) months of treatment.**

The drug can be administrated in daily or intermittent (2-3 times weekly) regimens in a wide range of clinical settings or at home, work, school.

♣-Corticosteroid therapy in tuberculosis:

The main **indications** for steroid therapy in tuberculosis are as follow:

1-In **seriously ill patient.**

2-To control **drug hypersensitivity** reactions.

3-Tuberculosis of the **serous sacs** as **pericarditis, peritonitis**, pleural **effusion** to **prevent fibrosis and adhesions and hasten absorption of fluid.**

4-TB **meningitis** to prevent morbidity and mortality due to adhesions.

5-In **genitourinary TB** to prevent the development of ureteric strictures.

The usual dose of prednisolone is 40mg daily in adult for 6 weeks reducing gradually thereafter over 4 weeks.

♣-Control and prevention of tuberculosis:

1-BCG vaccination:

- BCG was originally a **bovine strain** of tuberculosis that was attenuated by 230 passages through media containing glycerin and ox bile. First orally then intradermal injections.
- It doesn't prevent infection with tuberculosis but limits multiplication and dissemination of mycobacteria following infection. **So, it is used to protect against severe forms of TB e.g. protective effect against TB meningitis and disseminated TB (miliary disease) in children.**
- BCG given obligatory in Egypt in first 3 months of life.
- In developing countries with high incidence of TB the WHO has recommended administration of BCG to neonates or as early as possible to children.

♠-Technique: the vaccine is best given by **intradermal injection** of 0.1 ml (0.05 ml for the <3 months old) in the lower deltoid area. A papule appears in 3-4 weeks that usually remains for a number of weeks and may ulcerate slightly, and discharge and the tuberculin test should become positive within 3 months.

2-Chemoprophylaxis:

Chemoprophylaxis is the giving chemotherapy to prevent the development of tuberculous disease.

In chemoprophylaxis single drug can be used

☺ - Primary Chemoprophylaxis: involves giving chemotherapy **to individual who have not so far been infected (-ve tuberculin test)** i.e. **suckling infant cannot be separated from a TB mother** because of the risk of malnutrition, the child (infant) given

Drug used: Isoniazid until the mother is considered noninfectious.

☺ - Secondary Chemoprophylaxis: Is giving chemotherapy to:

Individuals with **positive tuberculin test, no clinical manifestations of TB disease and a normal chest film** who have been infected to **prevent development of clinical disease (active disease)**. **For example:**

- 1-Positive tuberculin test in special clinical situations with: silicosis, DM, Hodgkin disease, **patients on dialysis**
- 2-Household members with close contacts of newly diagnosed patients and newly infected persons
- 3-Children under the age of 5 years old with strongly positive tuberculin test reaction.

Drug used: ♠- **isoniazid (INH)** 300 mg daily for adult & 5 mg/kg up to 300mg/day in children with total duration of therapy of one year, many physicians prefer the use of two drugs

Extra-pulmonary tuberculosis: include

- 1-Tuberculous pleural disease
- 2-Central nervous system tuberculosis
- 3-superficial tuberculous lymphadenitis
- 4-Bone and joint tuberculosis
- 5-gastrointestinal tract and peritoneal tuberculosis
- 6-Genitourinary tuberculosis
- 7-Cutaneous tuberculosis
- 8-Tuberculous pericarditis
- 9-Tuberculosis of the breast

Interstitial Pulmonary Fibrosis

HAMMAN and RICH were the first to describe (in 1935 and 1944) four patients who died of rapidly progressive lung disease characterized by diffuse interstitial pneumonia and fibrosis.

Interstitium

Refers to the microscopic anatomic space bounded by the basement membrane of epithelial and endothelial cells.

Within this interstitial space, fibroblast like cells (mesenchymal and connective tissue cells) and extracellular matrix components (interstitial collagens, elastin, and proteoglycans) are present.

WHAT DOES THE TERM “INTERSTITIAL” MEAN?

It implies that the inflammatory process is limited specifically to the area between the alveolar epithelial and capillary endothelial basement membranes.

This group of pulmonary disorders frequently involves:

- 1-Alveolar epithelium
- 2-Alveolar space
- 3-Pulmonary microvasculature
- 4-Respiratory bronchioles

Definition:

Inflammatory-fibrotic infiltration of the interstitium resulting in profound effects on the capillary endothelium and the alveolar epithelial lining cells, producing a disease with similar clinical, radiographic, and physiologic features

Epidemiology:

ILDs constitute about 10% to 15% of the patients with respiratory diseases. About 50% of the ILDs is idiopathic in origin called Idiopathic pulmonary fibrosis (IPF).

Usual Interstitial pneumonitis (UIP) is commonest type of idiopathic pulmonary fibrosis.

Classification of interstitial lung diseases (ILD)

- 1- **Idiopathic pulmonary fibrosis (IPF)** (commonest type of ILD)
- 2- ILD caused by **Connective Tissue Diseases**
- 3- ILD caused by **Occupational factors**
- 4- **Drug-induced** ILD
- 5- **Unclassified** type of ILD

1-Idiopathic pulmonary fibrosis (IPF): *is commonest type of ILD.*

There is *no identifiable cause*. The affection is limited to the lungs.

Types of Idiopathic pulmonary fibrosis (IPF):

- 1-Usual Interstitial pneumonitis (UIP) is **commonest type of idiopathic pulmonary fibrosis**
- 2-Acute interstitial pneumonitis (Hamman-Rich syndrome)
- 3-Desquamative interstitial pneumonitis
- 4-Respiratory bronchiolitis
- 5-Cryptogenic organizing pneumonia
- 6-Nonspecific interstitial pneumonitis
- 7-Lymphocytic interstitial pneumonia

2-Connective tissue diseases (CTD) causing ILD:

- Rheumatoid arthritis (**commonest CTD cause ILD**)
- Systemic lupus erythematosus (**common** cause)
- Scleroderma (**common** cause)
- Polymyositis
- Dermatomyositis
- Mixed connective tissue disease
- Ankylosing spondylitis

3- Occupational factors causing ILD:

a. Organic dusts:

The lung disease that many farmers experience is often called **farmer's lung disease**. The most common cause is the result of **inhalation** of **agricultural biological dust** derived from straw, mold spores, domestic animals or other dust.

Farmer's lung includes number of diseases such as:

- Bird breeder's lung** (Chicken breeder's lung, pigeon breeder's lung),
- Mushroom worker's lung**,
- Cheese washer's lung**, and
- Coffee worker's lung**.
- Bagassosis** is caused by inhalation of the fibrous cane-sugar residue (bagasse).
- Also, certain types of molds called "**Aspergillus**" are important causes of Farmer's Lung.

b. Inorganic dusts:

- Silica (**commonest** inorganic dust causing ILD)
- Asbestos
- Aluminum
- Coal dust.

4- Drug-induced ILD: caused by

- Chemotherapeutic agents** (*common* causes of drug induced ILD): e.g. bleomycin, cyclophosphamide, methotrexate,
- Antibiotics**: Nitrofurantoin, sulfonamides, penicillin.
- Other drugs**: Amiodarone (**common** cause of drug induced ILD), procainamide, practolol, carbamazepine,
- Radiotherapy** (*common* cause of drug induced ILD): *for treatment of breast cancer or lung cancer*

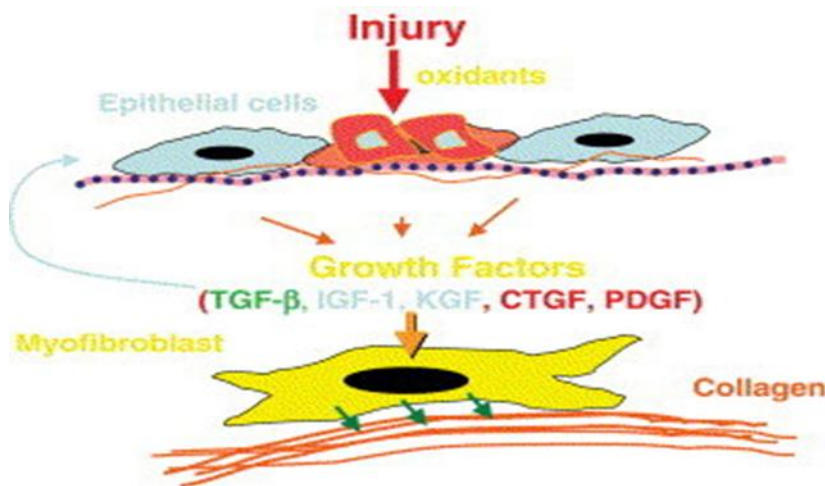
5-Unclassified group of ILD: caused by

- Sarcoidosis**: it is the *commonest cause in unclassified group*
- Idiopathic pulmonary hemosiderosis.
- Wegner's granulomatosis
- Histiocytosis-X.

Pathogenesis:

Epithelial cell apoptosis with loss of basement membrane integrity: lead to production of growth factors in response to alveolar epithelial injury with hyperplastic type II cells and myofibroblast recruitment (**with formation of collagen**

fibers), result in initiation and progression of fibrosis.



Evaluation of ILD:

History:

- **Dyspnea** is the first symptom of ILD, which is usually progressive, at first on exercise and then during rest.
- **Cough** is common, dry cough +/- little sputum, Wheezing is not a common complaint
- **Comorbid diseases**
 - Manifestations of connective tissue diseases: arthritis, rashes, dysphagia
 - History of drugs use
 - Environmental / occupational exposures: e.g. silica, Asbestos ...ect
 - Smoking status
 - Previous malignancies and treatment (chemotherapy drugs, irradiation)
 - Family history of lung diseases

Examination:

General examination:

- **Tachypnea**
- **Cyanosis**
- **Clubbing:** is a common finding occurring in about 50% of cases,
- Manifestations of connective tissue diseases e.g. joint affection, skin lesions

- **Lower limbs edema** (due to e.g. decompensated cor-pulmonal)

Chest and cardiac examination:

Auscultation of the chest:

- Intensity of the **breath sound** is normal.
- **Lung crackles (crepitations): end-inspiratory, fine dry** (Velcro” quality).
Crepitations are common, and sometimes are localized.
- Scattered late inspiratory bronchi may be heard.

Heart examination: manifestations of pulmonary hypertension and its sequelae, P2 might be accentuated, signs of right ventricular hyper-trophy and failure is usual

Investigations

Laboratory studies:

- **Complete blood count** with differential count
- **ESR.**
- **Renal & liver function tests.**
- **Serologic studies:** should be obtained if clinically indicated:
 - features suggestive of a **connective tissue disease:** Rheumatoid factor (RF), Antinuclear antibodies (ANA),
 - environmental exposure:** (hypersensitivity precipitin panel), **or**
 - systemic vasculitis:** [antineutrophil cytoplasmic antibodies (ANCA)].

Radiology:

- Usually, abnormal,
- but **Chest x-ray may be normal in about 10% of cases**
- **High-resolution computed tomography scanning (HRCT) of chest** is the imaging **modality of choice** for diagnosing IPF. It is a non-invasive method for diagnosis of ILD.

The common radiologic findings of IPF on HRCT are patchy, predominantly peripheral, subpleural, and bibasilar reticular abnormalities. Honeycomb appearance indicate late stage of ILD.

Pulmonary function testing:

Complete pulmonary function testing (PFT with spirometry, lung volumes,

diffusing capacity for carbon monoxide [DLCO]) is performed on all patients with suspected ILD.

- Measurement of lung function is helpful in **diagnosis** and **assessing the severity** of lung involvement in patients with ILD.
- Smoking history must be considered when interpreting the functional studies

Interstitial lung diseases have restrictive pattern include:

***Decreased FVC**

***FFV1 normal in early stage and decreased with disease progress** (in proportion to the decreased lung volumes)

***The FEV1/FVC ratio is usually normal or increased.**

***Reductions in ↓ total lung capacity (TLC), ↓functional residual capacity (FRC), and ↓ residual volume (RV).** The reductions in lung volumes become more pronounced as lung stiffness worsens with disease progression.

***Diffusing capacity of the lungs for carbon monoxide (DLCO) is reduced**

Arterial blood gas:

- The resting arterial blood gases may be normal or may reveal hypoxemia.
- *Importantly, a normal resting PaO₂ or O₂ saturation by oximetry does not rule out significant hypoxemia during exercise or sleep, which is common in ILD.*
- Respiratory failure caused by ILD commonly type I respiratory failure.
- *Carbon dioxide retention (and deterioration to type II respiratory failure) is manifestation of end-stage disease.*

Lung biopsy:

- It is not a routine method for diagnosis of all cases of ILD.

Methods of lung biopsy:

1-Bronchoscopy and Transbronchial lung biopsy: obtains a sample but it is **too small** to definitively diagnose ILD. This method results in higher risk of **biopsy associated complications**

2-Surgical lung biopsy (SLB),

Surgical lung biopsy whether done via:

a- Video-assisted thoracoscopic lung biopsy is the **preferred method** of obtaining lung tissue biopsy

b- Thoracotomy lung biopsy

Indications:

To provide a specific diagnosis in patients with atypically or progressive pattern

Contraindications of lung biopsy:

- + Serious cardiovascular disease
- + Radiographic evidence of diffuse end-stage disease (honey combing)
- + Severe pulmonary dysfunction

Treatment:

Lines of medical treatment of ILD:

- **Anti-inflammatory drugs:** Corticosteroids, Azathioprine, Cyclophosphamide
- **Anti-fibrotic agents:** *Colchicine,
*Pirfenidone (*Novel therapies for ILD*),
- **Anti-oxidant agents:** N-acetylcysteine
- **Supportive and symptomatic treatments:** Oxygen therapy, Diuretics (treatment of decompensated cor-pulmonal), Antibiotics (if infection)

Corticosteroids **Represent the classic treatment of ILD (first line treatment)** due to their anti-inflammatory and immunosuppressive effects.

Potential alternative treatments: Agents that alter collagen synthesis and fibrosis are considered second line drugs **or** first line drugs, if there is contraindication to corticosteroids: include

1-Colchicine (*Anti-fibrotic agents*)

2-Methotrexate: *has anti-inflammatory and immunosuppressant effects.*

3-Cyclophosphamide *has immunosuppressive effect*

4-Azathioprine: *has immunosuppressive effect*

Novel therapies in IPF: A number of agents that interfere with collagen synthesis such as: Pirfenidone (*novel Anti-fibrotic agents*)

Lung Transplantation:

-Transplantation should be considered for patients with progressive physiologic deterioration despite optimal medical management

-Single lung transplantation is currently the preferred surgical operation.

Complications of Interstitial Pulmonary Fibrosis:

- **Respiratory failure (commonest complication)**
- Cor pulmonale (common)
- Pneumonia
- Pneumothorax
- Side effects of drugs used for treatment of ILD
- Pulmonary embolism
- Bronchial carcinoma

Worse prognosis: following factors associated with poor prognosis:

- **Old age: is an important factor** associated with poor prognosis
- **Honeycombing radiological pattern: is an important factor** associated with poor prognosis
- **Pulmonary hypertension (PHT): is an important factor** associated with poor prognosis
- Male
- Smoking
- Lower DLCO < 40%
- Higher rate of acute exacerbations

Causes of death in ILD:

- **Pulmonary embolism (commonest cause of death in ILD)**
- Acute exacerbation of IPF
- Infection
- Coronary artery disease
- Lung cancer

Lung Cancer

Incidence:

Lung cancer is the **most common malignant neoplasm in men** throughout the world.

Etiological factors of lung cancer:

1-Tobacco smoking:

Cigarette smoking is by far the most common cause of lung cancer.

2- Occupational factors:

Proven occupational carcinogens for lung cancer include *asbestos*, *arsenic* and *ionizing radiation*.

Classification:

A- Non-small cell lung cancer (NSCLC): represents **80%** of lung cancer and includes squamous cell carcinoma, adenocarcinoma, large cell carcinoma, adeno-squamous or other mixed carcinomas and bronchioloalveolar carcinoma.

B-Small cell lung cancer (SCLC): accounts for **20%** of lung cancer and requires different staging and treatment protocols.

A-Non-small cell lung cancer (NSCLC) (80%)

1- Squamous cell carcinoma (epidermoid carcinoma): It account for **30%** of lung cancer, *strongly associated with smoking, common to cavitate* and *poor prognosis*, it arises from *bronchial epithelium* and tend to be found as *central masses*, **histologically** it is characterized by stratified layers of malignant cells with frequent mitotic figures and conglomerates of keratin called *epithelial pearls* or *cell nests*.

2-Adenocarcinoma (AC):

Most common cell type overall, most common in women and non-smokers but still most patients are smokers and peripheral. AC represents about **35%** of lung cancer and usually presents as **peripheral** nodules or masses on chest x-ray. **Histologically**, it tends to form glandular or acinar structures of mucin producing cuboidal or columnar cells.

3-Large cell carcinoma (LCC):

Peripherally located and very large, usually > 4 cm, accounting for **10 to 15%** of lung cancer. LCC is defined by the absence of pathologic characteristics such as gland formation or keratin it tends to arise peripherally **histologically** the cells are large and undifferentiated with abundant cytoplasm.

B-Small cell lung cancer (SCLC) (20%)

Small cell lung cancer (SCLC), also known as **oat cell lung cancer**, it has a unique presentation, imaging appearances, treatment, and prognosis. SCLCs are **neuroendocrine tumors** of the lung that **rapidly grow**, are **highly malignant**, widely **metastasize**, and, despite showing an initial response to chemotherapy and radiotherapy, have a **poor prognosis** and are **usually unresectable**.

It almost always in **smokers metastasizes** early and worst prognosis. It is the most common lung cancer subtype to produce **necrosis**, **superior vena cava (SVC) infiltration/SVC obstruction**, and **Para neoplastic syndrome** Approximately 90-95% of SCLCs occur **centrally**, usually arising **adjacent to a lobar or main bronchus**

Clinical presentation of lung cancer

Patients with lung cancer may be asymptomatic in up to 50% of cases.

A- Symptoms related to the primary tumor:

1-Cough: may be related to airway involvement, post obstructive pneumonitis or bronchorrhea

Bronchorrhea: is large amount of mucoid watery sputum in **bronchoalveolar carcinoma** (subtype of adenocarcinoma).

2-Hemoptysis: more common with central tumors (*red currant jelly sputum*)

3-Dyspnea: Dyspnea may be **caused by:**

***Endobronchial** obstruction or extrinsic airway compression, post obstructive atelectasis, infection, or

***Increased** airway secretions, or due to

*Metastatic disease including pleural effusion, lymphangitic tumor spread, pericardial effusion with tamponade, and

*Pulmonary thromboembolic.

4- Pleuritic chest pain: more common with *peripheral tumors*

B-Symptoms related to spread intrathoracic spread

Intrathoracic spread of lung cancer may occur by **direct extension** of the primary tumor or by **involvement of lymph nodes** and **other thoracic structures**.

1- **Chest pain:** is reported by up to 50% of patients at presentation and often indicates extension of tumor to the mediastinum, chest wall, or pleura.

2- **Hoarseness:** due to involvement of the left recurrent laryngeal nerve.

3- **Superior vena cava syndrome:** occurs more commonly with **small cell carcinoma**. Patients may present with a sensation of facial fullness, dyspnea, dysphagia, and headache or with actual swelling of the face, neck, and upper chest, and typically exhibit distended neck veins as well as dilated veins over the upper chest and shoulders.

4-Pancoast syndrome (Thoracic inlet or superior sulcus syndrome):

A Pancoast tumor is a tumor of the **pulmonary apex**. Most tumors are **NSCL (Squamous cell carcinoma)**.

Pancoast syndrome is caused by pressure on:

A-Sympathetic chain: Horner's syndrome (**miosis, anhidrosis, ptosis and pseudo-enophthalmos**)

B-Lower trunk of the brachial plexus: causing pain and weakness in the muscles of the arm and hand.

C-The recurrent laryngeal nerve: hoarseness of voice

D-Blood vessels: unequal pulse and blood pressure. Congested non pulsating neck veins and dilated veins over the chest.

E-Erosion of ribs: chest pain

5-Pericardial effusion: due to extension of tumor to the pericardium with or without tamponade, as well as arrhythmias .

6-Bulky mediastinal adenopathy itself causes compression symptoms:

- **Trachea:** cough and dyspnea
- **Esophagus:** dysphagia
- **Left recurrent laryngeal nerve:** hoarseness of voice

C-Symptoms related to metastatic spread:

Due to **hematogenous** spread to any organ or **lymphatic spread**. The presence of new symptoms related to any organ should heighten the clinical suspicion for metastatic disease.

D- Constitutional symptoms:

Including weight loss, fatigue, weakness and poor appetite.

E-Para neoplastic syndromes:

They are **endocrine/ metabolic** and **neurologic extra pulmonary syndromes not related to direct invasion or distant metastases of the tumor**. It occurs in approximately **10%** of patients with lung cancer. Endocrine syndromes associated with lung cancers are due to production of biologically active hormones by the tumor. **Most common with small cell carcinoma**.

Various Para neoplastic syndromes can arise in the setting of lung cancer:

a-Endocrine/metabolic:

1-The syndrome of inappropriate antidiuretic hormone secretion (SIADH or SIADH)

ADH: common with **small-cell subtype**. It is due to inappropriate (excessive) release of antidiuretic hormone (ADH), now known as arginine-vasopressin (AVP). The result is often dilutional hyponatremia in which the sodium remains normal but total body fluid increases.

2-ACTH secretion (Cushing syndrome): common with carcinoid and SCLC. In most cases, tumor cells express a precursor hormone that is cleaved to ACTH. A minority of

patients produce corticotropin releasing hormone, which stimulates ACTH production in the pituitary

3-Carcinoid syndrome: Carcinoid syndrome typically occurs in people who have carcinoid tumors (neuroendocrine tumors) that are advanced. It refers to a spectrum of symptoms that result from excessive hormone (mainly serotonin) secretion. Occurs equally between the sexes, most commonly in the 40–70-year age group. Diarrhea is the most common and earliest symptom but others include flushing, right heart failure, abdominal pain and bronchospasm.

4-Gynecomastia

5-Adrenal insufficiency (Addison disease): from bilateral metastases to adrenal glands. It refers to inadequate secretion of corticosteroids (glucocorticoids and mineralocorticoids).

6-Hyperparathyroidism: NSCLC can produce parathyroid hormone (extremely rare).

7-Hypocalcemia: occurs in the setting of skeletal metastases; especially associated with NSCLC.

8-Parathyroid hormone–related peptide (PTHrP) causing hypercalcemia: common with squamous cell carcinoma. It is usually related to the ectopic production of parathyroid hormone– related peptide (PTHrP)

b-Neurological: common with SCLC due to autoimmune mechanisms.

C-Others:

1-Finger clubbing, hypertrophic pulmonary osteoarthropathy (HPOA): commonly seen with adenocarcinoma. The pathogenesis of HPOA is unknown, it characterized by symmetric, painful arthropathy that typically involves the ankles, shins, knees, wrists, and elbows.

2-Nephrotic syndrome

3-Polymyositis

4-Dermatomyositis

5-Hematologic dyscrasias commonly occurring in patients with lung cancer include anemia, thrombocytosis, and leukocytosis.

Investigations for lung cancer diagnosis

The investigations to diagnose lung cancer may include:

1-Chest x-ray: A chest X-ray is usually the 1st test used to diagnose lung cancer. Most lung tumors appear on X-rays as a white-grey mass. However, chest X-rays cannot give a definitive diagnosis because they often cannot distinguish between cancer and other conditions, such as a lung abscess. It can detect mass, nodule, pleural effusion, lung collapse and malignant abscess

2-CT scan of the chest: A CT scan is more likely to show lung tumors than routine chest x-rays. It can also show the size, shape, and position of any lung tumors and can help find enlarged lymph nodes. It can also be used to look for metastases in the adrenal glands, liver, brain, and other organs that might be due to the lung cancer spread. CT-guided needle biopsy can be done.

3-Bronchoscopy and biopsy:

A bronchoscopy is used in central tumors and can be used for biopsy.

Criteria of inoperability by bronchoscopy:

- Paralysis of vocal cords
- Affection of the carina
- Tumor < 2 cm distal to carina

4-Endobronchial ultrasound (EBUS): A newer procedure which combines a bronchoscopy with an ultrasound scan. Like a bronchoscopy, an EBUS allows a doctor to see the inside of airways. However, the ultrasound probe on the end of the camera also allows the doctor to locate the lymph nodes in the mediastinum so they can take a biopsy from them.

5-Other types of biopsy:

By using thoracoscopy, mediastinoscopy, or a biopsy done using a needle inserted through the skin (percutaneous).

6-Sputum cytology: A sample of sputum is looked at in the lab to see if it has malignant cells. The best way to do this is to get early morning samples 3 days in a row. This test is more likely to be positive in central tumors, such as squamous cell carcinoma.

7-Pleural fluid analysis: to detect malignant cells in suspected malignant pleural effusion

8-Magnetic resonance imaging (MRI) scan: MRI scans are most often used to look for possible spread of lung cancer to the brain or spinal cord.

9-Positron emission tomography (PET) scan: For a PET scan, a slightly radioactive form of sugar (known as FDG) is injected into the blood and collects mainly in cancer cells.

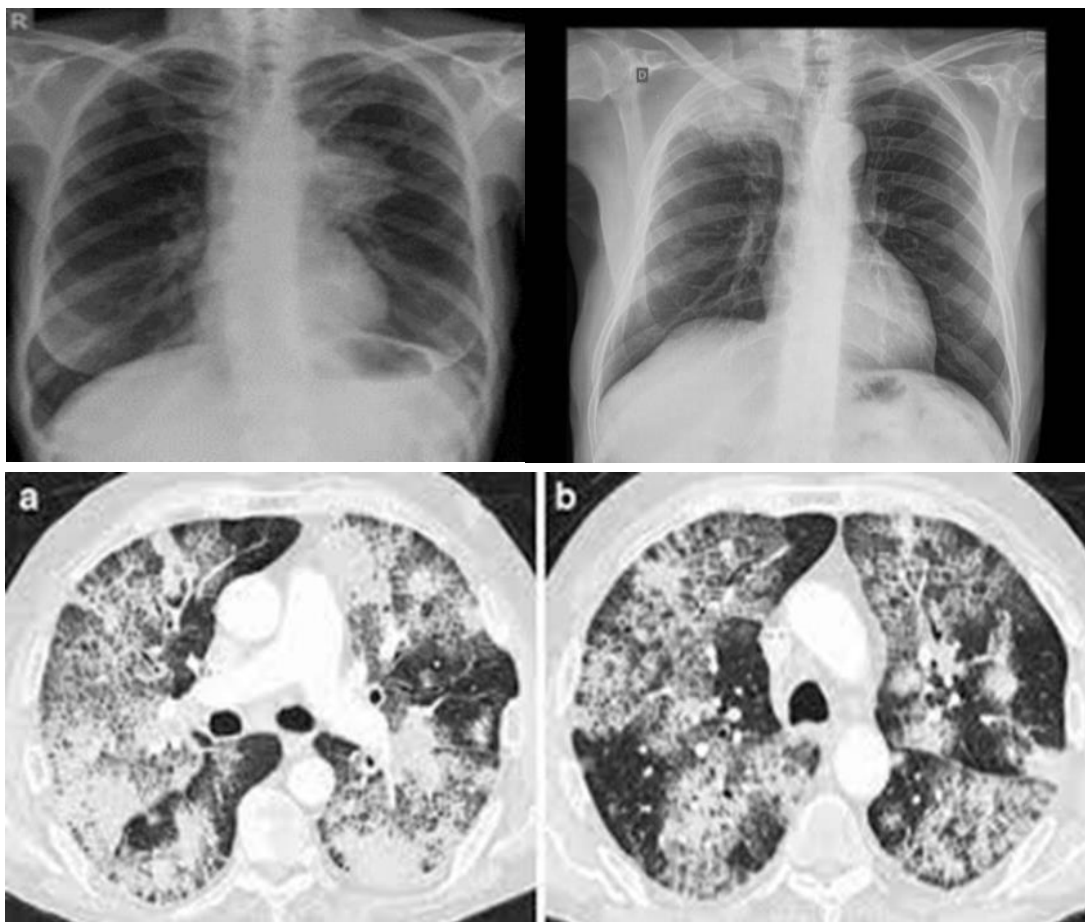
10-PET/CT scan: Often a PET scan is combined with a CT scan. This lets the doctor compare areas of higher radioactivity on the PET scan with a more detailed picture on the CT scan. PET/CT scans can be useful to detect spread of cancer to the liver, bones, adrenal glands, or some other organs. They are not useful for looking at the brain or spinal cord.

11-Bone scan to detect bone metastases

12-Further tests:

Other tests can give more information about the cancer and help doctors work out whether it has spread. These tests can guide treatment and may include:

Molecular tests: laboratory tests on the biopsy sample to identify particular genetic mutations in the cancer



Lung cancer (staging)

1-Non- Small cell lung cancer staging: through

TNM system:

T: primary tumor

N: regional lymph node involvement

M: distant metastasis

2-Small cell lung cancer staging:

The idea behind a different staging system for SCLC was on account of it being almost never resectable at the time of diagnosis, except in rare cases where it appears as a solitary pulmonary nodule. The staging was performed dividing the tumors as such:

A-Limited:

- Disease confined to one hemithorax
- Inclusive of hilar/supraclavicular lymph nodes
- Patients were suitable for radiation therapy as well as chemotherapy

B-Extensive:

- Disease extends beyond one hemi thorax
- Patients received only chemotherapy
- Accounted for 70% of cases

Treatment and prognosis:

Treatment and prognosis vary not only with stage but also with cell type. In general, surgery, chemotherapy, and radiotherapy are offered according to the stage, resect ability, operability, functional status of the lung and general condition of the patient.

****End bronchial treatment:**

The aim is to **achieve palliation of symptoms** such as hemoptysis and critical obstruction of main airway by establishing a patent airway.

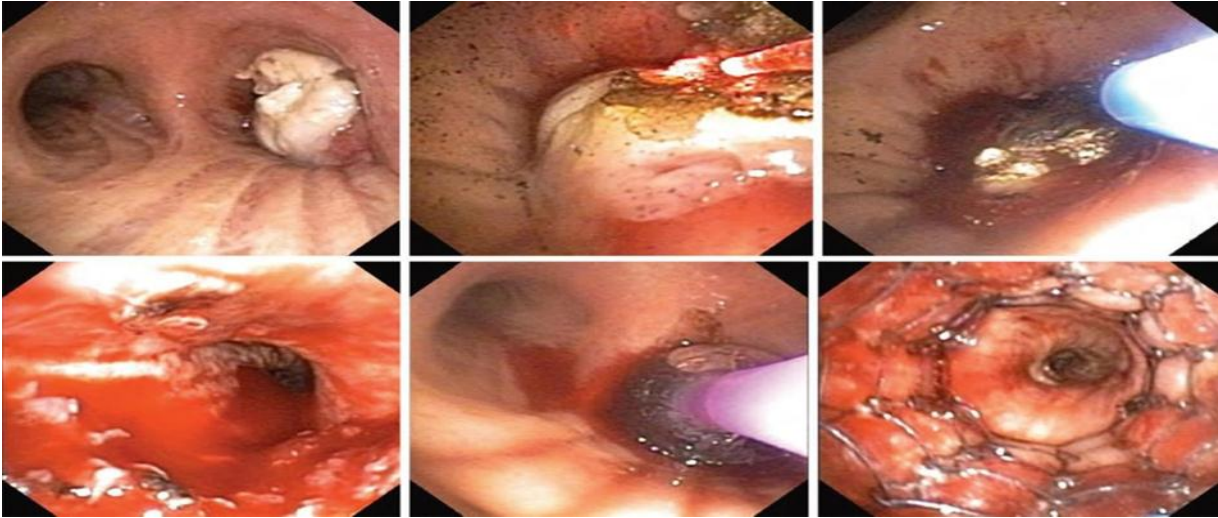
1-Photodynamic therapy (PDT): It kills cancer cells by using a combination of a light sensitizing drug and a very bright light.

2-End bronchial brachytherapy: is used to give an added radiation dose to lung cancers that are located in the major bronchi.

3-Laser therapy.

4-Electrocautery.

5-Cryotherapy



A, Pretreatment. B, Laser photoresection (note the laser fiber). C, Argon plasma coagulation (note the blue argon plasma coagulation catheter). D, After mechanical débridement. E, Balloon dilation. A silicone balloon is fully inflated in the right main-stem bronchus location. F, Stent placement. Note the self-expanding metallic

****Symptomatic treatment:**

- Analgesics for **chest pain**
- Antibiotics for **lung infections**
- Aspiration of **pleural effusion** and pleurodesis with cytotoxic drugs.

Respiratory Failure

Specific Learning Objective:

By the end of the lecture the student will be able to

- ✓ Define RF.
- ✓ Detect types of RF.
- ✓ Recognize diagnosis and management of RF.

Contents:

- ✓ Definition and classification.
- ✓ Pathogenesis and etiology.
- ✓ Clinical picture and blood gas criteria for each type.
- ✓ Management of each type of Respiratory Failure.

Definition: -

- The lungs cannot fulfill their primary function of maintaining adequate gas exchange at rest or during exercise. This results in an inability to maintain normal blood gases, so that the Pa O₂ is low, with or without a rise in Pa CO₂.
- Pa O₂ measured at rest at sea level less than 60 mmHg OR Pa CO₂ above 49mmHg.

Types: -

A- Type I respiratory failure

hypoxemia without CO₂ retention.

B- Type II respiratory failure

hypoxemia with hypercapnia

Causes:

Respiratory failure can occur if there is an abnormality with any component of the respiratory system. Components of the respiratory system include the upper and lower respiratory tracts, the central and peripheral nervous systems, in addition to the chest wall and muscles of respiration.

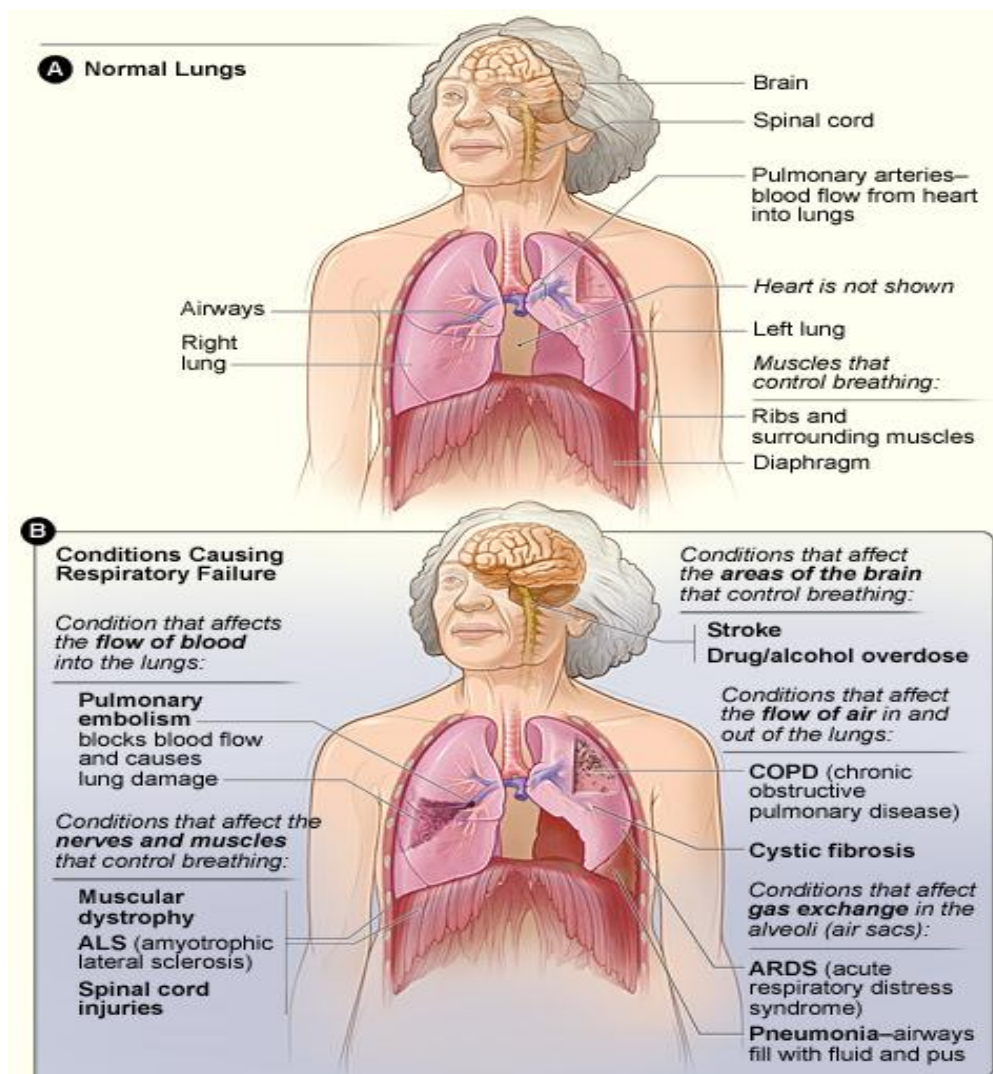
A- Type I (hypoxemic) respiratory failure:

- Diseases that affect the airway (the flow of air in and out of the lungs) including chronic obstructive pulmonary disease (COPD) and bronchial asthma.
- Diseases that affect the lung tissue, including pneumonia, acute respiratory distress syndrome (ARDS) and interstitial lung diseases (ILD).

- Diseases that affect the flow of blood into the lungs (e.g. pulmonary embolism).
- Others e.g. pleura (pneumothorax) and congestive heart failure (pulmonary congestion and pulmonary edema).

B- Type II (hypercapnic) respiratory failure

- The most common cause is chronic obstructive pulmonary disease (COPD).
- Chest-wall deformities (e.g. kyphoscoliosis).
- Respiratory muscle weakness (e.g. Guillain-Barre syndrome).
- Central depression of the respiratory center (e.g. heroin overdose).



Clinical features: -

A- Clinical features of hypoxemia:

1. Central cyanosis
2. Central nervous system
 - Irritability
 - Convulsions
 - Coma, death

3. Cardiovascular system

- Increase heart rate
- Increase COP
- Dilated peripheral vessels
- Cardiac dysrhythmia
- Pulmonary vasoconstriction, Pulmonary hypertension and cor-pulmonal

4. Secondary polycythemia due to increase production of erythropoietin.

B- Clinical features of hypercapnia:

1. Central nervous system

- Irritability → Confusion → Coma

2. Vascular system

- Cerebral vasodilation
- Increase intracranial tension
- Headache
- Papilledema
- Warm flushed skin
- Bounding pulse

3. Sympathetic stimulation

- Tachycardia → Sweating

4. Gastric dilatation and paralytic ileus

5. Headache on waking due to progressive increase in CO₂ retention during sleep

Diagnosis: -

- Arterial blood gases analysis

- Pa O₂ less than 60 mmHg
- Pa CO₂ more than 49 mmHg

- Respiratory failure may be

- Acute as in opiate overdose
- Chronic as in COPD

Treatment: -

- The primary aims of treatment are

1. To maintain a patent airway
2. Ensure adequate alveolar ventilation and oxygenation
3. Treat the primary cause

A- Type I respiratory failure

1. O₂ therapy in high concentration up to 50%
2. Indication of mechanical ventilation
 - Inadequate oxygenation despite of an increasing concentration of inspired O₂
 - Increased Pa CO₂ associated with decreased mental status of increasing fatigue
 - Failure to control secretions

B- Type II respiratory failure

1. Controlled oxygen therapy 24-28% oxygen using venturi mask or nasal prongs at flow of 1-3 L/min

N.B. Patients with COPD dependent on their hypoxic drive to breathing, thus removal of hypoxic drive to breathing with supplemental O₂ result in a decrease in ventilation and increase Pa CO₂

2. Respiratory stimulants as doxapram in patients with acidosis
3. If the patient's hydrogen ion concentration remains less than 7.26, ventilator support should be considered using NIPPV (**non-invasive positive pressure ventilation**)

4. Indication of mechanical ventilation

- Rapid increase in hypercapnia
- Mental confusion
- Tachypnea > 35 breath/min
- Exhaustion

Acute respiratory distress syndrome

Acute respiratory distress syndrome (ARDS) is a form of acute respiratory failure caused by **permeability pulmonary edema** resulting from **endothelial damage** due to a cascade of inflammatory events developing in response to an initiating injury or illness. **ARDS is a serious condition.**

Causes associated with development of ARDS:

ARDS develop in response to variety of injuries or illnesses that affect the lungs, either directly or indirectly.

Direct (Pulmonary) injury	Indirect (Extra pulmonary) injury
Common -Aspiration of gastric content -Severe pneumonia Less Common -Inhalation injury (smoke, chemicals) -Pulmonary contusion -Fat embolism -Near Downing	Common -Systemic sepsis -Major (multiple) trauma Less Common -Acute pancreatitis -Extensive burn -Shock -Multiple blood transfusion -Drug overdose -DIC

Pathogenesis of ARDS:

In ARDS, a cascade of inflammatory events arises over a period of hours from a focus of tissue damage. In particular, activated **neutrophils** aggregate and adhere to endothelial cells, releasing various toxins, oxygen radicals and mediators (e.g. arachidonic acid, histamine, kinins).

This systemic inflammatory response may be initiated by variety of injuries or illnesses and give rise to acute lung injury as one of its earliest manifestations, with the development of endothelial damage and increased alveolar capillary permeability (**inflammation-mediated disruptions in alveolar-capillary permeability**). **The alveoli become filled with a protein rich exudate** containing abundant neutrophils and other inflammatory cells, and the airspaces show a rim of proteinaceous material (**the hyaline membrane**).

Clinical features and investigations:

About 12-48 hours after initiating event (injuries or illnesses), the patient develops **respiratory distress, with increasing dyspnea and tachypnea**. Tachycardia, cyanosis, and chest examination: the crackles are diffuse but sometimes increase basally.

Arterial blood gases show **deteriorating hypoxemia**, which **responds poorly to oxygen therapy**.

Diffuse bilateral infiltrates develop on **chest X ray/ CT chest** in the absence of evidence of cardiogenic pulmonary edema.

Echocardiography: for cardiac evaluation (to exclude cardiogenic pulmonary edema).

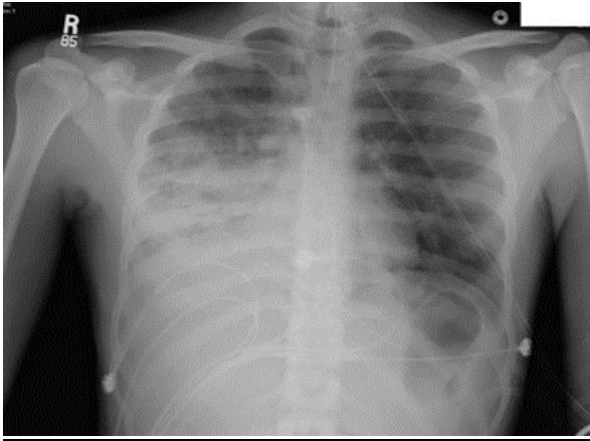
ARDS is characterized by the following features:

- 1-A history of an initiating injury or illness
- 2-Hypoxemia refractory to oxygen therapy
- 3-Bilateral diffuse infiltrates on chest X ray/ chest CT
- 4-No evidence of cardiogenic pulmonary oedema (e.g. pulmonary capillary wedge pressure in case of ARDS **< 18 mmHg**)

NB: The measurement of **pulmonary capillary wedge pressure** reflects left atrial pressure.

In ARDS, it is typically < 18 mmHg, whereas in **cardiogenic pulmonary edema (e.g. left side heart failure), it is elevated > 18 mmHg**.

Pulmonary capillary wedge pressure may be measured by passing a special balloon tipped pulmonary artery catheter (e.g. Swan-Ganz catheter) into a central vein through the right side of the heart to the pulmonary artery. The balloon of the catheter is then inflated and is carried forward in the blood flow until it wedges in a pulmonary capillary.



Therapeutic strategies

Although no specific therapy exists for ARDS, Therapeutic strategies include:

- Treatment of the underlying condition is essential,
- Mechanical ventilation (noninvasive ventilation or invasive ventilation) and
- Adjuvant treatment: e.g.
 - Corticosteroid
 - Inhaled nitric oxide (NO) (it is a potent pulmonary vasodilator)
 - Inhaled synthetic surfactant
- Conservative fluid management.

Pulmonary embolism

Definition:

Pulmonary embolism is a blockage of pulmonary artery by a substance that has moved from elsewhere in the body through the bloodstream. Most pulmonary emboli are multiple, **more commonly in the lower lobes**.

Lobar, segmental, and subsegmental PEs are clots located in the branches of the pulmonary artery corresponding to the anatomical lung segment.

Pulmonary Infarction: The pathological changes which develop in the lung because of pulmonary embolism.

Sources of emboli

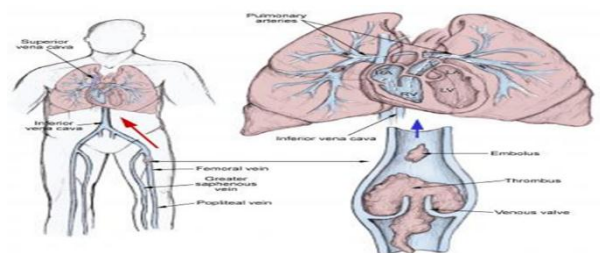
The most common source of PE is deep venous thrombosis of the lower extremity.

Other sources:

- a-the pelvic veins (pregnancy, pelvic infection, prostate disease, or recent pelvic surgery)
- b-Emboli may also originate from upper extremity thrombosis associated with central venous catheters
- c-the renal veins are less common source.

Pathogenesis:

Most PEs originate as thrombi in the deep veins of the lower extremities. Thrombosis begins in areas of decreased flow such as valve cusps and bifurcations and then propagates. Emboli detach from their point of origin and travel through the systemic venous system, through the right sided chambers of the heart, and lodge in the pulmonary arterial system.



Etiology

Three primary influences predispose a patient to thrombus formation; these form the so-called **Virchow triad**, which consists of the following:

- a. Endothelial injury
- b. Stasis or turbulence of blood flow
- c. Blood hypercoagulability

Risk factors for venous thromboembolism

Inherited risk factors:

- a. Factor V Leiden mutation
- b. Prothrombin gene mutation
- c. Antithrombin deficiency
- d. Protein C deficiency
- e. Protein S deficiency

Acquired risk factors:

Major risk factors:

- a. Surgery:** Major abdominal/pelvic surgery, Hip/knee replacement, Postoperative intensive care
- b. Obstetrics:** Late pregnancy, Puerperium
- c. Lower limb problems:** Fracture, Varicose veins
- d. Malignancy:** Abdominal/pelvic malignancies, metastatic
- e. Reduced mobility:** Hospitalization, Institutional care

Minor risk factors:

- a. Cardiovascular:** Congenital heart disease, Congestive cardiac failure,
- b. Estrogen therapy**
- c. Obesity, aging**

Clinical manifestation:

Clinical symptoms

- Dyspnea (most common symptoms)
- Pleuritic chest pain (66%)
- Cough (37%)
- Hemoptysis
- Low grade fever
- Symptoms of DVT as (Unilateral leg pain, Unilateral leg swelling)

physical signs:

- Tachypnea (respiratory rate >20/min)
- Rales
- Accentuated second heart sound
- Tachycardia (heart rate >100/min)
- Fever (temperature >37.8°C)
- Clinical signs and symptoms suggesting thrombophlebitis
- Signs of DVT as (Lower extremity edema)
- Cardiac murmur
- Cyanosis

CLINICAL ASSESSMENT

Wells and Geneva Scoring Systems Used in Risk Assessment for the Diagnosis of Pulmonary Embolism (PE)* Both scoring systems divide patients into groups with low, intermediate, and high clinical probability for diagnosis of PE

LABORATORY TESTS

1-Arterial Blood Gas Analysis: hypoxemia

2-D-Dimer Assay : D-dimer is a plasmin-derived fibrin degradation product.

D-dimer testing has proven to *be highly sensitive but not specific*.

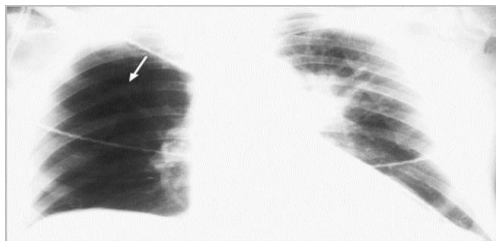
Blood D-dimer assay should only be considered following assessment of clinical probability. D-dimer assay should not be performed in those with high clinical probability of PE.

A negative D-dimer test reliably excludes PE in patients with low or intermediate clinical probability; such patients do not require imaging for VTE.

3-B-Type Natriuretic Peptide Assay, Troponin Assay

Chest Radiographs:

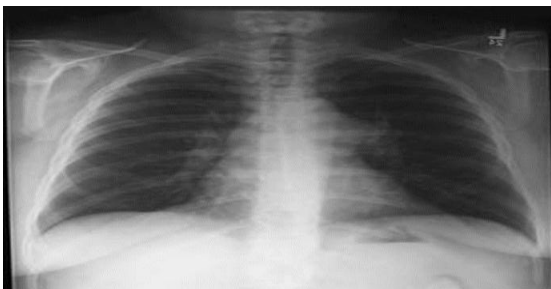
Classic abnormalities include:



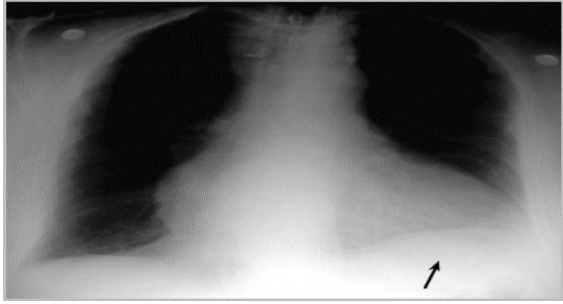
1-Westermark's Sign - focal oligemia (hypoperfusion of one lung secondary to massive pulmonary embolism)



2- Hampton's Hump - wedge shaped density indicates pulmonary infarction



3-Enlarged Right Descending Pulmonary Artery (Palla's sign) usually seen in the acute setting.



4-A diaphragm may be elevated, reflect in volume loss in the affected lung.

5-Infiltration, pleural effusion, atelectasis, rounded nodular opacity.

ECG: usually nonspecific- Rt. Ventricular strain or arterial arrhythmias

The main value of ECG is in excluding other potential diagnoses, such as myocardial infarction or pericarditis.

Venous Compression Ultrasonography (Doppler ultrasound):

Doppler ultrasound imaging frequently is used when the initial laboratory tests for PE are non-diagnostic. *A positive ultrasound test result confirms the need for anticoagulation and obviates the need for further diagnostic studies.*

Echocardiogram: Transthoracic and trans esophageal echocardiography have limited use in the diagnosis of PE. The sensitivity and specificity of these tests are inadequate for diagnosis because the offending emboli is proximal enough to be visualized.

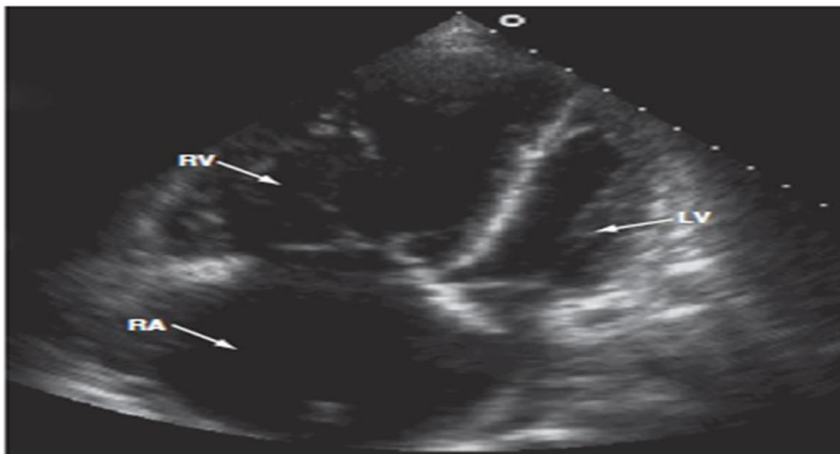
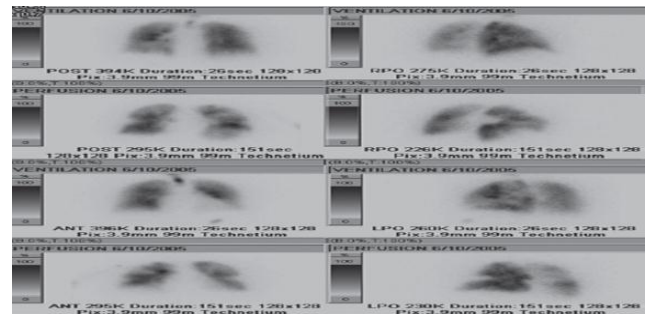


Figure 57-4 Echocardiogram of a patient with markedly dilated right ventricle (RV) and right atrium (RA) secondary to pulmonary thromboembolic disease. The interventricular septum is bowing into and compressing the left ventricle (LV).

Ventilation-Perfusion Lung Scan: For many years, the ventilation-perfusion lung scan was considered the imaging study of choice to evaluate for PE.

Recently, CT pulmonary angiography (CTA) has replaced the ventilation-perfusion lung scan as the predominant diagnostic test. However, the ventilation-perfusion scan maintains an important place in the evaluation of patients for thromboembolic disease **who have contraindications to CTA such as renal insufficiency or contrast allergy.**

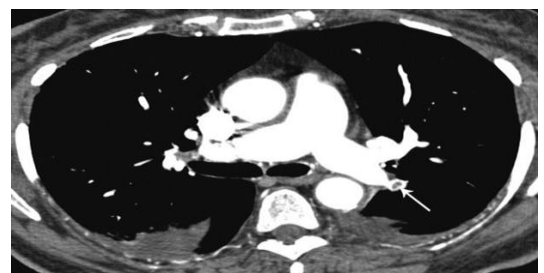


Computed Tomography Angiography (CTA):

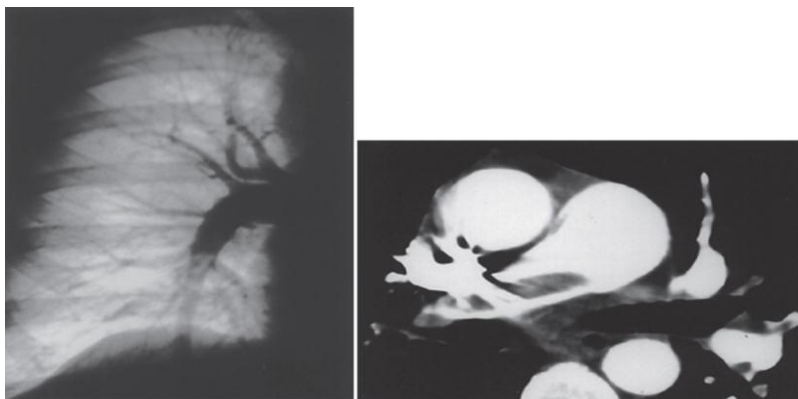
CT pulmonary angiography (i.e., CTA) has become a favored study for the diagnosis of PE.

CTA provides several potential advantages over other imaging modalities in the diagnosis of PE, including:

- (1) direct visualization of the embolus
- (2) the ability to assess for other potential causes for the patient's complaints such as pneumonia
- (3) imaging algorithms that scan through the pelvis and lower extremities, as well as the chest, allowing simultaneous evaluation for PE and for DVT



Pulmonary angiography: The diagnosis of acute PE is based on direct evidence of a thrombus in two projections, either as a filling defect or as amputation of a pulmonary arterial branch. Thrombi as small as 1–2 mm within the sub-segmental arteries can be visualized



Treatment:

Indication for hospitalization:

- 1-Older patients who may have less cardiopulmonary reserve, or significant coexisting illnesses
- 2-Those who may not be able to follow instructions or have adequate follow-up.
- 3-Hypoxemia, hypotension, hemodynamic instability

1-Heparin

Action: *The major anticoagulant effect of heparin is to reduce thrombus propagation and prevent embolic recurrence.* Choices include either intravenous unfractionated heparin (UFH) or subcutaneous low molecular weight heparin (LMWH) preparation.

Unfractionated heparin (UFH) should be considered (a) as a first dose bolus, (b) in massive PE, or (c) where rapid reversal of effect may be needed.

Dosing regimen: using an initial intravenous bolus of 80 units of heparin per kilogram followed by a continuous infusion initiated at 18 U/kg/h. *The heparin drip is adjusted based on monitoring of the aPTT, drawn 6 hours after the initial bolus dose, then 6 hours after each dose adjustment, with a target aPTT ratio of 1.5 to 2.5.*

Antidote: protamine sulphate

Duration of heparin therapy: for at least 5-7 days

The main complication of anticoagulation is intracranial bleeding may be life-threatening.

2-LMWH

LMWH preparations have displaced unfractionated heparin as the anticoagulant of choice in PE

Advantages of LMWH compared with UFH include

- (1) longer half-life and ease of use
- (2) ability to consistently achieve early therapeutic anticoagulation
- (3) no need to monitor anticoagulant effects
- (4) reduced incidence of major bleeding complications.

Dose regimen: Enoxaparin: 1 mg/kg every 12h

3-FACTOR Xa INHIBITORS

1. Fondaparinux:

Subcutaneous administration. Has a relatively long half-life (17 hours) and can be administered once daily

2. New oral anticoagulant (Rivaroxaban):

Their excellent bioavailability, rapid onset, predictable effect, and ease of use will greatly facilitate outpatient therapy. **Close monitoring of coagulation levels, as is required with the vitamin K antagonists, is unnecessary**, and the risk of over- or under anticoagulation is negligible. **One potential complicating factor is the lack of a specific antidote for rapid reversal of anticoagulation.**

4-Thrombolytic therapy

Thrombolytic drugs cause direct lysis of thrombi by increasing plasmin production through plasminogen activation. **Thrombolysis is the first line treatment for massive PE.**

(e.g., Streptokinase, alteplase, and urokinase). thrombolytic therapy recommended the use of thrombolytic therapy in patients with acute PE associated with hypotension and in a subgroup of patients who are hemodynamically stable at presentation but are at high risk for hypotension.

5-Adjuvants therapy: Oxygen therapy, Antibiotics (in case of association with infection), Rest in bed

6-Pulmonary embolectomy:

Consider surgical embolectomy:

- a. Patients with persistent hypotension, shock, or cardiac arrest
- b. Who either failed thrombolysis or have a contraindication to thrombolytic management.

LONG-TERM MANAGEMENT

Warfarin: is *Vitamin K antagonist*, is generally used for long-term treatment of venous thromboembolism because of its proven efficacy. Warfarin inhibits activation of coagulation factors II, VII, IX, and X as well as proteins C and S.

To maximize efficacy while minimizing side effects, an INR range between 2 and 3 is recommended for most patients

New oral anticoagulant: new oral anticoagulant, has been used successfully for long-term management of patients with venous thromboembolism. As noted previously, *rivaroxaban* used for initial and long- term treatment of PE.

Pulmonary hypertension

Definition

Pulmonary hypertension (PH) is defined by a mean pulmonary arterial pressure (mPAP) of **20 mmHg at rest** as assessed by right heart catheterization (RHC).

Epidemiology

- Prevalence of Pulmonary hypertension (PH) is 1% of the global population.
- Due to the presence of cardiac and pulmonary causes of PH, prevalence is higher in individuals aged **above 65 years**.
- **LHD (left sided heart diseases)** is the **most prevalent cause** of PH, accounting for 65–80% of cases
- **Lung disease**, especially chronic obstructive pulmonary disease (COPD), is the **second** most common cause.

Clinical classification of pulmonary hypertension

GROUP 1: Pulmonary arterial hypertension (PAH)

1.1 **Idiopathic** (Non-responders or acute responders at vasoreactivity testing)

1.2 Heritable

1.3 Associated with drugs and toxins e.g Aminorex

1.4 Associated with: **Connective tissue disease**, HIV infection, Portal hypertension, congenital heart disease and Schistosomiasis

1.5 PAH with features of venous/capillary (PVOD/PCH) involvement

1.6 Persistent PH of the newborn

GROUP 2: PH associated with left heart disease

2.1 **Heart failure: (with preserved or reduced ejection fraction)**

2.2 **Valvular heart disease**

2.3 Congenital/acquired cardiovascular conditions leading to post-capillary PH

GROUP 3: PH associated with lung diseases and/or hypoxia

3.1 **Obstructive lung disease e.g emphysema**

3.2 **Restrictive lung disease**

3.3 **lung disease with mixed restrictive/obstructive pattern**

3.4 Hypoventilation syndromes

3.5 Hypoxia without lung disease (e.g. high altitude)

GROUP 4: PH associated with pulmonary artery obstructions

4.1 Chronic thrombo-embolic pulmonary hypertension (CTPH)

GROUP 5: PH with unclear and/or multifactorial mechanisms.

5.1 Hematological disorders, Systemic disorders and Metabolic disorders

Pathophysiology

Each group has different characteristic pathological features.

➤ Pulmonary arterial hypertension (Idiopathic type):

An imbalance between nitric oxide (NO), prostacyclin (potent vasodilators) and thromboxane A₂ (a potent vasoconstrictor), resulting in vascular remodeling (persistent structural changes) with increasing pulmonary arterial pressure.

➤ Pulmonary hypertension due to left sided heart diseases:

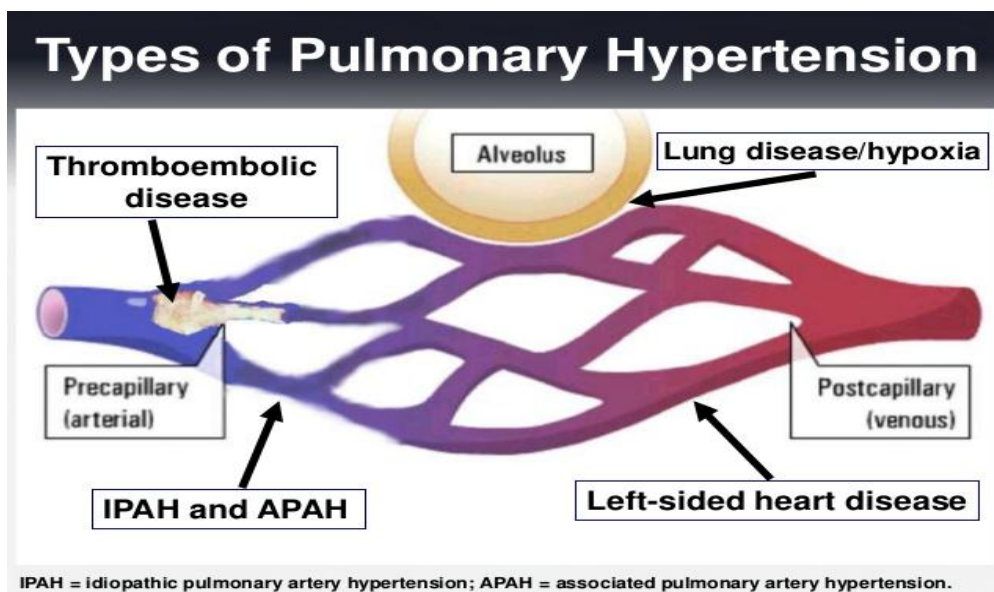
Left sided heart diseases (heart failure, valvular heart diseases) increase pulmonary venous congestion, leading to increase pulmonary venous pressure (**post capillary-PH**).

➤ Pulmonary hypertension due to lung diseases:

- Chronic lung diseases associated with **chronic hypoxemia** results in pulmonary **vasoconstriction** which leads to PH.
- A significant proportion of COPD patients will develop PHT, possibly up to 25%.
- Mechanisms of PH in COPD patients are: -
 - ✓ Chronic hypoxia.
 - ✓ Emphysematous changes lead to destruction of the vascular bed (capillary loss)
 - ✓ Cigarette smoke may have a toxic effect on the intrapulmonary vessels

➤ Chronic thrombo-embolic pulmonary hypertension (CTPH):

- Unresolved or incompletely resolved Pulmonary thromboembolic result in formation of endothelialized fibrotic obstructions of the pulmonary vascular bed lead to increase of pulmonary arterial pressure (**precapillary-PH**).
- It is an under-diagnosed complication of acute PE.
- It should be suspected in patients who remain dyspneic after pulmonary embolism (PE).



Presenting features

Symptoms

- Symptoms of PH are primarily due to right ventricle (RV) dysfunction.
- **Exertional dyspnea** is the cardinal symptom.
 - Chest pain (anginal pain in character)
 - Syncope due to a fall in systemic blood pressure (BP) on exercise
 - Palpitations
 - Lower limb swelling.
- Symptoms of the underlying cause of PH e.g symptoms of COPD, interstitial lung diseases, left heart diseases, connective tissue diseases,....etc

Signs

➤ Signs of Pulmonary hypertension

- **Inspection:** Pulsation in the 2nd left intercostal space.
- **Palpation:** Diastolic shock (palpable accentuated pulmonary component of the second heart sound).
- **Percussion:** Dullness in the 2nd left intercostal space.
- **Auscultation:**
 - ✓ Accentuated 2nd heart sound.
 - ✓ Ejection clicks.
 - ✓ Ejection systolic murmur of tricuspid regurgitation
 - ✓ Early diastolic murmur of pulmonary regurgitation.
 - ✓ S₄

➤ Signs of RV failure

- Congested pulsating internal jugular veins.
- Tender hepatomegaly.
- Ascites.
- Bilateral lower limb edema.

➤ Signs pointing to the underlying cause of PH

- Digital clubbing: Interstitial lung disease, bronchiectasis, congenital cyanotic heart diseases.
- Auscultatory findings (crackles or wheezing, murmurs): lung or heart disease.
- Signs of deep venous thrombosis (DVT): CTEPH
- Signs of connective tissue diseases.

Investigations

The investigations aim to: -

- Make a diagnosis of PH.
- Investigate any possible underlying cause.

1- **Chest radiograph (CXR):** may show

- Enlarged pulmonary arteries, with pruning (loss) of peripheral vessels.

- Signs of the underlying cause of PH, such as left sided heart diseases or lung disease, may be found.
- **Normal chest X-ray does not exclude PH.**

2- ECG:

- Peaked P wave (P pulmonale) due to right atrial dilatation.

3- Arterial blood gases (ABG):

- If PH is due to underlying lung diseases: ABG findings will show type 1 or type 2 respiratory failure, according to **type** and **severity** of the lung disease.
- **Normal blood gases don't exclude PH.**

4- Pulmonary function tests (PFTs):

- PFTs are abnormal if PH is due to underlying lung disease, for example **obstructive pattern** in COPD, while **restrictive pattern** in interstitial lung disease.

5- High resolution computed tomography (HRCT):

- To detect underlying lung diseases.

6- Echocardiogram (echo):

- The most useful **screening tool** in PHT.
- Detection of the cause.
- It shows dilated or hypertrophied RV, tricuspid regurgitation (TR), providing estimate of PAP (pulmonary artery pressure).

7- Right heart catheterization (RHC):

- **The 'gold standard' test to confirm the diagnosis**
- It is used to assess PAP and cardiac output by using a Swann–Ganz catheter

Treatment:

➤ Treatment of the cause:

treatment of the underlying cardiac or respiratory diseasesetc

➤ General management

- 1- **Anticoagulation:** All patients with CTEPH and most with IPAH should be commenced on anticoagulation.

- 2- **Long-term O₂ therapy:** O₂ therapy may reduce any further rise in PAP and improves exercise tolerance.
- 3- **Diuretics:** Right HF is associated with systemic venous congestion so we use diuretics such as loop diuretics.
- 4- **Immunization:** it is recommended that patients with PAH should be vaccinated at least against influenza, Streptococcus pneumonia and SARS-CoV-2 to reduce risk of recurrent respiratory infection.

➤ **Pharmacological treatments (Vasodilators).**

1- **Calcium channel blockers (CCBs)**

e.g. **nifedipine**.

2- **Phosphodiesterase type-5 inhibitors (PDE-5)**

- These augment the vasodilatory effects of nitric oxide, causing pulmonary vasodilatation e.g. **sildenafil**

3- **Prostanoids group.**

- Include prostacyclin, which is a potent vasodilator e.g. **Epoprostenol**.

4- **Endothelin receptor antagonists.**

- Endothelin is a powerful vasoconstrictor, so endothelin receptors antagonist drugs act as a vasodilator e.g. **Bosentan**.

➤ **Surgical treatments**

1- **Pulmonary thromboendarterectomy**

- This is the surgical removal of organized thrombotic material.
- Is the treatment of choice in CTEPH for **proximal** obstructive organized thrombotic material.

2- **Lung transplantation**

- Lung transplantation remains an important treatment option for patients with PAH **refractory to (resistant to)** optimized medical therapy.
- Improves survival and quality of life.

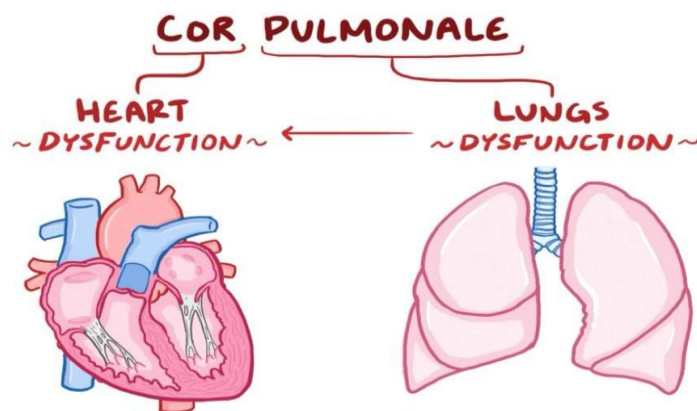
Corpulmonale

Definition:

Corpulmonale is **defined** as an alteration in the **structure** (e.g hypertrophy and/or dilatation) and **function** of the right ventricle (RV) of the heart **caused by** a primary disorder of the respiratory system **resulting in** pulmonary hypertension, “*after exclusion of left sided heart failure or congenital heart diseases*”.

N.B:-

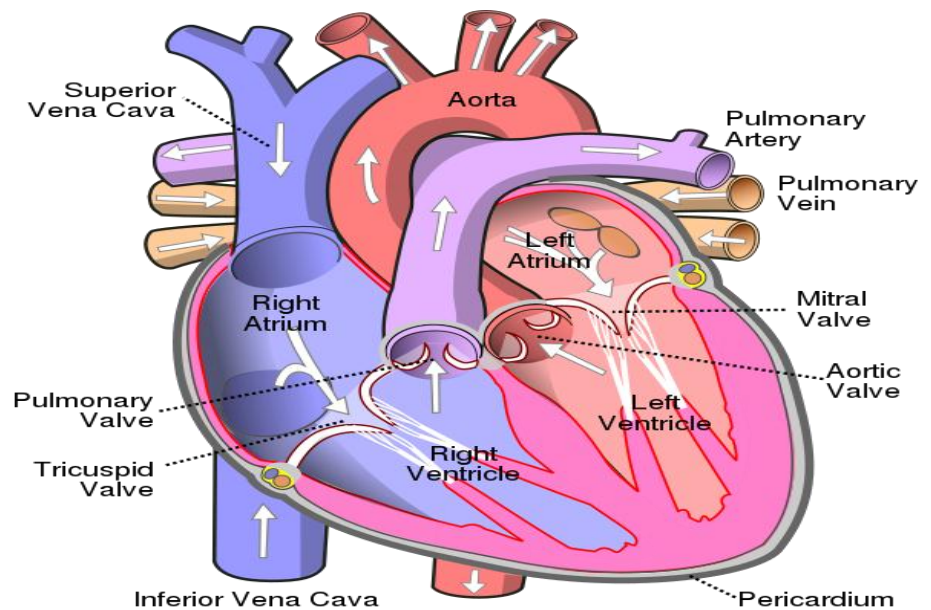
- Right-sided heart failure secondary to left-sided heart failure, or congenital heart disease **is not considered** cor pulmonale.
- Cor pulmonale is a Latin word that means "*Pulmonary heart*"



Pathophysiology:

- 1- Lung disease causes hypoxia
- 2- Hypoxia **acutely** causes pulmonary **vasoconstriction** and **chronically** causes pulmonary vascular **remodeling (structural changes)**, that causes elevation of pulmonary vascular resistance which rises the pulmonary arterial pressure and leads to “pulmonary hypertension”.
- 3- Pulmonary hypertension leads to increase “pressure afterload” on RT ventricle
- 4- This firstly leads to “**RT ventricular hypertrophy**” which is called (*Compensated corpulmonale*).

5- Lastly, **RT ventricle dilatation and failure** occurs which is called (*Decompensated cor-pulmonale*)



Etiology:

1- **Acute Corpulmonale** :-

- Acute massive pulmonary embolism.

2- **Chronic Corpulmonale**:-

- Parenchymal lung diseases e.g COPD, ILD.

N.B: **COPD** is the most common cause of chronic cor pulmonale.

- Pulmonary vascular diseases e.g primary pulmonary hypertension, chronic thromboembolic pulmonary hypertension and Bilharziasis.
- Thoracic wall diseases: kyphoscoliosis, obesity hypoventilation syndrome.

N.B: kyphoscoliosis or obesity causes corpulmonale by their **extrapulmonary restriction** of the lungs.

Clinical features:-

- 1- Features of the underlying cause e.g. COPD, Bronchiectasis, Bilharziasis, etc.
- 2- Epigastric pulsation (RT ventricular hypertrophy).
- 3- Congested pulsating neck veins.
- 4- Lower limb oedema.
- 5- Enlarged, tender liver.

Investigation:-

- 1- **CXR**— enlarged pulmonary arteries/ underlying lung disease.
- 2- **ECG**— may indicate right axis deviation (RAD), p pulmonale (right atrial hypertrophy), and right bundle branch block (RBBB).
- 3- **Echocardiography**— dilated or hypertrophied RV, tricuspid regurgitation (TR), providing estimate of PAP (pulmonary artery pressure).

Management

- 1- **Treatment of the cause** is the main treatment in patients with cor pulmonale.
- 2- Long- term oxygen therapy improves survival.
- 3- Diuretics.
- 4- Elevate legs when sitting for patients with L.L edema.

Pleural effusion

Definition of pleural effusion: it is the collection of fluid in the pleural sac.

Normally, there is a small amount of pleural fluid that lubricates the space and facilitates normal lung movements during respiration.

-Normally pleural fluid is→ **in dynamic state**

-Pleural fluid originates from the vasculature of parietal pleura surfaces and is absorbed back by lymphatics of parietal pleura. Also, visceral pleura has a role in fluid absorption

-The balance of fluid is maintained by the oncotic and hydrostatic pressure and lymphatic drainage; disturbances in any one of these systems can lead to accumulation of the pleural fluid .

Etiology of pleural effusion:

1- **Transudative pleural effusion (hydrothorax)**

2- **Exudative pleural effusion**

NB: although there are more than 60 recognized causes of PE, 4 etiologies account for about 75% of cases: **heart failure, pneumonia, cancer, and tuberculosis.**

Pleural fluid is classified as a transudate or exudate based on Light's criteria. Pleural fluid is considered an exudative effusion if **at least one of the criteria is met:**

1-Pleural fluid protein **more than 3 g/dL**

2-Pleural fluid protein/serum protein ratio of **more than 0.5**

3-Pleural fluid lactate dehydrogenase (LDH)/serum LDH **ratio of more than 0.6**

4-Pleural fluid LDH is **more than two-thirds** of the upper limits of normal laboratory value for serum LDH.

5-Absolute pleural fluid LDH **more than 200 IU/L**

Transudative pleural effusion (hydrothorax)

Mechanism of transudative pleural effusion:

1-Formed through a normal pleural capillary membrane.

2-Occurs due to disturbance of the relation between capillary hydrostatic pressure and colloid osmotic pressure so that fluid formation at pleural surface exceeds the fluid reabsorptive capacity.

3-Characterized by **small protein** content.

4-In most cases, effusions are **bilateral**.

Causes of transudative pleural effusion:

A- Increased hydrostatic pressure:

1-***Congestive heart failure*** (the most common cause of effusion) (the most common cause of transudative effusion): it starts unilateral on the right side but it is the most frequent cause of bilateral effusion.

2-***Pulmonary embolism***

3-Constrictive pericarditis

4-Restrictive cardiomyopathy

B-Decrease capillary osmotic pressure (hypoproteinemia):

Liver cirrhosis, nephrotic syndrome, chronic renal failure ect

C-Transmission of fluid from the peritoneum:

The ascetic fluid traverse the diaphragm through the normal pleuroperitoneal communications causing pleural effusion, for example: peritoneal dialysis....ect

D- Other rare causes such as **Meigs syndrome** (Triad of benign ovarian fibroma, ascites and pleural effusion which may be a **transudate or exudate**)

Exudative pleural effusion

Mechanism of exudative pleural effusion:

-Increased mesothelial and capillary permeability or impaired lymphatic drainage usually causes exudates. It has a **high protein** content

Causes:

The commonest causes are parapneumonic effusion, TB, malignancy, pulmonary embolism and infraction, and connective tissue diseases.

A-Malignant pleural effusion:

-Metastatic malignant tumors are more common than the primary pleural tumors (as mesothelioma)

-Common causes are lung cancer, breast cancer, gastrointestinal cancer, lymphoma

B-Infections:

1-Parapneumonic effusion:

Definition: any pleural effusion associated with **bacterial pneumonia, lung abscess, or bronchiectasis**. **Parapneumonic effusion is the commonest cause of exudative effusion.**

- At least 40-60% of patients with bacterial pneumonia will develop a pleural effusion of varying severity.
- Parapneumonic pleural effusion, starts out as reactive effusion precipitated by pulmonary infections (uncomplicated effusion).
- Progress from **uncomplicated** (free flowing, high WBC) to **complicated** effusion (*pleural pH* <7.2, LDH >1000, glucose <60 mg/dL, *loculations*) to **empyema** (frank pus or bacteria detected in Gram stain and /or culture of the pleural fluid)

2-Tuberculosis PE:

Pleural effusion may occur as a result of:

a- Hypersensitivity to TB bacilli

b- Actual infection of the pleura by TB bacilli (rupture of TB cavity into the pleura in case of post primary pulmonary TB)

3- Viral infections: rare, usually diagnosed by exclusion.

4- Fungal infections: very rare, occur in immunosuppressed patients.

5- Subphrenic infections: subphrenic abscess.

6- Liver abscess: mostly right sided effusion (for example, amebic liver abscess).

C- Pulmonary embolism and infarction: Pleural fluid caused by pulmonary emboli is usually exudative but occasionally transudative. 20–55% of patients with pulmonary embolism have a pleural effusion.

D- Connective tissue diseases: Systemic lupus erythematosus (SLE), rheumatoid arthritis

E-Other causes:

1-Pancreatitis: usually left sided effusion

2-Drugs: amiodarone, methotrexate, phenytoin

3-Eosinophagic perforation: eg. during upper GIT endoscopy in case of esophageal cancer

4-Dressler syndrome consists of inflammatory pericarditis and pleurisy following a myocardial infarction, cardiac surgery, or cardiac trauma (**post pericardiotomy syndrome**). It's believed that, this inflammation occurs as a result of immune system response to damaged heart tissue or pericardium.

Clinical manifestations (effusion + cause):

Symptoms:

1-No symptoms: small effusions are often symptomless

2-Pleuritic pain:

- ✓ **Pleuritic chest pain** is characterized by (sharp/stitching, localized, and made worse by deep inspiration or coughing, sneezing....)
- ✓ **Pleuritic chest pain** is caused by inflammation of the **parietal** pleura and can be triggered by a variety of causes (bacterial infection, such as pneumonia& pulmonary embolism....)
- ✓ **Pain diminishes in the intensity** as the fluid accumulates and separates the pleural layers (falsely implying an improvement in condition).

3-Dyspnea: It is the usual symptom of large effusion especially if rapidly accumulated.

4- Cough: Dry cough is frequently present especially if the fluid has accumulated rapidly, but cough is productive in cases such as bacterial pneumonia

5-Symptoms of the cause

Signs (pleural effusion, cause):

Signs of pleural effusion **depend on the size and site** of the pleural effusion.

1-Inspection:

- 1-Restriction of respiratory movement (same side)
- 2-Bulge (same side)

Retraction → if there are associating malignancy with proximal bronchial obstruction or fibrosis in the lung in the same side.

2-Palpation:

- 1-Diminished respiratory movement (same side)
- 2-Diminished TVF (same side)
- 3-Shift of the trachea to the opposite side (massive effusion)

3-Percussion: Stony dullness on percussion (same side)

4-Auscultation:

1-Diminished or absent breath sound (same side)

2-Decreased vocal resonance (same side)

3-Sometimes, bronchial breathing is heard at the upper level of the fluid

5-signs of the cause

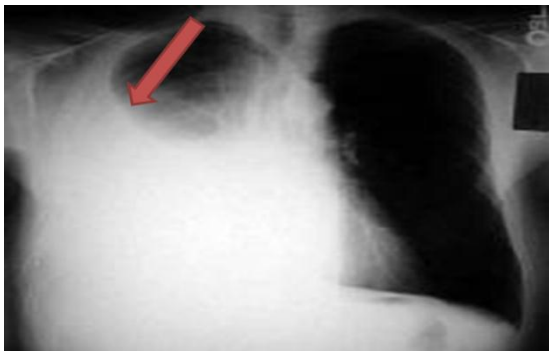
Investigations:

1-Plain X-ray chest: Postero-anterior (PA) view, lateral view

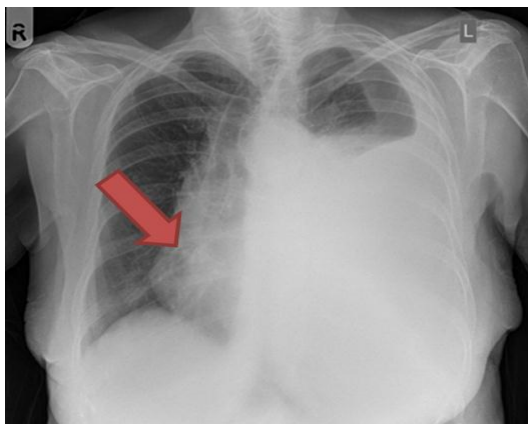
To detect: radiological picture of pleural effusion, cause and complications (such as iatrogenic pneumothorax,)



1-Blunting of the costophrenic angle
due to free fluid even in small amount.

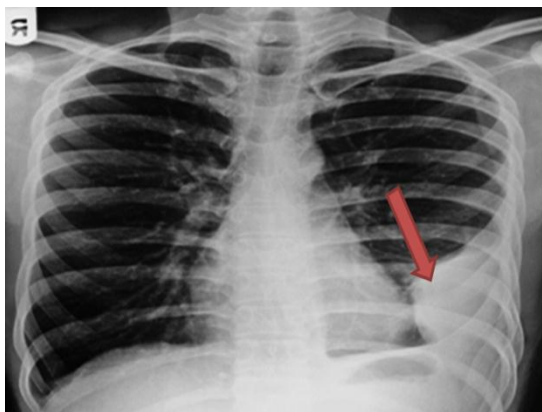


2-Larger amount of effusion →
homogenous opacity obliterating the costophrenic angle and raised towards the axilla.



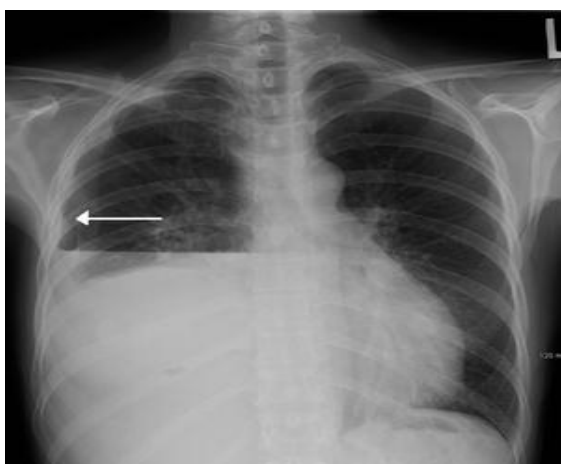
3-Massive pleural effusion causes
massive opacification of one hemithorax with shift of the mediastinum to the other side

➤ except if there is underlying lung collapse due to bronchial obstruction → *shift of mediastinum towards the ipsilateral side*



4-Loculated effusion.

(NB: loculated effusions occur when there are adhesions between visceral and parietal pleurae)



5-Hydropeumothorax shows the characteristic fluid level.

2-Pleural fluid aspiration (thoracocentesis)

Indications:

- 1-Diagnostic aspiration for analysis to detect the cause of effusion.
- 2-Therapeutic aspiration to:
 - a-Relieve dyspnea caused by effusion
 - b-Drainage of parapneumonic effusion

Complications:

- 1-Iatrogenic pneumothorax (commonest complication)
- 2- Introduction of infection
- 3- Re-expansion pulmonary edema
- 4- Bleeding/ Hemothorax.
- 5-Injury of the diaphragm, lung, spleen or kidney

Items of pleural fluid analysis:

- | | |
|----------------------------|---|
| 1- Macroscopic appearance | 2 - Microscopic appearance |
| 3- Biochemical examination | 4- Immunological study |
| 5- Tumor markers | 6- Bacteriological examination |
| 7- PCR for diagnosis of TB | 8- Cytological examination of the fluid |

A- Macroscopic appearance:

1-Transudate fluid is clear, with only a slightly yellow tint.

2-Exudate fluid is straw colored and may be turbid if the cell count is high.

3-Hemorrhagic effusion (uniform blood staining) is usually due to pleural malignant tumors, pulmonary embolism and infraction

NB. Pus → empyema

Milky fluid → chylothorax

Frank blood → hemothorax → (e.g. trauma)

B- Microscopic examination:

1-Pleural fluid total leucocytic count: Marked increase with exudative effusion

2-Differential leucocytic count in pleural fluid:

- **Polymorph** predominant effusion in case of parapneumonic effusion

- **Lymphocyte** predominant effusion in case of TB, malignancy, collagen diseases,ect

C-Biochemical examination of pleural fluid (PF):

1-Pleural fluid Protein:

<3g/dl with transudative effusion

>3g/dl with exudative effusion

2- PF pH: normally pH of the pleural fluid equal to the plasma pH.

* pH <7.2 → indicate complicated pleural effusion with high risk of empyema development **so** drainage of the fluid is indicated.

* pH <6.8 → indicates empyema **so** drainage is indicated

3- PF Lactic acid dehydrogenase (LDH):

- * Normal PF LDH is < 50% plasma level LDH
- * Increase with exudative effusion (such as parapneumonic effusion).

4- PF Glucose:

- * Normally pleural fluid glucose content is equal to that in the plasma.
- * Decrease pleural fluid glucose content detected in case of parapneumonic effusion, TB, malignancy, rheumatoid arthritis

5-High level of PF adenosine deaminase (ADA):

Level of ADA in pleural fluid >40 IU/L can indicate TB pleural effusion

6-High level of pleural fluid Amylase:

High level of *pancreatic* amylase with pancreatitis

High level of *salivary* amylase with esophageal perforation

D-Immunological study: for diagnosis of autoimmune diseases:

Rheumatoid factor, antinuclear antibody...ect

E-Tumor markers: according to suspected type of malignancy

F-Bacteriological Examination:

1-Gram stain for PF

and culture (for aerobic and anaerobic bacteria) and sensitivity

Detection of bacteria by stain and/or culture diagnose empyema

2-Direct smear and culture for TB bacilli: Samples can be used are **pleural fluid, sputum, pleural biopsy (PB) sample**

NB: Negative pleural fluid or sputum smear for TB bacilli does not exclude diagnosis of TB effusion

3-Direct smear and culture for other organisms (if suspected) e.g. fungal infection....ect

G- PCR for diagnosis of TB:

Used samples pleural fluid, sputum, pleural biopsy (PB)

H-Cytological examination of the pleural fluid for malignant cells which indicates tumor involvement of the parietal or the visceral pleura

3- Chest sonography: (simple, inexpensive):

- 1-**Confirm** diagnosis of effusion
- 2-Detection of loculated effusion
- 3-Detect the etiology of effusion (malignancy.)
- 4- Sonographic guided diagnostic& therapeutic aspiration
- 5- Sonographic guided biopsy.
- 6- Help in positioning of chest tube (in case of loculated effusion)

4- CT chest:

- 1-Detection of effusion 2- Detection of cause of effusion (mass, pneumonia)
- 3- detection of complications (iatrogenic pneumothorax...)
- 4-CT guided biopsy

5- Biopsy for histopathology:

1-Pleural biopsy (percutaneous pleural biopsy):

- Sonographic guided or CT guided** needle biopsy of the pleura
- It is **indicated if** the pleural fluid analysis failed to establish the etiology of exudative effusion.

2-Thoracoscopy and thorascopic biopsy:

Thoracoscopy should be considered when less invasive investigations have failed to detect the diagnosis

3-Thoracotomy and open biopsy are considered when less invasive investigations have failed to detect the diagnosis

6- Other investigation→ to detect the cause:

- 1-Abdominal sonography...(to detect liver cirrhosis, liver abscess...)
- 2- Echocardiography
- 3-Bronchoscopy → for diagnosis of suspected lung cancer

NB: Routine flexible bronchoscopy is not recommended for all cases of pleural effusion

Treatment of pleural effusion: Depends on the cause of the effusion:

- 1- Treatment of cause
- 2- Treatment of effusion

Treatment of parapneumonic effusion:

a- Treatment of the cause (infection) → by proper antimicrobial agents (proper dose, duration):

Start empirical antibiotic treatment and modify according to the result of culture and sensitivity

- **Duration of treatment** for self-resolving **uncomplicated** bacterial parapneumonic effusions, treatment may last one to two weeks,
- while two to three weeks for a **complicated** parapneumonic effusion
- and four to six weeks for **empyema**.

b- Treatment of parapneumonic effusion:

1- Nonspecific treatment:

Minimal amount of parapneumonic effusions resolve without specific therapy directed towards the effusion (only treatment of the cause).

2- Therapeutic repeated pleural fluid aspiration (thoracocentesis):

- For treatment of considerable amount of effusion
- It is used if the fluid is accessible and thin to allow adequate removal of the fluid.
- Sonographic guided aspiration can be used

3-Intercostal tube drainage (underwater seal):

Indicated if pleural fluid aspiration failed due to presence of thick fluid or loculations

4-In patients with complicated PE or empyema, intercostal tube drainage with intrapleural fibrinolytic therapy (streptokinase, or urokinase) was associated with a reduced need for surgery and decrease treatment failure.

Indication: if simple intercostal tube drainage failed.

5-Video-assisted thoracoscopy,

Indication: It is an option for patients with septated effusion if chest tube failed as it can be used for pleural debridement (helps control sepsis and allow subsequent lung re-expansion) and place chest tube under vision.

6- Decortication: indicated for treatment of **organizing empyema** with extensive pleural fibrosis

Treatment TB effusion:

Same treatment regimen as pulmonary TB.

Steroids treatment is indicated with anti TB to speed re-absorption of fluid and prevent pleural fibrosis.

Treatment of Malignant effusion:

a- Treatment of effusion:

Treatment of effusion depends on the rate of accumulation, prognosis and general condition of the patient:

1-Repeated therapeutic pleural fluid aspiration

Used to relieve dyspnea, is indicated in debilitated patient in whom short survival is expected due to late malignancy.

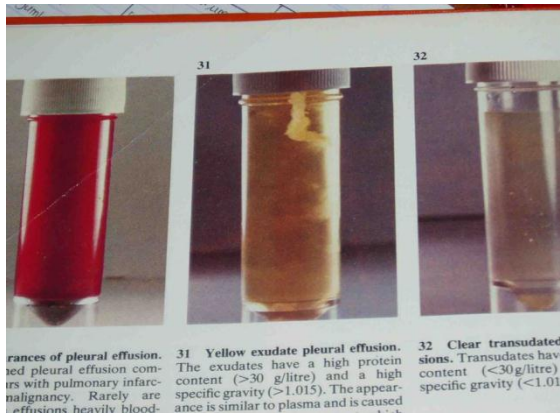
2-Chest tube (intercostal chest tube) drainage and chemical pleurodesis:

Indication: It indicated when effusion is rapidly accumulated and the underlying lung can be fully re-expanded.

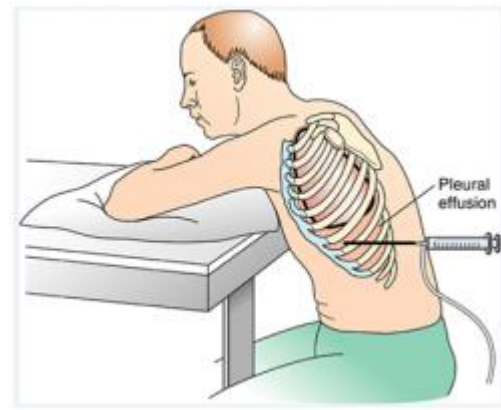
Mechanism: It induces a sterile pleuritis and adhesion of the two layers of the pleura → obliterating the pleural space and preventing formation of further effusion.

Sclerosing agents include bleomycin, concentrated glucose or talc

b- Treatment of malignancy: chemotherapy, radiotherapy (according to type of malignancy)



Macroscopic appearance of Pleural fluid



Pleural fluid aspiration



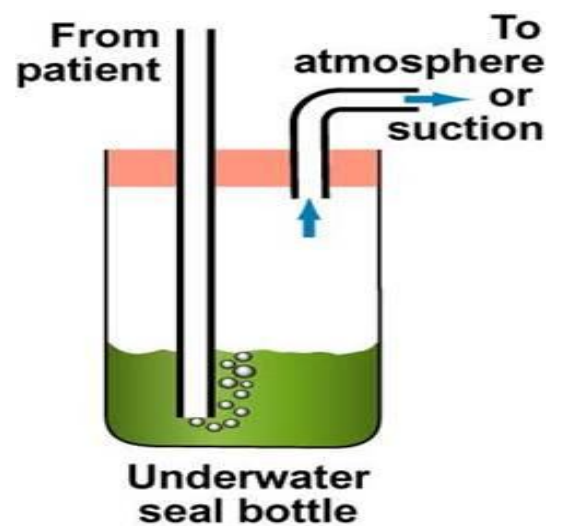
Sonographic guided pleural fluid aspiration



Purulent pleural fluid



Intercostal tube



Empyema

Definition of pleural empyema: Collection of pus in the pleural cavity, or presence of microorganism in the pleural fluid.

- It is not a primary disease, it is a complication of other diseases

Causes (local, systemic):

✓ **Local causes:**

1-Chest wall causes:

- Osteomyelitis of ribs, thoracic vertebrae, thoracic wall abscess,
- Penetrating wounds (stab wound...)

2-Pleural causes: secondary to bacterial infection of effusion, hemothorax

3-Pulmonary causes: pneumonia, lung abscess, bronchiectasis, pulmonary TB

4-Sub-diaphragmatic causes: liver abscess, subphrenic abscess,

5-Iatrogenic causes: pleural fluid aspiration, pleural biopsy, lung biopsy, esophageal perforation during esophagoscopy, post-pneumectomy, post-thoracotomy

✓ **Systemic causes:** septicemia

Bacteriology:

1-Aerobic Gram-positive organisms: *e.g.* Staphylococcus aureus, Streptococcus pneumoniae

2-Aerobic Gram-negative organisms: *e.g.* Klebsiella, Pseudomonas, and Hemophilus species

3-Anaerobic organisms: *e.g.* Bacteroides

4-M. Tuberculosis

4-Others: ...for example, amebic infection [as a complication for Amebic liver abscess which is caused by **Entamoeba histolytica** (parasite)]

NB: The bacteriology of empyema may differ depending the cause of empyema on whether the infection was community acquired or hospital acquired.

- 1-Gram-positive bacteria, particularly Streptococcus species, are most common in **community-acquired empyema**.
- 2-Pseudomonas and Staphylococcus aureus (methicillin-resistant S. aureus (MRSA) are most particularly prevalent in **hospital-acquired empyema**.
- 3-Anaerobes are common with aspiration pneumonia, lung abscess, sub diaphragmatic abscess.
- 4- Staphylococcus aureus is also the most common organism in **trauma and surgery**.

Pathological classification of empyema

- 1- Exudative (early) empyema
- 2- Fibrino-purulent (established) empyema
- 3- Organizing empyema

Stage I: Exudative (early) stage:

- This is an inflammatory process in which there is an increase in permeability of small blood vessels leading to exudation of fluid in the pleural cavity
- The fluid is very thin with low cellular content and underlying lung that re-expands readily

Stage II: Fibrino-purulent (established) stage:

- **Bacterial invasion of the pleural space**
- Large number of polymorphonuclear leucocytes
- Deposition of fibrin on both visceral and parietal surfaces of the involved pleura
- Tendency towards loculation

Stage III: Organizing stage:

- The fluid is very thick
- In this case fibroblasts appear in the heavier fibrin coating of the pleural membranes

Clinical picture of empyema:

Symptoms:

- Symptoms of the cause, such as pneumonia.....
- Manifestations of toxemia→ fever, anorexia....
- Chest pain, cough, dyspnea
- Expectoration of pus as a result of bronchopleural fistula

Signs:

- 1- Signs of the cause
- 2- Same chest finding as pleural effusion, but in chronic empyema (pleural fibrosis), there is retraction and crowding of the ribs (interspace between ribs is narrower compared to the opposite side)

NB: finger clubbing is detected in case of chronic neglected empyema

Investigations:

1- Chest x-ray → resemble pleural effusion

2- Hematological investigations: anemia, leukocytosis, ESR (↑)

3- Pleural fluid aspiration and analysis

Microbiology investigations for pleural fluid and sputum:

Pleural fluid:

- 1- Gram stain & **culture (aerobic and anaerobic) and sensitivity**
- 2- Direct smear and culture for AFB

- 3- Microbiological investigations (stain, culture) according to the suspected other micro-organism.

Sputum:

- 1- Gram stain & culture and sensitivity
- 2- Direct smear and culture for AFB
- 3- Microbiological investigations (stain, culture) according to the other suspected micro-organism.

4- Chest US:

- 1- Help the diagnosis of empyema
- 2- Detection of adhesions, loculations
- 3- May detect the cause → pneumonia...
- 4- Sonographic guided diagnostic and therapeutic aspiration

5-Abdominal US: to detect abdominal lesions such as liver abscess...

6-CT chest:

- 1- Detect the pleural fluid and loculations
- 2- Detect airway or parenchymal abnormality e.g. the presence of lung abscess, endobronchial obstruction ...

Treatment (cause, empyema):

A- Control the infection by appropriate antimicrobial drugs

Antibiotics → proper (type -dose –duration). Start empirical antibiotic treatment and modify according to the result of culture and sensitivity

B-Evacuation of pleural fluid →

- 1-to allow re-expansion of the underlying lung to restore function,
- 2- to prevent complications

NB: Method of evacuation depends on the stage of the empyema

1- Repeated Pleural fluid aspiration (thoracocentesis): It may be adequate only in exudative stage (thin fluid).

2-Surgical treatment:

a-Intercostal tube drainage (underwater seal drainage): This is done if the fluid (pus) in the pleural sac is thicker to be evacuated by simple needle aspiration. If this treatment not sufficient for drainage of the fluid, intrapleural fibrinolytic agent through the tube is injected.

Intrapleural fibrinolytic agents have a fibrinolytic activity so dissolution of fibrin strands increases drainage of pleural fluid, which facilitate resolution of the infection and lung re-expansion

b-Decortication: thoracotomy is done and peel out the cortical layer over the parietal and visceral surfaces.

Complications of empyema:

1-Local:

- Lung collapse, respiratory failure
- Bronchopleural fistula
- Empyema necessitans (**Empyema necessitans** is the extension of an empyema through the parietal pleura into surrounding tissue, most commonly the chest wall to form a subcutaneous abscess that may eventually rupture through the skin).

2- Systemic: -Septicemia -Septic shock

3-Complications of drainage:

- Lung injury
- Hemorrhage,
- Subcutaneous emphysema
- Pneumothorax.....

4-Re-expansion pulmonary edema

5-Amyloidosis (in chronic neglected cases)

Pneumothorax

Definition: Pneumothorax can be defined as air in the pleural cavity

There are two classifications for pneumothorax:

- 1- Etiological classification, according to the cause of pneumothorax
- 2- Pathological classification

Etiological classification of pneumothorax:

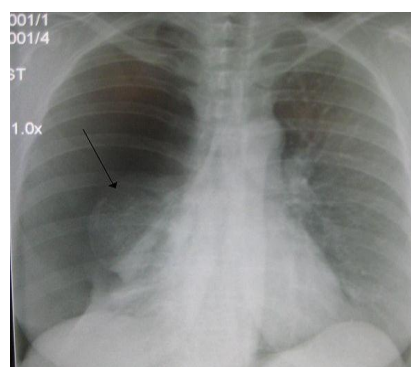
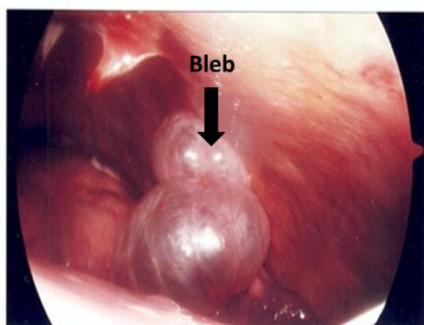
- 1-Spontaneous pneumothorax: primary, secondary
- 2-Traumatic pneumothorax: iatrogenic, non-iatrogenic

1-Spontaneous pneumothorax: (*primary, secondary*)

Any pneumothorax occurring in absence of trauma

A- Primary spontaneous pneumothorax :

- Pneumothorax most commonly occurs in apparently healthy subject with **no history of pre-existing lung disease.**
- Pneumothorax results from the rupture of a pulmonary bleb (a small air-filled space) into the pleural cavity.
- Tall and thin adolescent males are at great risk for primary spontaneous pneumothorax.
- Smoking is a risk factor.



B- Secondary spontaneous pneumothorax:

Secondary spontaneous pneumothorax occurs due to underlying chest diseases, such as:

- 1- **Chronic bronchitis and emphysema (most common causes of secondary spontaneous pneumothorax),**

2-Asthma

3- Pulmonary tuberculosis

4-Necrotizing pneumonia (such as staph and Klebsiella pneumonia), because pneumatoceles rupture into pleural cavity

5-Rare lung causes such as occupational lung diseases (silicosis), rheumatoid arthritis affects the lungs.

2-Traumatic pneumothorax: (iatrogenic, non-iatrogenic)

A- Non-iatrogenic pneumothorax:

1- **Open chest trauma** such as (gun shot, stab wound) and **closed chest trauma** (blunt trauma) such as motor car accident

2-**Barotrauma** (such as diving).

B-Iatrogenic pneumothorax:

Iatrogenic pneumothorax: it is a traumatic pneumothorax secondary to invasive procedure or surgery. Such as the following causes:

1- Pleural fluid aspiration (thoracocentesis) is **most common cause of iatrogenic pneumothorax**

2- Pleural biopsy

3- Lung biopsy.

5-Central venous cannulation

6-Needle biopsy of the mediastinum, breast or liver

5-Barotrauma (due to positive pressure during mechanical ventilation)

Pathologically, pneumothorax classified into :

1-Open pneumothorax:

The pleura is connected with the atmosphere as a result of an open chest wound.

The pleural cavity pressure is equal to the atmospheric pressure.

2-Closed pneumothorax:

The pleura is not connected with the atmosphere. A type of pneumothorax in which air can enter the pleural space but cannot escape via the route of entry.

The pleural cavity pressure is less than the atmospheric pressure

3-Tension pneumothorax:

It is a severe condition develops when air continuously enters the pleura without evacuation [air enters into the pleural cavity during inspiration but will not allow any air to escape during expiration (one-way valve)], resulting in displacement of the mediastinal structures, and disturbing cardiopulmonary function.

The pleural cavity pressure is greater than the atmospheric pressure.

Clinical picture (pneumothorax + cause):

Symptoms:

The most common mode of presentation is for previously fit young adult (primary spontaneous type) → patient is thin, tall and smoker.

The most common symptoms are chest pain and shortness of breath

1-Sudden unilateral sharp, stabbing chest pain. The pain increases by deep inspiration and coughing and may radiate to the ipsilateral shoulder.

2-Dyspnea depends on amount of pneumothorax and presence of underlying lung disease.

3-Cough: dry cough. But cough is productive in cases such as bacterial pneumonia

4-Symptoms of the cause.

Signs:

A- General examination:

Tension pneumothorax: patient is anxious, restless, tachypneic and cyanosed. Unless the tension is relieved the patient worsen, developing a rapid, small volume pulse and hypotension.

B-Chest examination:

1-Inspection: diminished chest movement and bulge (same side).

2-Palpation: diminished respiratory movement and TVF (same side). Trachea is shifted to the opposite side

3-Percussion: hyper-resonant note (same side).

4-Ascultation: diminished, or absent breath sound (same side), amphoric breathing with tension pneumothorax.

5-Signs of an associated pleural effusion or hemothorax are likely to be overlooked unless it is of moderate size (**shifting dullness**).

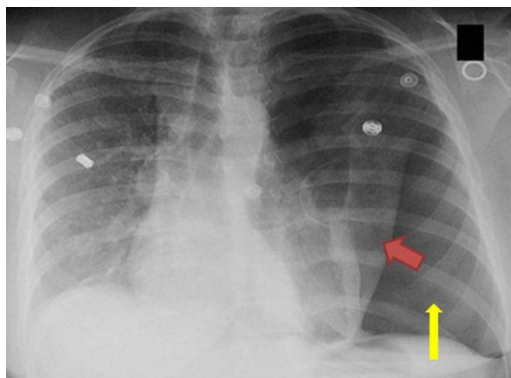
NB: Subcutaneous emphysema may be present

6-Signs of the cause.

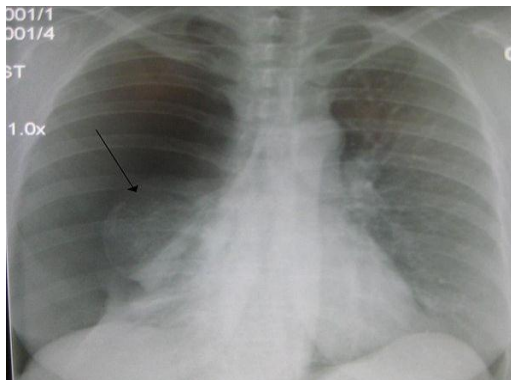
Investigations:

1-X-ray chest: first investigation performed to assess pneumothorax, because it is simple, inexpensive, rapid, and noninvasive

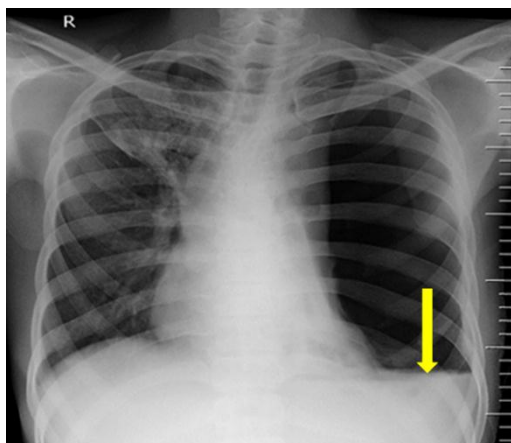
Radiological finding of the pneumothorax, the cause and complications.



- *Air in the pleura* appears as jet black area devoid of lung marking separating the collapsed lung from the bony cage (**characteristic finding**).



- *Lung* →partially or completely collapsed. In severe cases lung appears as globular mass at the hilum& displacement of the mediastinum to the opposite side.



- When there is also fluid as well as air in the pleural cavity it appears as completely horizontal line with no rise to the axilla (**fluid level**).

2-Blood gases analysis:

In severe cases → type 1 respiratory failure.

But in presence of associated COPD type 2 respiratory failure may be present.

3-Chest Ultrasound:

Advantage:

- 1- Ultrasound examination can be done more quickly than a chest x-ray
- 2- The lack of ionizing radiation.

Disadvantage: is lack of accuracy in the evaluation of pneumothorax size

4-CT chest: is the established *gold standard investigation* for the diagnosis of pneumothorax, but it is *not recommended for routine* use in pneumothorax

Treatment of the pneumothorax:

The main aim is to relieve the pressure on the lung and allow it to expand

-Treatment of pneumothorax depends upon:

- a-Volume of pneumothorax
- b-Presence of underlying lung disease associated with a pre-existing impairment of lung function so that even a small volume of pneumothorax may be serious

-Treatment of the cause (don't forget treatment of the cause)

Treatment of pneumothorax (conservative, active treatment)

A- Conservative treatment of pneumothorax:

Indicated in case of small volume of pneumothorax (< 20%)

- **Oxygen therapy:** Absorption of gas from pneumothorax space is fasted by breathing a **high concentration of oxygen**. Provided that the pneumothorax remains closed, the lung will re-expand slowly.
 - High concentration of oxygen should be avoided in patient with COPD (stable COPD and exacerbation), because it can cause an increase in PaCO₂ because of reduction in respiratory drive
- **Rest** (avoid heavy physical exertion).

B- Active treatment:

Indication: Volume of the pneumothorax **moderate or large** (>20%)

1- Intercostal tube drainage (underwater seal) → in which an intercostal tube is inserted into the pleural space and connected to an underwater seal bottle.

2-Intercostal tube drainage and chemical pleurodesis

For **recurrent spontaneous pneumothorax** (by concentrated glucose 50%, or tetracycline). The used substance introduced into pleural cavity through the chest tube, in order to produce a sterile pleurisy, with the subsequent formation of adhesions between the visceral and parietal pleural layers to prevent recurrence of pneumothorax.

3-Thoracoscopic treatment (video thoracoscopy): pleural abrasion

4- Thoracotomy: pleural abrasion, or parietal pleurectomy

Indicated if the lung fails to expand fully with tube drainage provided that the patient's general condition permits thoracoscopy or thoracotomy

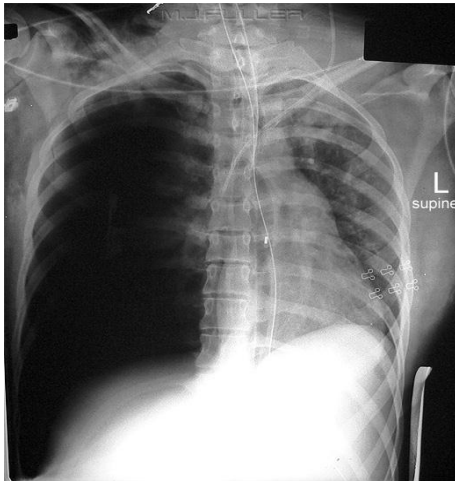
Tension pneumothorax

- It is **life threatening condition**.
- The pleural pressure is **more than** the atmospheric pressure.
- It is more common with: mechanical ventilation & traumatic pneumothorax.
- Clinical manifestations of pneumothorax, associated with clinical manifestations of circulatory collapse (tachycardia, hypotension).

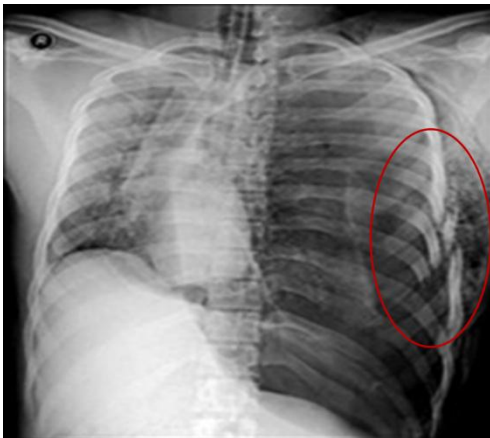
Radiological manifestations

- Large pneumothorax
- Lung collapse
- Mediastinal shift (to opposite side) &
- Flattening or depression of the hemidiaphragm (same side)

Chest X-ray of right tension pneumothorax



CXR of left tension pneumothorax with fracture ribs and subcutaneous emphysema



Management of tension pneumothorax

In the absence of hemodynamic compromise, it is prudent to wait for the results of an emergent chest X-ray prior to intervention.

1-Intercostal tube drainage (underwater seal) is the definitive treatment (after x-ray)

2- Prehospital treatment:

In clear cut cases:

1-Shock with 2-Distended neck veins 3-Reduced breath sounds

4-

Deviated trachea, it could be lifesaving to treat with:

a- Needle thoracostomy (prehospital setting).

b-Surgical thoracostomy without chest tube placement have been described in the pre-hospital setting.

Also, oxygen therapy used with the previous treatment lines

Complications of pneumothorax:

1-Failure of lung re-expansion → due to persistent air leak.

2-Respiratory failure

3-Shock, cardiac arrest

4-Recurrence in same side or opposite side

5-Pyopneumothorax, following penetrating trauma, suppurative pneumonia, rupture lung abscess or complication of intercostal tube.

6-Haemopneumothorax in case of trauma

Indication of intercostal tube in treatment of pleural diseases:

1-Empyema

2-Pneumothorax (moderate, large or tension)

3- Pneumothorax in patient on mechanical ventilation

3-Heamothorax

4-Pyopneumothorax

5-Heamo-pneumothorax

6-Malignant effusion (large amount, rapidly accumulating)

NB:

Needle Thoracostomy Emergent chest decompression (in the prehospital setting)→ with needle thoracostomy. A 14-16G intravenous cannula is inserted into the second intercostal space in the mid-clavicular line

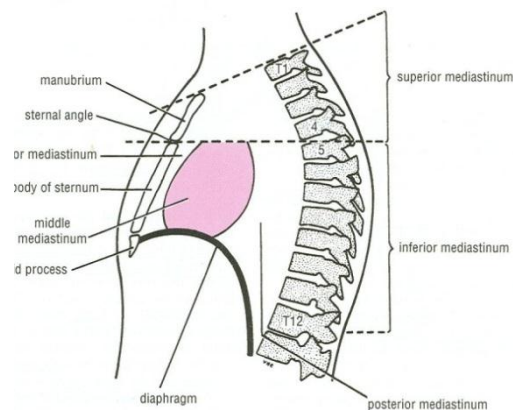


Mediastinum and Mediastinal Diseases

Anatomy of the mediastinum:

Mediastinum is the central space within the thoracic cavity **bounded by:**

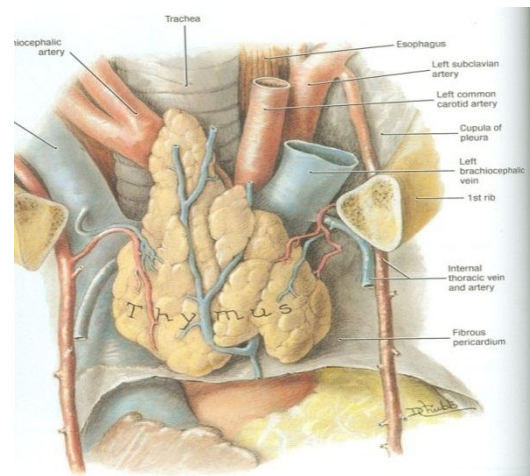
- Sternum anteriorly
- Lungs and parietal pleura laterally
- The vertebral column posteriorly
- The thoracic inlet superiorly
- The diaphragm inferiorly



Divisions of the mediastinum

(1) Superior mediastinum: it contains

- Aortic arch & its 3 branches
- Superior vena cava (S.V.C.) & its 2 innominate veins
- Trachea, esophagus, thoracic duct
- Vagus, phrenic nerves, left recurrent laryngeal nerve, and sympathetic chain.
- Lymph nodes & thymus.



(2) Anterior mediastinum:

Boundary:

- Anterior → Sternum
- Posterior → Pericardium

Contents:

- Thymus
- Lymph nodes
- Fatty tissue

(3) Middle mediastinum: Boundary: By the 3 divisions.

Contents:

- Heart & pericardium
- Ascending aorta, superior vena cava (S.V.C) & inferior vena cava (I.V.C.)
- Pulmonary arteries & veins
- Tracheal bifurcation
- Phrenic nerves

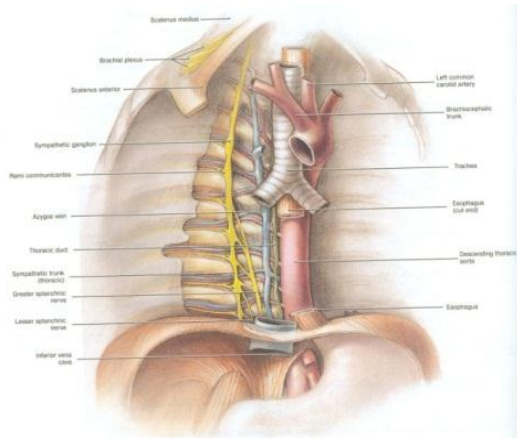
(4) Posterior mediastinum:

Boundary: Anterior → pericardium & diaphragm

Posterior → lower 8 thoracic vertebrae

Contents:

- Descending aorta
- Esophagus
- Sympathetic chain & vagus nerve
- Lymph nodes
- Thoracic duct



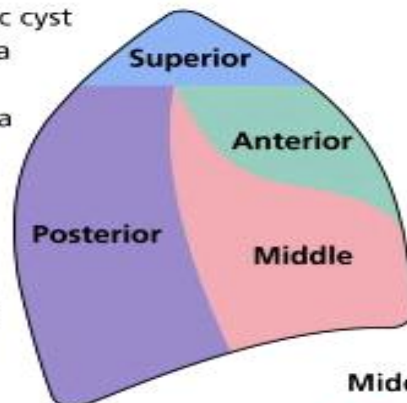
Mediastinal masses

Superior

- Thymoma and thymic cyst
- Malignant lymphoma
- Thyroid lesions
- Parathyroid adenoma

Posterior

- Neurogenic tumours
 - Schwannoma
 - Neurofibroma
 - Ganglioneuroma
 - Ganglioneuroblastoma
 - MPNST
 - Neuroblastoma
 - Paraganglioma
- Gastroenteric cyst



Anterior

- Thymoma and thymic cyst
- Germ cell tumors
- Thyroid lesions
- Parathyroid adenoma
- Malignant lymphoma
- Paraganglioma
- Hemangioma
- Lipoma

Middle

- Pericardial cyst
- Bronchial cyst
- Malignant lymphoma

Fig. 8.1 Location of most common lesions of mediastinum.

Location of common lesions of mediastinum

(A) Superior mediastinum:

- Thymoma and thymic cyst
- Malignant lymphoma
- Parathyroid adenoma

(B) Anterior mediastinum: (30% of mediastinal masses) (4 T's)

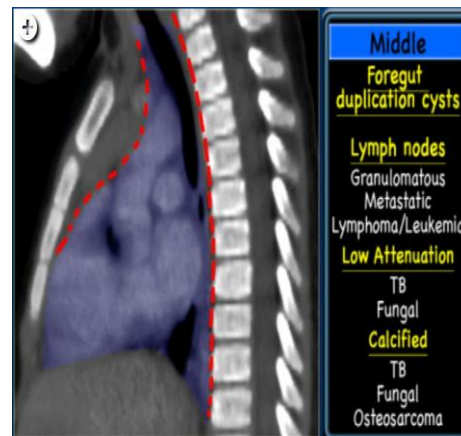
- **Thymoma:** The **most common** primary mediastinal neoplasm in adults and the **most frequent** tumor of the anterior mediastinum. It is a malignant tumor originating from the epithelial cells of the thymus gland.
- **(Terrible) Lymphoma.**
- **Teratoma:** The most common mediastinal germ cell neoplasm.
- **Thyroid** lesions.

NB: Germ cell tumours include:

- (1) Teratoma
- (2) Seminomas
- (3) Nonseminomatous germ cell neoplasms

(C) Middle mediastinum: (30% of mediastinal masses)

- **Lymphadenopathy:** Benign, secondaries, lymphoma and leukemia.
- **Cysts:** Bronchogenic and enteric.
- **Vascular lesions:** Aortic and pulmonary artery aneurysms.



(D) Posterior mediastinum:

- **Neurogenic tumors:** The **most common** cause of posterior mediastinal mass.
- **Spinal lesions:** Meningocele.
- **Esophageal lesions:** Esophageal neoplasms, esophageal duplication cyst.

Mediastinal Syndrome

Compression of mediastinal structures by any mass gives rise to a group of symptoms known as mediastinal syndrome.

Manifestations depend on: * Site of lesion

* Structure involved

Pressure symptoms:

- **Esophagus** → dysphagia.
- **Trachea & bronchi** → brassy cough, stridor, obstructive emphysema due to partial obstruction (valve like) or atelectasis due to complete obstruction
- **Arteries** of upper limbs → **unequal** pulse, ischemic manifestations: pallor, pain
- **Veins:** usually superior vena cava → congested non pulsating neck veins, dilatation of collaterals.
- **Nerves:**
 - * **Sympathetic chain** → Horner's syndrome (ptosis, miosis, anhydrosis and enophthalmos).
 - * **Left recurrent laryngeal nerve** → hoarseness of voice.
 - * **Phrenic nerve** → diaphragmatic paralysis.

Diagnostic Procedures

1-Computed tomography (CT)

CT has become the initial imaging procedure for evaluation of the mediastinum.

2-Magnetic Resonance Imaging (MRI)

MRI is a useful diagnostic tool for evaluating mediastinal masses that appear indeterminate on CT.

MRI is a useful diagnostic tool for differentiating cystic from solid lesions and evaluating invasiveness into adjacent structures.

3-Positron Emission Tomography (PET scan)

PET scan is a useful addition to traditional imaging approaches. It can decrease the need for invasive diagnostic procedures. PET imaging is used to stage, plan treatment, and monitor the response to treatment and follow up.

4-Mediastinoscopy

Allows direct inspection and biopsy of lymph nodes or other masses in the mediastinum.

Mediastinoscopy remains the **gold standard** for invasively staging the mediastinal lymph nodes.

Tissue Diagnosis

The biopsy is taken by one of the following procedures:

1- Radiological procedures: Sonographic or CT-guided biopsy.

2-Transbronchial biopsy.

3-Transesophageal biopsy.

4-Surgical procedures and biopsy: Mediastinoscopy, thoracoscopy and mediastinotomy.

Treatment according to cause:

- **Surgery.**
- **Chemotherapy.**
- **Radiotherapy.**

Malignant Pleural Mesothelioma

The **most common** primary malignant tumor of the pleura is malignant mesothelioma, an insidious neoplasm with a **poor** prognosis, arising from the mesothelial surfaces of the pleural and peritoneal cavities, as well as from the tunica vaginalis and pericardium. **Eighty percent** of all cases of mesothelioma are pleural in origin.

Malignant pleural mesothelioma (MPM) incidence is now declining, secondary to control of exposure to asbestos. Median survival is roughly 1 year. MPM predominantly affects males.

Etiology

A- Asbestos Exposure

- Inhalational exposure to **asbestos (carcinogenic)** has been clearly established as the predominant cause of malignant mesothelioma in humans. (Approximately 70 percent of cases of pleural mesothelioma are associated with documented asbestos exposure).
- In addition to asbestos miners and workers, other occupations at especially high risk include plumbers, mechanical engineers, ship and boat building and repairing.
- The latency period from asbestos exposure to the development of mesothelioma ranges from approximately 20 to 50 years. Despite extensive investigatory effort, the exact mechanisms of asbestos carcinogenesis have not yet been fully elucidated.

B- Other Etiologic Factors

- The development of malignant pleural mesothelioma has also been associated in rare cases with other etiologic factors such as **therapeutic irradiation**.

Pathology

Gross Pathology

1-Diffuse malignant mesothelioma. The vast majority of malignant mesotheliomas involving the pleura are those tumors that diffusely involve the pleura and are properly termed “diffuse malignant mesothelioma”.

Diffuse malignant mesothelioma begins as multiple discrete nodules that, in earlier stages, tend to preferentially involve the parietal pleura over the visceral pleura. In time, these nodules tend to coalesce on the visceral and parietal pleural surfaces with subsequent fusion of the pleurae.

2-Localized malignant mesothelioma. A **rare** localized gross variant of malignant mesothelioma that forms a single mass attached to the pleura but is otherwise microscopically identical to diffuse malignant mesothelioma has been described and termed “localized malignant mesothelioma”.

Histology

There are three different pathological types of MPM: the epithelioid, the sarcomatoid and the biphasic type.

1-The epithelioid MPM: is the **most common** type, with proliferation of oval or polygonal tumor cells. The prognosis of the epithelioid type is better than the non-epithelioid.

2-The sarcomatoid type is mainly characterized by the proliferation of spindle cells with oval nuclear formation. The sarcomatoid type has the worst prognosis.

3-The biphasic (mixed) type has both epithelioid and sarcomatoid components. Each component should represent at least 10 percent of the tumor for the designation of biphasic. Biphasic mesotheliomas represent about 30 percent of cases.

Clinical manifestations

- Malignant pleural mesothelioma most commonly presents in the fifth to seventh decades of life. Most patients diagnosed with mesothelioma have a history of asbestos exposure.

The most frequent presenting symptoms of pleural mesothelioma are

1-Chest pain (60 to 70 percent of patients) which tends to be dull or boring in nature.

2-Dyspnea (25 percent), and

3-Cough (20 percent).

Some patients are asymptomatic at diagnosis, with unilateral pleural effusions found incidentally on routine chest radiographs

Mesothelioma is typically a unilateral disease—only 10 percent of patients with mesothelioma have bilateral involvement at presentation.

Physical findings may include

1-unilateral dullness to percussion throughout the hemithorax

2- palpable chest wall masses

3-scoliosis toward the side of the malignancy.

Diagnosis

Laboratory Studies

Pleural effusion associated with mesothelioma is **exudative effusion**.

Tumor markers:

1-Mesothelin

There is increasing evidence for the clinical utility for a monoclonal antibody-based (**serum assay**) for the diagnosis of MPM.

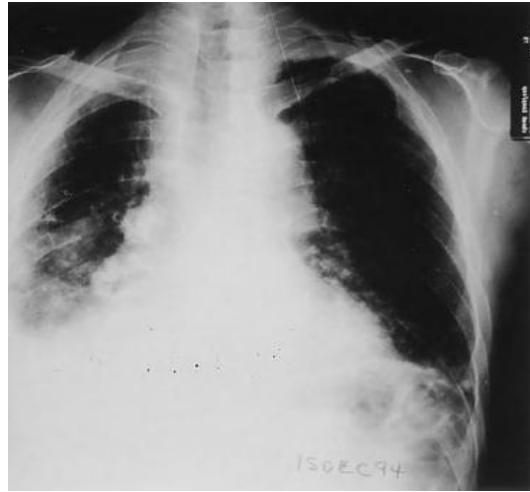
2-Immunohistochemical markers

The selection of immunohistochemical markers (**tissue assay**) depends on the differentiated tumors.

Imaging

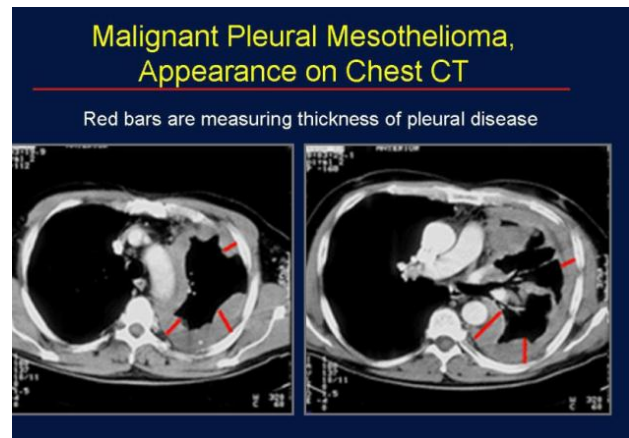
The imaging diagnosis of MPM is based on X-ray, CT, PET/CT and MRI.

1-X-ray is the most simple and practical method, which manifests pleural thickening, pleural nodules or pleural effusion. The X-ray **can't make** a confirmed diagnosis for MPM.



2-Chest ultrasonography can demonstrate pleural thickening or effusion in patients with mesothelioma. Sonographic-guided biopsy can be obtained.

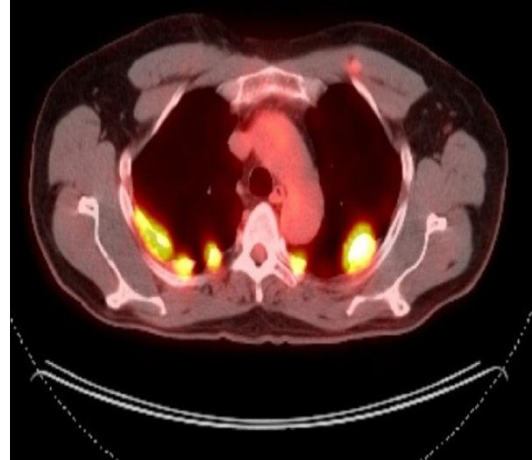
3-Chest CT is commonly used in the diagnosis of MPM. The chest CT mainly manifests a nodular or cyclic thickening of the pleura. But it is still difficult in the tumor staging and distinguishing the diffuse pleural thickening from MPM. CT-guided biopsy can be obtained.



4-MRI: The imaging features of MRI are similar to the chest CT.

MRI is mainly used for the preoperative assessment of mediastinal structures, chest wall and diaphragm involvement, which is not used as a routine examination.

5-PET/CT can be used for the diagnosis of MPM as well using 18F-fluorodeoxyglucose (18FDG). The characteristic images of MPM is the concentration of 18FDG in lesion, which can make up the deficiency of CT images in MPM staging and play an important role in the diagnosis, follow up and estimating prognosis.



Pleural biopsy:

The diagnosis of MPM should always be based on histopathology of pleural biopsy; **Sonographic or CT-guided biopsy**, or more invasive methods if these fail e.g., **thoracoscopic pleural biopsy**.

Treatment

MPM is characterized by being **highly malignant and aggressive**. Most patients died of the local invasion of the tumor or the contralateral metastasis.

A variety of treatments have been taken and the comprehensive treatment of resecting the visible tumor as much as possible, combined with radiotherapy, chemotherapy and immunotherapy is the most promising strategy up to date.

Unfortunately, the efficacy of treatment is **very limited** for the tumor, usually diagnosed too late.

1-Surgical treatment

The purpose of surgical treatment is to remove the visible tumor, to eliminate the pleural effusion, to relieve symptoms such as dyspnea, to ease the pain, and to increase the efficacy of adjuvant therapy

2-Radiotherapy

Mesothelioma is **not sensitive** to radiotherapy and the radiotherapy effect is generally unsatisfactory. However, radiotherapy has a role in relieving the symptoms, especially easing the pain.

3-Chemotherapy

The purpose of chemotherapy is to kill the rapid proliferating tumor cells, so as to prolong the survival and improve the quality of life. Chemotherapy for MPM can be used alone or combined with surgical treatment.

4-Immunotherapy

There is a new hope in the treatment of MPM through modulation of T cell immunity.

5-Palliative treatment

Palliative treatment is advised to patients with **symptomatic malignant pleural effusion** and include surgical procedures like: **Pleurodesis or pleurectomy**.

Sleep-disordered breathing

Introduction

- Physiological changes during sleep:- normally respiratory function is significantly altered **by Sleep**.
- Following sleep onset, central motor neuron output falls. This means that diminished upper airway dilator muscle function leads to a narrower upper airway, while reduced inspiratory muscle activity causes a shallower breathing pattern compared to the awake state.

Risk factors

These physiological changes occur in everyone, but may become problematic if combined with one or more of these risk factors: -

1. **Obesity** (Trunkal obesity)
2. **Enlarged neck circumference.**
3. Obstruction of the upper airway e.g. **enlarged tonsils, adenoids, or polyps**
4. **Retrognathia**; is term used to describe the mandible that is posterior to and behind where it should be. OSA is more common with retrognathia as the airway is more closed and constricted.
5. **Acromegaly**; OSA is more common in patients with acromegaly due to enlargement of the tongue and thickening of the tissues surrounding the upper airway.

Sleep-disordered breathing (SDB)

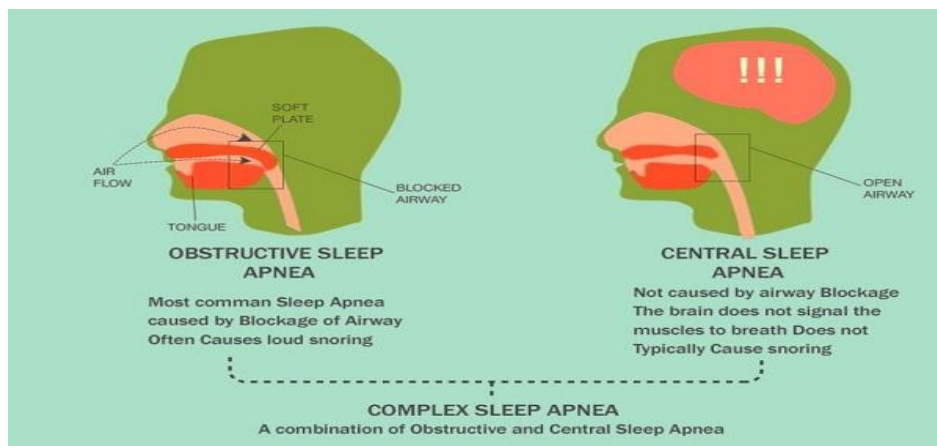
Definition

Any abnormal breathing **at night** leading to symptoms, sleep disturbance, or long-term complications.

Classifications: -

There are different types of SDB:

1. Obstructive sleep apnea (OSA).
2. Central sleep apnea (CSA).
3. Complex sleep apnea.
4. Obesity hypoventilation syndrome (OHS).

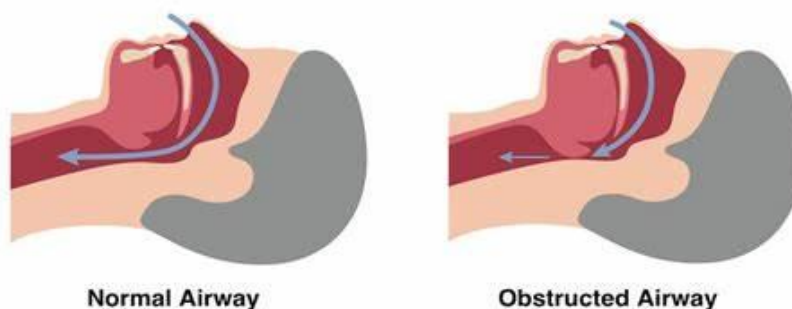


1-Obstructive sleep apnea (OSA)

Definition

Repetitive upper airway obstruction, causing **recurrent arousals** lead to highly **fragmented sleep**, which result in significant **daytime symptoms** but with ***maintained breathing effort.***

- OSA is characterized by **upper airway obstruction** during sleep
- OSA is the **most common** type of SDB accounting about 85% of the cases.



Clinical picture

1. Excessive daytime sleepiness
2. Loud snoring and apneic episodes recognized by the bed partner
3. Waking up with choking from time to time
4. Poor concentration and poor memory
5. Mood changes and depression
6. Unrefreshing sleep
7. Nocturia
8. Insomnia
9. Reduced libido

Serious consequences to undiagnosed or untreated OSA:

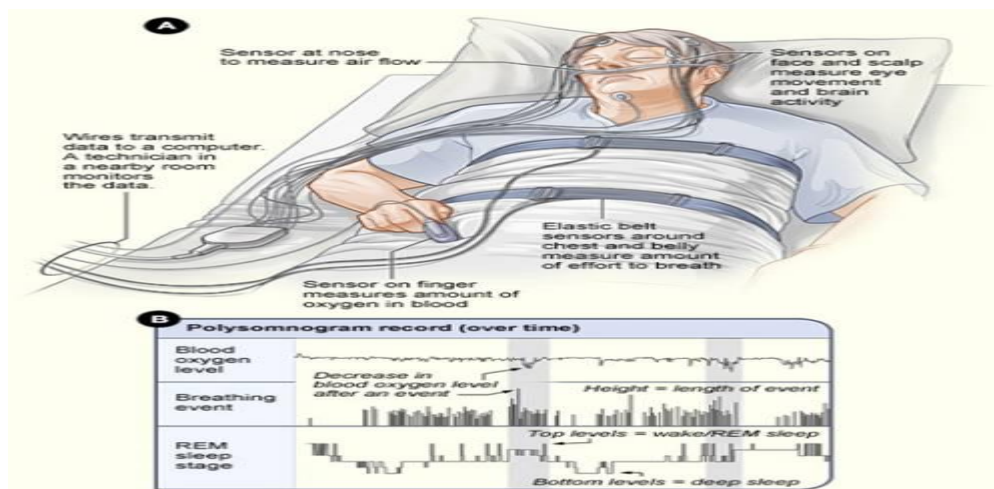
1. Cardiovascular disease; hypertension, myocardial infarction.
2. Type 2 diabetes
3. Stroke
4. Cognitive impairment
5. Mood disorders e.g. depression.
6. Gastro-esophageal reflux disease.
7. Motor car accidents due to OSA-related daytime sleepiness.

Investigation

Sleep study (Polysomnography) is the main diagnostic test.

- **Sleep study parameters are: -**
 1. **Apneas** are **defined** as cessation in respiratory airflow greater than 90% for a duration greater than 10 seconds.
 2. **Hypopneas** are **defined** as reductions in inspiratory airflow of greater than 30%, for duration >10 seconds, accompanied by arousal or reduction in oxygen saturation of at least 3%.
- **OSA severity is assessed by apnea-hypopnea index (AHI)**

- **AHI** represents the total apnea and hypopnea episodes per hour of sleep.
- **AHI thresholds of disease severity are:** -
 - AHI <5: normal
 - AHI 5–15: mild OSA
 - AHI 15–30: moderate OSA
 - AHI >30: severe OSA



Treatment

A- General therapeutic recommendations.

1. Weight loss
2. Positional therapy: avoidance of sleeping in a supine posture, instead lateral decubitus with the bedhead elevated is recommended.
3. Avoidance of sedative drugs such as benzodiazepines; to avoid respiratory center depression.
4. Avoidance of alcohol.

B- Specific therapy (Medical/ surgical)

Indicated to control and optimize upper airway patency while asleep

A- Medical treatment

1. **Continuous positive airway pressure (CPAP).**
2. **Bi-level positive airway pressure (BiPAP).**
3. **Mandibular advancement device.**

1- Continuous positive airway pressure (CPAP)

- CPAP delivers a **single** and **constant (fixed)** level of pressure during both inhalation and exhalation.
- CPAP is the **most common** and **reliable** method of treating OSA.

It is **highly effective** therapy due to **its benefits**;

1. Reduction the number of apneic episodes during sleep.
2. Improvement of sleep fragmentation.
3. Resolution of daytime sleepiness.
4. Improvement quality of life.

- **Indications of CPAP therapy:**

1. **Moderate or severe** sleep apnea (i.e., an AHI ≥ 15 events/h)
2. **Mild OSA** with **associated symptoms**.

- **A CPAP machine consists of:**

A blower (**pressure generator**) that sits by the patient's bed and is connected to a **mask** by a large-bore **tube**. **The masks** are either **nasal** or cover the nose and mouth (**full face mask**). The blower **raises** the pressure (between 4-20 cmH₂O pressure) to **splint** the pharynx and prevent its collapse.

2- Bi-level positive airway pressure (BiPAP).

- It is a type of **non-invasive ventilation** (NIV). BiPAP delivers **two** levels of air pressure; one level during inhalation and another

lower one during exhalation, so the BiPAP makes it **easier to breathe out** than CPAP.

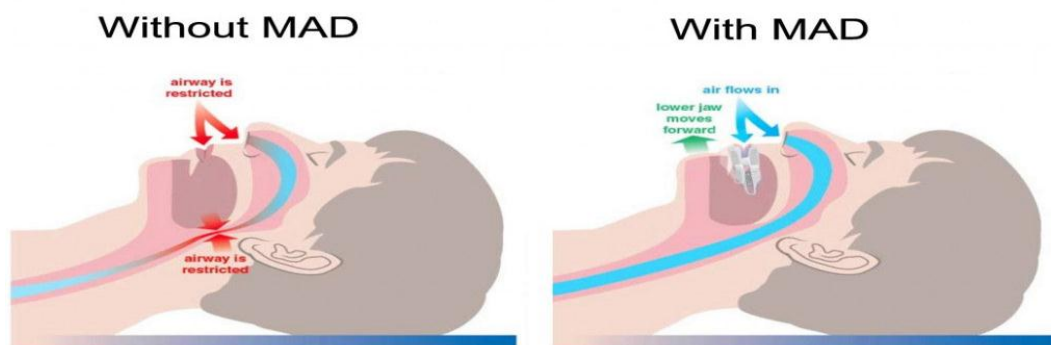
➤ It is **indicated in** cases of OSA associated with “**CO₂ retention**” for example these following cases: -

1. Severe OSA in case of very obese patient.
2. OSA combined with COPD (**overlap syndrome**).
3. OSA combined with obesity hypoventilation (**Pickwickian syndrome**).



3- Mandibular advancement device (MAD)

- A device worn in the mouth during sleep to keep the soft tissue in the airway from collapsing.
- **Indicated in** patients with mild to moderate OSA who are **unable** to tolerate CPAP.



B- Surgical treatment

1. **Pharyngeal surgery** e.g. to excise enlarged tonsils, polyps, and adenoids.
2. **Bariatric (obesity) surgery:** indicated if BMI >35 kg/m².
3. **Tracheostomy:** indicated for patients with **severe OSA** who **have failed** medical or surgical therapy.

2-Central sleep apnea (CSA)

Definition:

CSA is episodic reduced ventilation during sleep. It occurs because the brain fails to send proper signals to the muscles that control breathing function.

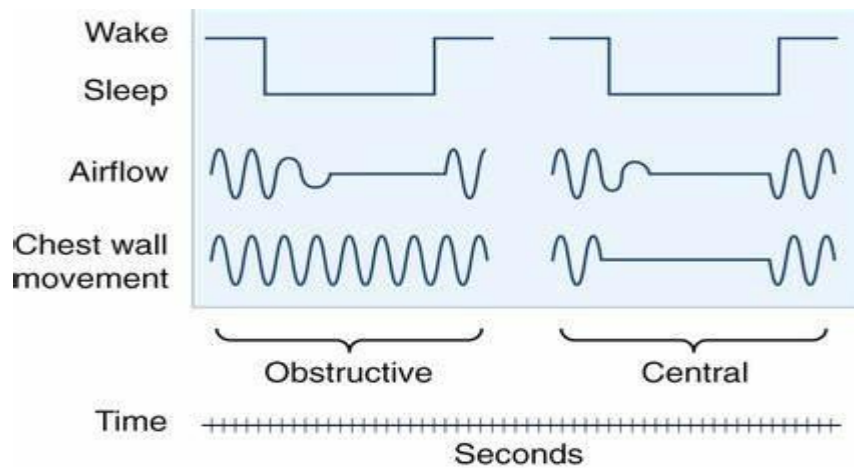
- CSA differs from OSA in that respiratory distress is not due to airway obstruction but it happens as there is no central inspiratory effort.
- CSA is less common than OSA.

Causes:

1. Heart failure (Cheyne–Stokes respiration)
2. Acute or chronic neurological conditions affecting brainstem function (e.g. stroke, tumors)
3. Medication use: Opiates, Sedatives (e.g. benzodiazepines)
4. Alcohol consumption.

Treatment:

Improvement or removal of the underlying causal factors, whether related to heart failure or medication use.



3-Complex sleep apnea

Complex sleep apnea is combination of OSA and CSA.

4- Obesity hypoventilation syndrome (Pickwickian syndrome)

Obesity hypoventilation syndrome (OHS) is **defined as** a combination of

- 1- Obesity (body mass index ≥ 30)
- 2- Daytime hypercapnia (arterial carbon dioxide tension ≥ 45 mmHg)
- 3- Sleep disordered breathing usually OSA

After **exclusion** of other disorders that may cause **hypoventilation** e.g COPD.

Pickwickian Syndrome

Obstructive sleep apnea was called the Pickwickian syndrome in the past because Joe the Fat Boy who was described by Charles Dickens in the Pickwick papers had typical features with snoring, obesity, sleepiness.

Joe from The Pickwick Papers
by Charles Dickens